Chapter 1

Operations Research: An Introduction

"The first rule of any technology used in a business is that automation applied to an efficient operation will magnify the efficiency. The second is that automation applied to an inefficient operation will magnify the inefficiency."

– Bill Gates

PREVIEW

This chapter presents a framework of a possible structural analysis of problems pertaining to an organization in order to arrive at an optimal solution using operations research approach.

LEARNING OBJECTIVES

After reading this chapter you should be able to

- understand the need of using operations research – a quantitative approach for effective decision-making.
- know the historical perspective of operations research approach.
- know various definitions of operations research, its characteristics and various phases of scientific study.
- recognize, classify and use of various models for solving a problem under consideration.
- be familiar with several computer software available for solving an operations research model.

CHAPTER OUTLINE

1.1 Operations Research – A Quantitative Approach to Decision-Making
1.2 The History of Operations Research
1.3 Definitions of Operations Research
1.4 Features of Operations Research Approach
1.5 Operations Research Approach to Problem Solving
- Conceptual Questions A

1.6 Models and Modelling in Operations Research
1.7 Advantages of Model Building
1.8 Methods for Solving Operations Research Models
1.9 Methodology of Operations Research
1.10 Advantages of Operations Research Study
1.11 Opportunities and Shortcomings of the Operations Research Approach
1.12 Features of Operations Research Solution
1.13 Applications of Operations Research
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1.15 Computer Software for Operations Research
- Conceptual Questions B

❑ Chapter Summary
❑ Chapter Concepts Quiz
❑ Case Study
❑ Puzzles in Operations Research

1.1 OPERATIONS RESEARCH – A QUANTITATIVE PERSPECTIVE TO DECISION-MAKING

Knowledge, innovations and technology are changing and hence decision-making in today's social and business environment has become a complex task due to little or no precedents. High cost of technology, materials, labour, competitive pressures and so many economic, social, political factors and viewpoints, have greatly increase the complexity of managerial decision-making. To effectively address the broader tactical and strategic issues, also to provide leadership in the global business environment, decision-makers cannot afford to make decisions based on their personal experiences, guesswork or intuition, because the consequences of wrong decisions can prove to be serious and costly. For example, entering the wrong markets, producing the wrong products, providing inappropriate services, etc., may cause serious financial problems for organizations. Hence, an understanding of the use of quantitative methods to decision-making is desirable to decision-makers.

Quantitative analysis is the scientific approach to decision-making.

Few decision-makers claim that the OR approach does not adequately meet the needs of business and industry. Lack of implementation of findings is one of the major reasons. Among the reasons for implementation failure is due to creative problem solving inabilities of the decision-maker. The implementation process presumes that the definition, analysis, modeling, and solution phases of a project have been performed as per prescribed guidelines.

Operations research approach helps in the comparison of all possible *alternatives* (*courses of action* or *acts*) with respect to their potential outcomes and then sensitivity analysis of the solution to changes or errors in numerical values. However, this approach (or technique) is an aid to the decision-makers's judgement not a substitute for it.

While attempting to solve a real-life problem, the decision-maker must examine the given problem from both quantitative as well as qualitative perspective. For example, consider the problem of investments in three alternatives: Stock Market, Real Estate and Bank Deposit. To arrive at any decision, the investor needs to examine certain quantitative factors such as financial ratios from the balance sheets of companies whose stocks are under consideration; real estate companies' cash flows and rates of return for investment in property; and how much investment will be worth in the future when deposited at a bank at a given interest rate for a certain number of years? Also, certain qualitative factors, such as weather conditions, state and central policies, new technology, political situation, etc.?

The evaluation of each alternative can be extremely difficult or time consuming for two reasons: First, the amount and complexity of information that must be processed, and second the availability of large number of alternative solutions. For these reasons, decision-makers increasingly turn to quantitative factors and use computers to arrive at the optimal solution for problems.

Decision maker should consider both qualitative and quantitative factors while solving a problem.

There is a need for *structural analysis* using operations research/quantitative techniques to arrive at a holistic solution to any managerial problem. This can be done by critically examining the levels of interaction between the *application process* of operations research, and *various systems* of an organization. Figure 1.1 summarizes the conceptual framework of organizational/management structures and operations research application process.

This book introduces a set of operations research techniques that should help decision-makers in making rational and effective decisions. It also gives a basic knowledge of the use of computer software needed for computational purposes.

1.2 THE HISTORY OF OPERATIONS RESEARCH

Operations research approach considers environmental influences along with both organization structures, and managerial behaviour, for decision-making.

It is generally agreed that operations research came into existence as a discipline during World War II when there was a critical need to manage scarce resources. However, a particular model and technique of OR can be traced back as early as in World War I, when Thomas Edison (1914–15) made an effort to use a tactical game board for finding a solution to minimize shipping losses from enemy submarines, instead of risking ships in actual war conditions. About the same time AK Erlang, a Danish engineer, carried out experiments to study the fluctuations in demand for telephone facilities using automatic dialling equipment. Such experiments were later on used as the basis for the development of the waiting-line theory.

Since World War II involved strategic and tactical problems that were highly complicated, to expect adequate solutions from individuals or specialists in a single discipline was unrealistic. Thus, groups of

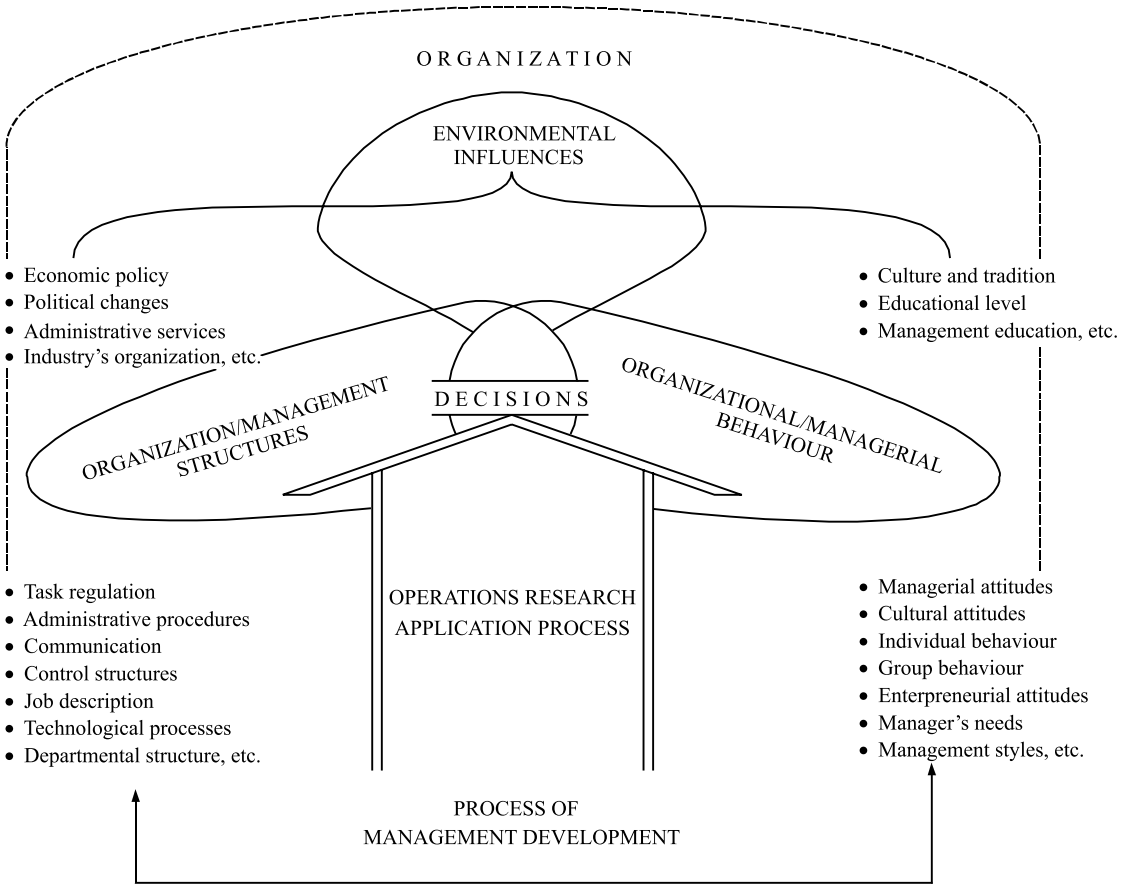


Fig. 1.1 Conceptual Framework of an Organization and OR Application Process

individuals who were collectively considered specialists in mathematics, economics, statistics and probability theory, engineering, behavioural, and physical science, were formed as special units within the armed forces, in order to deal with strategic and tactical problems of various military operations.

Such groups were first formed by the British Air Force and later the American armed forces formed similar groups. One of the groups in Britain came to be known as Blackett's Circus. This group, under the leadership of Prof. P M S Blackett was attached to the Radar Operational Research unit and was assigned the problem of analyzing the coordination of radar equipment at gun sites. Following the success of this group similar mixed-team approach was also adopted in other allied nations.

After World War II, scientists who had been active in the military OR groups made efforts to apply the operations research approach to civilian problems related to business, industry, research, etc. The following three factors are behind the appreciation for the use of operations research approach:

(i) The economic and industrial boom resulted in mechanization, automation and decentralization of operations and division of management functions. This industrialization resulted in complex managerial problems, and therefore the application of operations research to managerial decision-making became popular.

(ii) Continued research after war resulted in advancements in various operations research techniques. In 1947, G B Dantzig, developed the concept of *linear programming*, the solution of which is found by a method known as *simplex method.* Besides linear programming, many other techniques of OR, such as statistical quality control, dynamic programming, queuing theory and inventory theory were well-developed before 1950's.

(iii) The use of high speed computers made it possible to apply OR techniques for solving real-life decision problems.

The term '**operations research**' was coined as a result of conducting research on military operations during World War II

During the 1950s, there was substantial progress in the application of OR techniques for civilian problems along with the professional development. Many colleges/schools of engineering, public administration, business management, applied mathematics, computer science, etc. Today, however, service organizations such as banks, hospitals, libraries, airlines, railways, etc., all recognize the usefulness of OR in improving efficiency. In 1948, an OR club was formed in England which later changed its name to the

Operational Research Society of UK. Its journal, *OR Quarterly* first appeared in 1950. The Operations Research Society of America (ORSA) was founded in 1952 and its journal, *Operations Research* was first published in 1953. In the same year, The Institute of Management Sciences (TIMS) was founded as an international society to identify, extend and unify scientific knowledge pertaining to management. Its journal, *Management Science,* first appeared in 1954.

In India, during same period, Prof R S Verma set up an OR team at Defence Science Laboratory for solving problems of store, purchase and planning. In 1953, Prof P C Mahalanobis established an OR team in the Indian Statistical Institute, Kolkata for solving problems related to national planning and survey. The OR Society of India (ORSI) was founded in 1957 and it started publishing its journal *OPSEARCH* 1964 onwards. In the same year, India along with Japan, became a member of the International Federation of Operational Research Societies (IFORS) with its headquarters in London. The other members of IFORS were UK, USA, France and West Germany.

A key person in the post-war development of OR was George B Dantzig

A year later, project scheduling techniques – Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) were developed for scheduling and monitoring lengthy, complex and expensive projects of that time.

The American Institute for Decision Sciences came into existence in 1967. It was formed to promote, develop and apply quantitative approach to functional and behavioural problems of administration. It started publishing a journal, *Decision Science,* in 1970.

Because of OR's multi-disciplinary approach and its application in varied fields, it has a bright future, provided people devoted to the study of OR help to meet the needs of society. Some of the problems in the area of hospital management, energy conservation, environmental pollution, etc., have been solved by OR specialists. This is an indication of the fact that OR can also contribute towards the improvement of the social life and of areas of global need.

In India, operations research came into existence in 1949, when an OR unit was established at Regional Research Laboratory, Hyderabad, for planning and organizing research.

1.3 DEFINITIONS OF OPERATIONS RESEARCH

The wide scope of applications of operations research encouraged various organizations and individuals to define it as follows:

- *Operations research is the application of the methods of science to complex problems in the direction and management of large systems of men, machines, materials and money in industry, business, government and defence. The distinctive approach is to develop a scientific model of the system incorporating measurements of factors such as chance and risk, with which to predict and compare the outcomes of alternative decisions, strategies or controls. The purpose is to help management in determining its policy and actions scientifically.*

  – Operational Research Society, UK

- *The application of the scientific method to the study of operations of large complex organizations or activities. It provides top level administrators with a quantitative basis for decisions that will increase the effectiveness of such organizations in carrying out their basic purposes.*

  – Committee on OR National Research Council, USA

The definition given by Operational Research Society of UK has been criticized because of the emphasis it places on complex problems and large systems, leaving the reader with the impression that it is a highly technical approach suitable only to large organizations.

Operations research is concerned with scientifically deciding how to best design and operate man-machine systems that usually require the allocation of scarce resources.

- *Operations research is the systematic application of quantitative methods, techniques and tools to the analysis of problems involving the operation of systems.*

  – Daellenbach and George, 1978

- *Operations research is essentially a collection of mathematical techniques and tools which in conjunction with a system's approach, are applied to solve practical decision problems of an economic or engineering nature.* – Daellenbach and George, 1978

These two definitions imply another view of OR – being the collection of models and methods that have been developed largely independent of one another.

- *Operations research utilizes the planned approach (updated scientific method) and an interdisciplinary team in order to represent complex functional relationships as mathematical models for the purpose of providing a quantitative basis for decision-making and uncovering new problems for quantitative analysis.* – Thierauf and Klekamp, 1975

- *This new decision-making field has been characterized by the use of scientific knowledge through interdisciplinary team effort for the purpose of determining the best utilization of limited resources.* – H A Taha, 1976

These two definitions refer to the interdisciplinary nature of OR. However, there is nothing that can stop one person from considering several aspects of the problem under consideration.

- *Operations research, in the most general sense, can be characterized as the application of scientific methods, techniques and tools, to problems involving the operations of a system so as to provide those in control of the operations with optimum solutions to the problems.* – Churchman, Ackoff and Arnoff, 1957

This definition refers operations research as a technique for selecting the best course of action out of the several courses of action available, in order to reach the desirable solution of the problem.

- *Operations research has been described as a method, an approach, a set of techniques, a team activity, a combination of many disciplines, an extension of particular disciplines (mathematics, engineering, economics), a new discipline, a vocation, even a religion. It is perhaps some of all these things.* – S L Cook, 1977
- *Operations research may be described as a scientific approach to decision-making that involves the operations of organizational system.* – F S Hiller and G J Lieberman, 1980
- *Operations research is a scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control.* – P M Morse and G E Kimball, 1951
- *Operations research is applied decision theory. It uses any scientific, mathematical, or logical means to attempt to cope with the problems that confront the executive, when he tries to achieve a thorough-going rationality in dealing with his decision problems.* – D W Miller and M K Star, 1969
- *Operations research is a scientific approach to problem-solving for executive management.* – H M Wagner

Operations research is the art of winning wars without actually fighting them.

Operations research is the art of finding bad answers to problems which otherwise have worse answers.

As the discipline of operations research grew, numerous names such as *Operations Analysis, Systems Analysis, Decision Analysis, Management Science, Quantitative Analysis, Decision Science* were given to it. This is because of the fact that the types of problems encountered are always concerned with 'effective decision'.

1.4 FEATURES OF OPERATIONS RESEARCH APPROACH

OR utilizes a *planned approach following a scientific method and an interdisciplinary team, in order to represent complex functional relationship as mathematical models, for the purpose of providing a quantitative basis for decision-making and uncovering new problems for quantitative analysis*. This definition implies additional features of OR approach. The broad features of OR approach in solving any decision problem are summarized as follows:

Interdisciplinary approach For solving any managerial decision problem often an interdisciplinary teamwork is essential. This is because while attempting to solve a complex management problem, one person may not have the complete knowledge of all its aspects such as economic, social, political, psychological, engineering, etc. Hence, a team of individuals specializing in various functional areas of management should be organized so that each aspect of the problem can be analysed to arrive at a solution acceptable to all sections of the organization.

Operations research uses:
(i) interdisciplinary,
(ii) scientific,
(iii) holistic, and
(iv) objective-oriented approaches to decision making

Scientific approach *Operations research is the application of scientific methods, techniques and tools to problems involving the operations of systems so as to provide those in control of operations with optimum solutions to the problems* (Churchman et al.). The scientific method consists of observing and defining the problem; formulating and testing the hypothesis; and analysing the results of the test. The data so obtained is then used to decide whether the hypothesis should be accepted or not. If the hypothesis is accepted, the results should be implemented, otherwise not.

Holistic approach While arriving at a decision, an operations research team examines the relative importance of all conflicting and multiple objectives. It also examines the validity of claims of various departments of the organization from the perspective of its implications to the whole organization.

Objective-oriented approach An operations research approach seeks to obtain an optimal solution to the problem under analysis. For this, a measure of desirability (or effectiveness) is defined, based on the objective(s) of the organization. A measure of desirability so defined is then used to compare alternative courses of action with respect to their possible outcomes.

Illustration The OR approach attempts to find a solution acceptable to all sections of the organizations. One such situation is described below.

A large organization that has a number of management specialists is faced with the basic problem of maintaining stocks of finished goods. To the marketing manager, stocks of a large variety of products are purely a means of supplying the company's customers with what they want and when they want it. Clearly, according to a marketing manager, a fully stocked warehouse is of prime importance to the company. But the production manager argues for long production runs, preferably on a smaller product range, particularly if a significant amount of time is lost when production is switched from one variety to another. The result would again be a tendency to increase the amount of stock carried but it is, of course, vital that the plant should be kept running. On the other hand, the finance manager sees stocks in terms of capital that is unproductively tied up and argues strongly for its reduction. Finally, there appears the personnel manager for whom a steady level of production is advantageous for having better labour relations. Thus, all these people would claim to uphold the interests of the organization, but they do so only from their own specialized points of view. They may come up with contradictory solutions and obviously, all of them cannot be right.

Operations research attempts to resolve the conflicts of interest among various sections of the organization and seeks to find the optimal solution that is in the interest of the organization as a whole.

In view of such a problem that involves every section of an organization, the decision-maker, irrespective of his/her specialization, may require to seek assistance from OR professionals.

Remark A *system* is defined as an arrangement of components designed to achieve a particular objective or objectives according to plan. The components may either be physical or conceptual or both, but they all share a unique relationship with each other and with the overall objective of the system.

1.5 OPERATIONS RESEARCH APPROACH TO PROBLEM SOLVING

The most important feature of operations research is the use of the scientific method and the building of decision models. The operations research approach to problem solving is based on three phases, namely (i) Judgement Phase; (ii) Research Phase, and (iii) Action Phase.

Judgement phase This phase includes: (i) identification of the real-life problem, (ii) selection of an appropriate objective and the values of various variables related to this objective, (iii) application of the appropriate scale of measurement, i.e. deciding the measures of effectiveness (desirability), and (iv) formulation of an appropriate model of the problem and the abstraction of the essential information, so that a solution to the decision-maker's goals can be obtained.

Research phase This phase is the largest and longest amongst all the phases. However, even though the remaining two are not as long, they are also equally important as they provide the basis for a scientific method. This phase utilizes: (i) observations and data collection for a better understanding of the problem, (ii) formulation of hypothesis and model, (iii) observation and experimentation to test the hypothesis on the basis of additional data, (iv) analysis of the available information and verification of the hypothesis using pre-established measures of desirability, (v) prediction of various results from the hypothesis, and (iv) generalization of the result and consideration of alternative methods.

Operations research approach to decision making is based on three phases:
(i) Judgement,
(ii) Research, and
(iii) Action

Action phase This phase consists of making recommendations for implementing the decision. This decision is implemented by an individual who is in a position to implement results. This individual must be aware of the environment in which the problem occurred, be aware of the objective, of assumptions behind the problem and the required omissions of the model.

CONCEPTUAL QUESTIONS A

1. Briefly trace the history of operations research. How did operations research develop after World War II?
2. Is operations research (i) a discipline (ii) a profession (iii) a set of techniques (iv) a philosophy or a new name for old things? Discuss.
3. Discuss the following:
   (a) OR as an interdisciplinary approach.
   (b) Scientific method in OR.
   (c) OR as more than a quantitative analysis of the problem.
4. What are the essential characteristics of operations research? Mention different phases in an operations research study. Point out its limitations, if any.
5. Define operations research as a decision-making science.
   (a) Give the main characteristics of OR.
   (b) Discuss the scope of OR.
6. (a) Outline the broad features of the judgement phase and the research phase of the scientific method in OR. Discuss in detail any one of these phases.
   (b) What are various phases of the OR problem? Explain them briefly.
   (c) State the phases of OR study and their importance in solving problems.
7. What is the role of operations research in decision-making? [*Punjab, BE* (*Mech. Engg.*), *2000*]
8. Define operations research. Explain critically the limitations of various definitions as you understand them.
9. Give any three definitions of operations research and explain them.
10. (a) Discuss various phases of solving an OR problem.
    (b) State phases of an OR study and their importance in solving problems.
11. Discuss the role and scope of quantitative methods for scientific decision-making in a business environment. [*Delhi Univ., MBA, 2003*]
12. Discuss the advantage and limitations of operations research. [*AMIE, 2004*]
13. Write a critical essay on the definition and scope of operations research. [*Meerut, MSc* (*Maths*), *2001*]
14. What post-World War II factors were important in the development of operations research?
15. Does the fact that OR takes the organizational point of view instead of the individual problem-centered point of view generate constraints on its increased usage?
16. What were the significant characteristics of OR applications during World War II? What caused the discipline of OR to take on these characteristics during that period?
17. Comment on the following statements:
    (a) OR is the art of winning war without actually fighting it.
    (b) OR is the art of finding bad answers where worse exist.
    (c) OR replaces management by personality. [*Delhi Univ., MBA* (*HCA*), *2006*]
18. Explain critically the limitations of any three definitions of OR as you understand them.
19. Discuss the significance and scope of operations research in modern management. [*Delhi Univ., MBA* (*HCA*), *2005*]
20. Quantitative techniques complement the experience and judgement of an executive in decision-making. They do not and cannot replace it. Discuss. [*Delhi Univ.*, *MBA, 2004*]
21. Explain the difference between scientific management and operations research.
22. Decision-makers are quick to claim that quantitative analysis talks to them in a jargon that does not sound like English. List four terms that might not be understood by a manager. Then explain, in non-technical terms, what each term means.
23. It is said that operations research increases the creative capabilities of a decision-maker. Do you agree with this view? Defend your point of view with examples.
24. (a) Why is the study of operations research important to the decision-maker?
    (b) Operations research increases creative and judicious capabilities of a decision-maker. Comment. [*Delhi Univ., MBA, 2007*]
25. (a) 'Operations research is the application of scientific methods, techniques and tools to problems involving the operations of a system so as to provide those in control of the system with optimum solutions to the problem.' Discuss.
    (b) 'Operations research is an aid for the executive in making his decisions by providing him with the needed quantitative information, based on the scientific method analysis.' Discuss this statement giving examples of OR methods that you know.
26. Comment on the following statements.
    (a) OR is a bunch of mathematical techniques.
    (b) OR advocates a system's approach and is concerned with optimization. It provides a quantitative analysis for decision-making.
    (c) OR has been defined semi-facetiously as the application of big minds to small problems.
27. Discuss the points to justify the fact that the primary purpose of operations research is to resolve the conflicts resulting from the various subdivisions of the functional areas like production, marketing, finance and personnel in an optimal, near optimal or satisfying way.
28. How far can quantitative techniques be applied in management decision-making? Discuss, in detail, with special reference to any functional area of management pointing out their limitations, if any.

1.6 MODELS AND MODELLING IN OPERATIONS RESEARCH

Models do not, and cannot, represent every aspect of a real-life problem/system because of its large and changing characteristics. However, a model can be used to analyze, understand and describe certain aspects (key features) of a system for the purpose of improving its performance as well as to examine changes (if any) without disturbing the ongoing operations. For example, to study the flow of material in a manufacturing firm, a scaled diagram (descriptive model) on paper showing the factory floor, position of equipment, tools, and workers can be constructed. It would not be necessary to give details such as the colour of machines, the heights of the workers, or the temperature of the building.

The key to model-building lies in abstracting only the relevant variables that affect the criteria of the measures-of-performance of the given system and in expressing the relationship in a suitable form. However, a model should be as simple as possible so as to give the desired result. On the other hand, oversimplifying the problem can also lead to a poor decision. Model enrichment is done by changing value of variables, and relaxing assumptions. The essential three qualities of any model are:

- *Validity* of the model – model should represent the critical aspects of the system/problem under study,
- *Usability* of the model – a model can be used for the specific purposes, and
- *Value* of the model to the user.

Besides these three qualities, other consideration of interest are, (i) cost of the model and its sophistication, (ii) time involved in formulating the model, etc.

A **model** is an approximation or abstraction of reality which considers only the essential variables (or factors) and parameters in the system along with their relationships.

An informal definition of model that applies to all of us is a tool for thinking and understanding features of any problem/system before taking action. For example, a model tends to be formulated when (a) we think about what someone will say in response to our act(s), (b) we try to decide how to spend our money, or (c) we attempt to predict the consequences of some activity (either ours, someone else's or even a natural event). In other words, we would not be able to derive or take any purposeful action if we do not form a model of the activity first. OR approach uses this natural tendency to create models. This tendency forces to think more rigorously and carefully about the models we intend to use.

In general models are classified in eight categories as shown in Table 1.1. Such a classification provides a useful frame of reference for practioners/researchers.

Table 1.1 General Classification of Models

1. Function	4. Degree of certainty	7. Degree of closure
• Descriptive	• Certainty	• Closed
• Predictive	• Conflict	• Open
• Normative	• Risk	8. Degree of quantification
2. Structure	• Uncertainty	• Qualitative
• Iconic	5. Time reference	▪ Mental
• Analog	• Static	▪ Verbal
• Symbolic	• Dynamic	• Quantitative
3. Dimensionality	6. Degree of generality	▪ Statistical
• Two-dimensional	• Specialized	▪ Heuristic
• Multidimensional	• General	▪ Simulation

A summary classification of OR models based on different criteria is given in Fig. 1.2 and few of these classifications are discussed below:

1.6.1 Classification Based on Structure

Physical models These models are used to represent the physical appearance of the real object under study, either reduced in size or scaled up. Physical models are useful only in design problems because they are easy to observe, build and describe. For example, in the aircraft industry, scale models of a proposed new aircraft are built and tested in wind tunnels to record the stresses experienced by the air frame.

Physical model represents the physical appearance of the real object under study, either reduced in size or scaled up.

Physical models cannot be manipulated and are not very useful for prediction. Problems such as portfolio selection, media selection, production scheduling, etc., cannot be analysed with the help of these models.

Physical models are classified into two categories.

(i) ***Iconic Models*** *An iconic model is a scaled (small or big in size) version of the system. Such models retain some of the physical characteristics of the system they represent.*

Examples of iconic model are, blueprints of a home, maps, globes, photographs, drawings, air planes, trains, etc.

An iconic model is used to describe the characteristics of the system rather than explaining the system. This means that such models are used to represent system's characteristics that are not used in determining or predicting effects that take place due to certain changes in the actual system. For example, (i) colour of an atom does not play any vital role in the scientific study of its structure, (ii) type of engine in a car has no role to play in the study of the problem of parking, etc.

(ii) ***Analogue Models*** *An analogue model does not resemble physically the system they represent, but retain a set of characteristics of the system. Such models are more general than iconic models and can also be manipulated.*

For example, (i) oil dipstick in a car represents the amount of oil in the oil tank; (ii) organizational chart represents the structure, authority, responsibilities and relationship, with boxes and arrows; (iii) maps in different colours represent water, desert and other geographical features, (iv) Graphs of time series, stock-market changes, frequency curves, etc., represent quantitative relationships between any two variables and predict how a change in one variable effects the other, and so on.

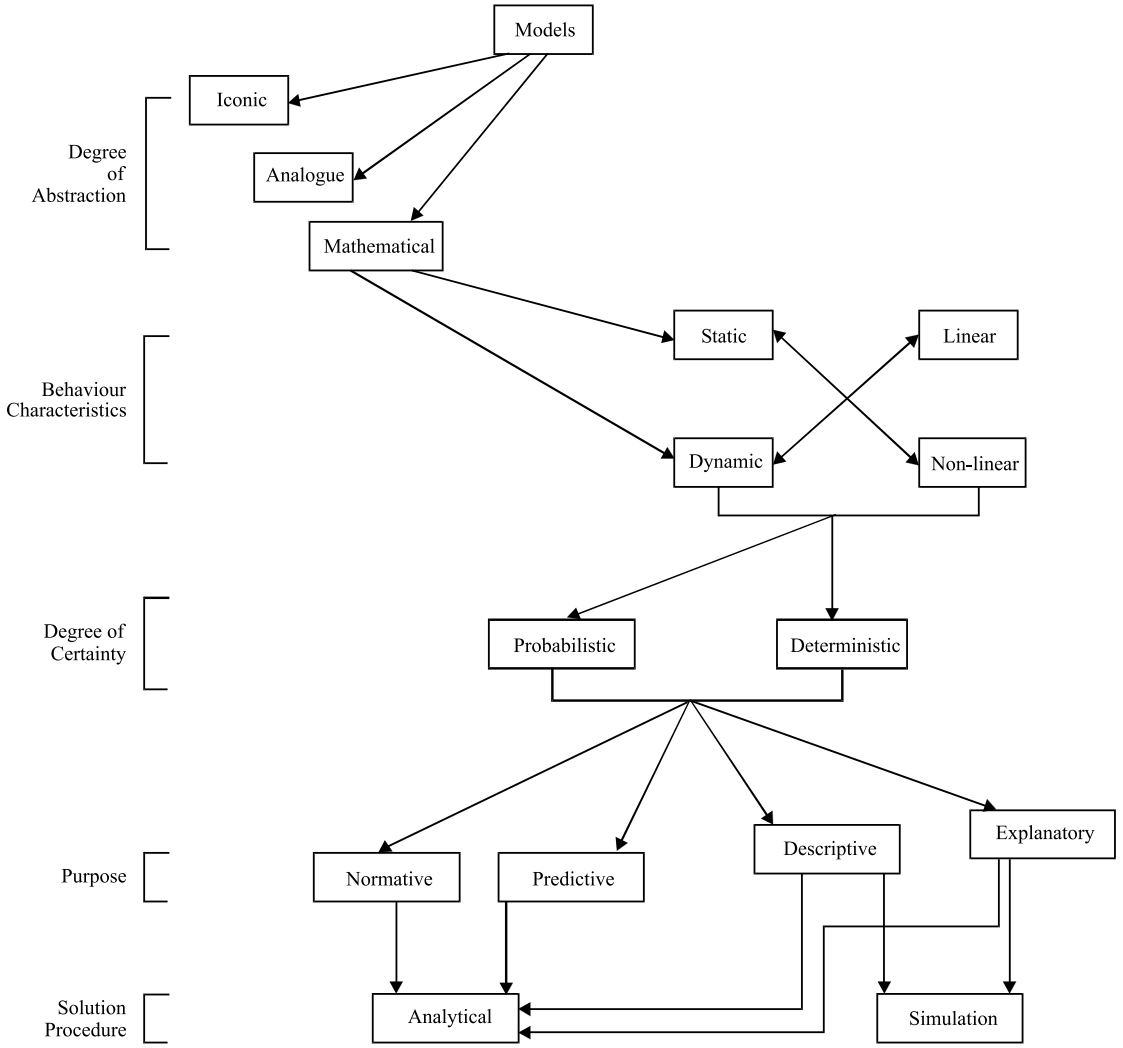


Fig. 1.2 Classification of Models

Symbolic models These models use algebraic symbols (letters, numbers) and functions to represent variables and their relationships for describing the properties of the system. Such relationships can also be represented in a physical form. Symbolic models are precise and abstract and can be analysed by using laws of mathematics.

Mathematical model uses mathematical equations and statements to represent the relationships within the model.

Symbolic models are classified into two categories.

(i) ***Verbal Models*** *These models describe properties of a system in written or spoken language. Written sentences, books, etc., are examples of a verbal model.*

(ii) ***Mathematical Models*** *These models use mathematical symbols, letters, numbers and mathematical operators (+, –, ÷, ×) to represent relationships among variables of the system to describe its properties or behaviour. The solution to such models is obtained by applying suitable mathematical technique.*

Few examples of mathematical model are (i) the relationship among velocity, distance and acceleration, (ii) the relationship among cost-volume-profit, etc.

1.6.2 Classification Based on Function (or Purpose)

Descriptive models These models are used to investigate the outcomes or consequences of various alternative courses of action (strategies, or actions). Since these models evaluate the consequence based on a given condition (or alternative) rather than on all other conditions, there is no guarantee that an alternative selected is optimal. These models are usually applied (i) in decision-making where optimization models are not applicable, and (ii) when objective is to define the problem or to assess its seriousness rather than to select the best alternative. These models are especially used for predicting the behaviour of a particular system under various conditions.

Simulation is an example of a descriptive model for conducting experiments with the systems based on given alternatives.

Predictive models These models represent a relationship between dependent and independent variables and hence measure 'cause and effect' due to changes in independent variables.

These models do not have an objective function as a part of the model of evaluating decision alternatives based on outcomes or pay off values. Also, through such models decision-maker does not attempt to choose the best decision alternative, but can only have an idea about the possible alternatives available to him.

For example, the equation $S = a + bA + cI$ relates dependent variable (S) with other independent variables on the right hand side. This can be used to describe how the sale (S) of a product changes with a change in advertising expenditure (A) and disposable personal income (I). Here, *a, b* and *c* are parameters whose values must be estimated. Thus, having estimated the values of *a, b* and *c*, the value of advertising expenditure (A) can be adjusted for a given value of I, to study the impact of advertising on sales. Also, through such models decision-maker does not attempt to choose the best decision alternative, but can only have an idea about the possible alternative available to him.

Normative (or Optimization) models These models provide the 'best' or 'optimal' solution to problems using an appropriate course of action (strategy) subject to certain limitations on the use of resources. For example, in mathematical programming, models are formulated for optimizing the given objective function, subject to restrictions on resources in the context of the problem under consideration and non-negativity of variables. These models are also called *prescriptive models* because they prescribe what the decision-maker ought to do.

1.6.3 Classification Based on Time Reference

In a **deterministic model** all values used are known with certainty.

Static models Static models represent a system at a particular point of time and do not take into account changes over time. For example, an inventory model can be developed and solved to determine an economic order quantity assuming that the demand and lead time would remain same throughout the planning period.

Dynamic models Dynamic models take into account changes over time, i.e., time is considered as one of the variables while deriving an optimal solution. Thus, a sequence of interrelated decisions over a period of time are made to select the optimal course of action in order to achieve the given objective. Dynamic programming is an example of a dynamic model.

1.6.4 Classification Based on Degree of Certainty

Deterministic models If all the parameters, constants and functional relationships are assumed to be known with certainty when the decision is made, the model is said to be deterministic. Thus, the outcome associated with a particular course of action is known, i.e. for a specific set of input values, there is only one output value which is also the solution of the model. Linear programming models are example of deterministic models.

In a **Probabilistic model** all values used are not known with certainty; and are often measured as probability values.

Probabilistic (Stochastic) models If at least one parameter or decision variable is random (probabilistic or stochastic) variable, then the model is said to be probabilistic. Since at least one decision variable is random, the independent variable, which is the function of dependent variable(s), will also be random. This means consequences (or payoff) due to certain changes in the independent variable(s) cannot be predicted with certainty. However, it is possible to predict a pattern of values of both the variables by their probability distribution.

Insurance against risk of fire, accidents, sickness, etc., are examples where the pattern of events is studied in the form of a probability distribution.

1.6.5 Classification Based on Method of Solution or Quantification

Heuristic models If certain sets of rules (may not be optimal) are applied in a consistent manner to facilitate solution to a problem, then the model is said to be Heuristic.

Analytical models These models have a specific mathematical structure and thus can be solved by the known analytical or mathematical techniques. Any optimization model (which requires maximization or minimization of an objective function) is an analytical model.

Simulation models These models have a mathematical structure but cannot be solved by the known mathematical techniques. A simulation model is essentially a computer-assisted experimentation on a mathematical structure of a problem in order to describe and evaluate its behaviour under certain assumptions over a period of time.

Simulation models are more flexible than mathematical models and can, therefore, be used to represent a complex system that cannot be represented mathematically. These models do not provide general solution like those of mathematical models.

1.7 ADVANTAGES OF MODEL BUILDING

Models, in general, are used as an aid for analysing complex problems. However, general advantages of model building are as follows:

1. A model describes relationships between various variables (factors) present in a system more easily than what is done by a verbal description. That is, models help the decision-maker to understand the system's structure or operation in a better way. For example, it is easier to represent a factory layout on paper than to construct it. It is cheaper to try out modifications of such systems by rearrangement on paper.
2. The problem can be viewed in its entirety, with all the components being considered simultaneously.
3. Models serve as aids to transmit ideas among people in the organization. For example, a process chart can help the management to communicate better work methods to workers.
4. A model allows to analyze and experiment on a complex system which would otherwise be impossible on the actual system. For example, the experimental firing of INSAT satellite may cost millions of rupees and require years of preparation.
5. Models considerably simplify the investigation and provide a powerful and flexible tool for predicting the future state of the system (or process).

1.8 METHODS FOR SOLVING OPERATIONS RESEARCH MODELS

In general, the following three methods are used for solving OR models, where values of decision variables are obtained that optimize the given objective function (a measure of effectiveness).

In general, operations research models are solved using either
(i) analytical,
(ii) numerical, or
(iii) Monte Carlo method

Analytical (or deductive) method In this method, classical optimization techniques such as calculus, finite difference and graphs are used for solving an OR model. The analytical methods are non-iterative methods to obtain an optimal solution of a problem. For example, to calculate economic order quantity (optimal order size), the analytical method requires that the first derivative of the mathematical expression

$$TC = (D/Q)\, C_0 + (Q/2)\, C_h$$

where TC = total variable inventory cost; D = annual demand;
C_0 = ordering cost per order; C_h = carrying cost per time period;
Q = size of an order

be taken and equated to zero as the first step towards calculating, $EOQ(Q) = \sqrt{2\,DC_0 / C_h}$. This is because of the concept of maxima and minima used for optimality.

Numerical (or iterative) method When analytical methods fail to obtain the solution of a particular problem due to its complexity in terms of constraints or number of variables, a numerical (or iterative) method is used to find the solution. In this method, instead of solving the problem directly, a general algorithm is applied for obtaining a specific numerical solution.

The numerical method starts with a solution obtained by trial and error, and a set of rules for improving it towards optimality. The solution so obtained is then replaced by the improved solution and the process of getting an improved solution is repeated until such improvement is not possible or the cost of further calculation cannot be justified.

Monte Carlo method This method is based upon the idea of experimenting on a mathematical model by inserting into the model specific values of decision variables for a selected period of time under different conditions and then observing the effect on the criterion chosen. In this method, random samples of specified random variables are drawn to know how the system is behaving for a selected period of time under different conditions. The random samples form a probability distribution that represents the real-life system and from this probability distribution, the value of the desired random variable can then be estimated.

1.9 METHODOLOGY OF OPERATIONS RESEARCH

Every OR specialist may have his/her own way of solving problems. However, the effective use of OR techniques requires to follow a sequence of steps. Each of these steps is described in detail in Fig. 1.3:

A problem is the statement that indicates an objective/goal that needs to be achieved.

Step 1: Defining the problem Problem definition involves the process of identifying, understanding, and describing the problem being faced by an organization. It helps to define the objective(s) to be achieved, and the alternative courses of action.

The procedure begins with gathering information (data) of the organizational structure, communication and control system, objectives and expectations. Such information will help in assessing the difficulty likely to be faced in terms of costs, time, resources, probability of success of the study, etc.

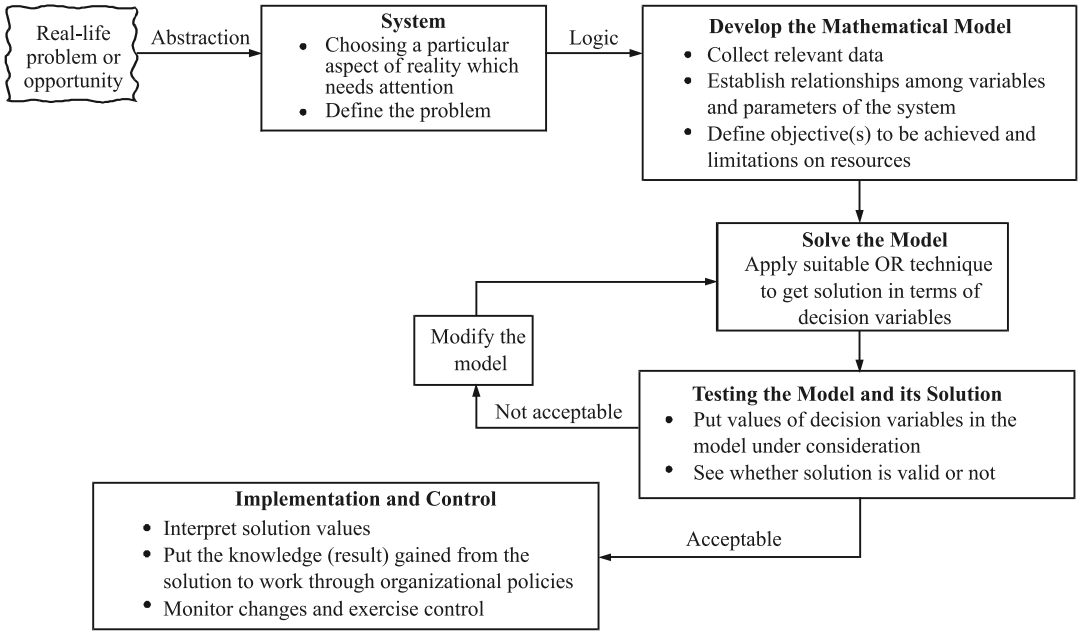


Fig. 1.3 Methodology of Operations Research

The major components involved in problem formulation are as under:

Decision-maker's Point of View The first component of the problem formulation is to know the point of view of the decision-maker who is not satisfied with the existing state of affairs. Such interaction with the decision-maker will help to understand whether he has already obtained the solution of the problem and wants to retain it, or wants to improve it to next level of satisfaction. If the decision-maker has conflicting multiple objectives, he may be advised to rank his objectives in order of preference; overlapping among several objectives may be eliminated.

All view points should be considered before defining the problem to be solved in order to arrive at an optimal solution.

Decision Environment It is desirable to know the details about resources (such as manpower, material, money, etc.) that are required to attain objectives of the organization while keeping in mind the social norms in which the organization need to function. Knowledge of such factors will help in modifying the initial set of the decision-maker's objectives.

Alternative Courses of Action If several courses of action are available for solution to a problem, then an exhaustive list of such courses of action should be prepared. Courses of action that are not feasible with respect to objectives and resources may be ruled out.

Measure of Effectiveness A measure of effectiveness (or performance) is required in order to evaluate the merit of the several available courses of action. The effectiveness can be measured in different units such as rupees (net profits), percentage (share of market desired) or time dimension (service or waiting time).

Step 2: Formulating a mathematical model After the problem is clearly defined and understood, the next step is to collect required data and then *formulate a mathematical model*. Model formulation requires to define relationships among decision variables. Certain basic components required in every decision problem model are:

Controllable (Decision) Variables These are certain factors (or variables) associated with the problem whose values are to be determined by solving the model. The possible values assigned to these variables are called decision alternatives. For example, in *queuing theory*, the number of service facilities is the decision variable.

Uncontrollable (Exogenous) Variables The values of these variables are not under the control of the decision-maker and are also termed as *state of nature*.

Exogenous variables are those whose numerical value depends upon the external environment that influences an organization.

Objective Function It represents the criterion of evaluating alternative courses of action in terms of value of decision variables so as to optimize (minimized or maximized) the desired performance.

Policies and Constraints There are certain constraints (limitations) on the use of resources, and such constraints arise due to organizational policy, legal restraints or limited resources such as space, money, manpower, material, etc. The constraints on the use of resources are expressed either in the form of equations or inequalities.

Functional Relationships In a decision problem, the decision variables in the objective function and in the constraints are connected by a specific functional relationship. A general decision problem model might take the following form:

Optimize (Max. or Min.) $Z = f(\mathbf{x})$

subject to the constraints

$$g_i(\mathbf{x}) \{\leq, =, \geq\} b_i; \quad i = 1, 2, \ldots, m$$

and $\mathbf{x} \geq 0$

where, $\mathbf{x}$ = a vector of decision variables $(x_1, x_2, \ldots, x_n)$

$f(\mathbf{x})$ = criterion or objective function to be optimized

$g_i(\mathbf{x})$ = the ith constraint

b_i = fixed amount of the ith resource

A model is referred to as a linear model if all functional relationships among decision variables $x_1, x_2, \ldots, x_n$ in $f(x)$ and $g(x)$ are linear. But if one or more of the relationships are non-linear, the model is said to be a non-linear model.

Parameters These are constants in the functional relationships among decision variables. Parameters can either be deterministic or probabilistic in nature. A deterministic parameter is one whose value is assumed to occur with certainty. Otherwise, it is probabilistic.

Optimization methods yield the best value for the decision variables both for unconstrained and constrained problems.

Step 3: Solving the mathematical model Once a mathematical model of the problem has been formulated, the next step is to obtain numerical values of decision variables. Obtaining these values depends on the specific form or type of mathematical model. In general, the following two categories of methods are used for solving an OR model.

(i) *Analytical Methods* These methods are applied to solve both constrained and unconstrained mathematical models. In constrained problems, value of decision variables satisfy all the constraints simultaneously and provide an optimal value for the objective function.

However, in unconstrained problems, value of decision variables provide an acceptable value for the objective function.

(ii) *Heuristic Methods* (also referred as *rules of thumb which work*). These methods are used when obtaining optimal solution is either very time consuming or the model is too complex. A commonly used heuristic is '*stand in the shortest line*',

Step 4: Validating the solution After solving the mathematical model, it is important to review the solution to see whether value of variables make sense and that the resulting decisions can be implemented. Some of the reasons for validating the solution are:

(i) Mathematical model may not have considered all constraints present in study because few constraints may either be omitted or incorrect.
(ii) Certain aspects of the problem may have been overlooked, omitted or simplified.
(iii) The data may have been incorrectly estimated or recorded.

Step 5: Implementing the solution Before implementing the solution, the decision-maker should ensure to select alternatives that are capable of being implemented. Further, it is also important to ensure that any solution implemented is continuously reviewed and updated in the light of a changing environment. In any case, the decision-maker who is in the best position to implement solution (or result), must be aware of the objective, assumption, omissions and limitations of the model.

Step 6: Modifying the model For a mathematical model to be useful, the degree to which it actually represents the system or problem being modelled must be established. If during validation the solution cannot be implemented, then identify the constraint(s) that were omitted during the mathematical model formulation or find out whether some of the original constraints were incorrect. In all such cases, return to the problem formulation step and carefully make the appropriate modifications to represent the given problem more accurately.

Heuristic Methods yield values of the variables that satisfy all the constraints, but do not necessarily provide optimal solution.

A model must be applicable for a reasonable time period and should be updated from time to time, taking into consideration the past, present and future aspects of the problem.

Step 7: Establishing controls over the solution The changes within the environment (society or business) influence the continuing validity of models and their solutions. Thus, a control procedure needs to be developed for detecting significant changes in decision variables of the problem, without having to build a model every time a significant change occurs.

1.10 ADVANTAGES OF OPERATIONS RESEARCH STUDY

In addition to learning about various OR models that are useful to arrive at an optimal decision, the knowledge of operations research techniques can also help the decision-maker to attain several other advantages as listed below.

1. **Structured approach to problems:** A substantial amount of time and effort can be saved in developing and solving OR models if a logical and consistent approach is followed. This implies that the decision-maker has to be careful while defining variables, availability of resources, and functional relationships among variables in the objective function and constraints. This will also reduce the chance of conceptual and computational errors. Any such error can also be detected easily and corrected at an early stage.
2. **Critical approach to problem solving:** The decision-maker will come to understand various components of the problem and accordingly select a mathematical model for solving the given problem. He will become aware of the explicit and implicit assumptions and limitations of such models. Problem solutions are examined critically and the effect of any change and error in the problem data can be studied through sensitivity analysis techniques.

1.11 OPPORTUNITIES AND SHORTCOMINGS OF THE OPERATIONS RESEARCH APPROACH

The use of quantitative methods is appreciated to improve managerial decision-making. However, besides certain opportunities OR approach has not been without its shortcomings. The main reasons for its failure are due to unawareness on the part of decision-makers about their own role, as well as the avoidance of behavioural/organizational issues while constructing a decision model. A few opportunities and shortcomings of the OR approach are listed below.

Opportunities

The OR approach of problem solving facilitates.

- decision-maker to define explicitly his/her objectives, assumptions and constraints.
- decision-maker to identify variables that might influence decisions.

- in identifying gaps in the data required to support solution to a problem.
- the use of computer software to solve a problem.

Shortcomings

- The solution to a problem is often derived by simplifying assumptions. Consequently, such solutions are not optimal.
- Models may not represent the realistic situations in which decisions are made.
- Often the decision-maker is not fully aware of the limitations of the models.

1.12 FEATURES OF OPERATIONS RESEARCH SOLUTION

A solution that is expensive as compared to the potential savings from its implementation, should not be considered. Even, if a solution is well within the budget but does not fulfill the objective is also not acceptable. Few requirements of a good solution are:

1. **Technically Appropriate:** The solution should work technically, meet the constraints and operate in the problem environment.
2. **Reliable:** The solution must be useful for a reasonable period of time under the conditions for which it was designed.
3. **Economically Viable:** Its economic value should be more than what it costs to develop it and it should be seen as a wise investment in hiring OR talent.
4. **Behaviourally Appropriate:** The solution should be behaviourally appropriate and must remain valid for a reasonable period of time within the organization.

A solution needs to be: (i) technically appropriate, (ii) reliable, and (iii) economically viable

1.13 APPLICATIONS OF OPERATIONS RESEARCH

Some of the industrial/government/business problems that can be analysed by the OR approach have been arranged by functional areas as follows:

Finance and Accounting

- Dividend policies, investment and portfolio management, auditing, balance sheet and cash flow analysis
- Claim and complaint procedure, and public accounting
- Break-even analysis, capital budgeting, cost allocation and control, and financial planning
- Establishing costs for by-products and developing standard costs

Marketing

- Selection of product-mix, marketing and export planning
- Advertising, media planning, selection and effective packing alternatives
- Sales effort allocation and assignment
- Launching a new product at the best possible time
- Predicting customer loyalty

Purchasing, Procurement and Exploration

- Optimal buying and reordering with or without price quantity discount
- Transportation planning
- Replacement policies
- Bidding policies
- Vendor analysis

Production Management

Facilities planning

- Location and size of warehouse or new plant, distribution centres and retail outlets
- Logistics, layout and engineering design
- Transportation, planning and scheduling

Manufacturing

- Aggregate production planning, assembly line, blending, purchasing and inventory control
- Employment, training, layoffs and quality control
- Allocating R&D budgets most effectively

Maintenance and project scheduling

- Maintenance policies and preventive maintenance
- Maintenance crew size and scheduling
- Project scheduling and allocation of resources

Personnel Management

- Manpower planning, wage/salary administration
- Designing organization structures more effectively
- Negotiation in a bargaining situation
- Skills and wages balancing
- Scheduling of training programmes to maximize skill development and retention

Techniques and General Management

- Decision support systems and MIS; forecasting
- Making quality control more effective
- Project management and strategic planning

Government

- Economic planning, natural resources, social planning and energy
- Urban and housing problems
- Military, police, pollution control, etc.

1.14 OPERATIONS RESEARCH MODELS IN PRACTICE

There is no unique set of problems that can be solved by using OR techniques. Several OR techniques can be grouped into some basic categories as given below. In this book, a large number of OR techniques have been discussed in detail.

- **Allocation models** Allocation models are used to allocate resources to activities so as to optimize measure of effectiveness (objective function). Mathematical programming is the broad term for the OR techniques used to solve allocation problems.

  If the measure of effectiveness such as profit, cost, etc., is represented as a linear function of several variables and limitations on resources (constraints) are expressed as a system of linear equalities or inequalities, the allocation problem is classified as a linear programming problem. But, if the objective function or all constraints cannot be expressed as a system of linear equalities or inequalities, the allocation problem is classified as a non-linear programming problem.

  When the value of decision variables in a problem is restricted to integer values or just zero-one values, the problem is classified as an integer programming problem or a zero-one programming problem, respectively.

  A problem having multiple, conflicting and incommensurable objective functions (goals) subject to linear constraints is called a goal programming problem. If values of decision variables in the linear programming problem are not deterministic, then such a problem is called a stochastic programming problem.

  If resources (such as workers, machines or salesmen) have to be assigned to perform a certain number of activities (such as jobs or territories) on a one-to-one basis so as to minimize total time, cost or distance involved in performing a given activity, such problems are classified as assignment problems. But if the activities require more than one resource and conversely if the resources can be used for more than one activity, the allocation problem is classified as a transportation problem.

- **Inventory models** These models deal with the problem of determination of how much to order at a point in time and when to place an order. The main objective is to minimize the sum of three conflicting inventory costs: the cost of holding or carrying extra inventory, the cost of shortage or delay in the delivery of items when it is needed and the cost of ordering or set-up. These are also useful in dealing with quantity discounts and selective inventory control.

- **Waiting line (or Queuing) models** These models establish a trade-off between costs of providing service and the waiting time of a customer in the queuing system. A queuing model describes: Arrival process, queue structure and service process and solution for the measure of performance – average length of waiting time, average time spent by the customer in the line, traffic intensity, etc., of the waiting system.

- **Competitive (Game Theory) models** These models are used to characterize the behaviour of two or more competitors (called players) competing to achieve their conflicting goals. These models are classified according to several factors such as number of competitors, sum of loss and gain, and the type of strategy which would yield the best or the worst outcomes.

- **Network models** These models are applied to the management (planning, controlling and scheduling) of large-scale projects. PERT/CPM techniques help in identifying delay and project critical path. These techniques improve project coordination and enable the efficient use of resources. Network methods are also used to determine time-cost trade-off, resource allocation and help in updating activity time.
- **Sequencing models** These models are used to determine the sequence (order) in which a number of tasks can be performed by a number of service facilities such as hospital, plant, etc., in such a way that some measure of performance (such as total time to process all the jobs on all the machines) is optimized.
- **Replacement models** These models are used to calculate optimal time to replace an equipment when either its efficiency deteriorates with time or fails immediately and completely.
- **Dynamic programming models** These models are used where a problem requires optimization of multistage (sequence of interrelated decisions) decision processes. The method starts by dividing a given problem into stages or sub-problems and then solves those sub-problems sequentially until the solution to the original problem is obtained.
- **Markov-chain models** These models are used for analyzing a system which changes over a period of time among various possible outcomes or states. That is, these models describe transitions in terms of transition probabilities of various states.
- **Simulation models** These models are used to evaluate alternative courses of action by experimenting with a mathematical model of the problems with random variables. Thus, repetition of the process by using the simulation model provides an indication of the merit of alternative course of action with respect to the decision variables.
- **Decision analysis models** These models deal with the selection of an optimal course of action given the possible payoffs and their associated probabilities of occurrence. These models are broadly applied to problems involving decision-making under risk and uncertainty.

1.15 COMPUTER SOFTWARE FOR OPERATIONS RESEARCH

Many real-life OR models require long and complex mathematical calculations. Thus, computer software packages that are used to do these calculations rapidly and effectively have become a part of OR approach to problem solving. Computer facilities such as spread sheets or statistical and mathematical software packages that make such analysis readily available to a decision-maker.

Prentice-Hall distributes some of the leading software packages for quantitative approach to decision-making. The features of a few such software packages are explained as follows.

- *Quantitative Systems for Business Plus (QSB+), Version 3.0,* by Yih-long Chang and Robert S Sullivan, is a software package that contains problem-solving algorithms for operations research/management science, as well as modules on basic statistics, non-linear programming and financial analysis.

  QSB+ gives tips for getting started, presents tutorials on how to use it, details the special functions and steps involved in using each program. Because of the friendliness of this microcomputer software, the user will not encounter much difficulty.
- *Quantitative Systems for Operations Management (QSOM),* by Yih-long Chang, is an interactive user-friendly system. It contains problem-solving algorithms for most of the important operations management problems and associated information systems.
- *Value STORM: MS Quantitative Modelling for Decision Support,* by Hamilton Emmons, A D Flowers, Chander Shekhar, M Khot and Kamlesh Mathur, is a special version of Personal STORM Version 3.0 developed for use in the operations research/management science.
- *Excel 97* by Gene Weiss Kopf and distributed by BPB Publications, New Delhi, is an easy-to-use task-oriented guide to Excel Spread sheet applications. It displays data in a wide variety of charts, creates, maintains and sorts databases, while taking advantage of improved file-sharing capabilities.
- *Linear Interactive Discrete Optimization (LINDO)* is also a popular package for solving LP models on personal computers. It was developed by Linus Schrage Lindo in his book: *An Optimization Modeling System*, 4th ed. (Palo Alto, CA: Scientific Press, 1991)

CONCEPTUAL QUESTIONS B

1. Model building is the essence of the operations research approach. Discuss.
2. What is meant by a mathematical model of a real situation? Discuss the importance of models in the solution of OR problems. [*AMIE*, *2009*]
3. What is the purpose of a mathematical model? How does a model achieve this purpose? In your answer consider the concept of 'a model an abstraction of reality.'
4. Discuss in detail the three types of models with special emphasis on important logical properties and the relationship the three types bear to each other and to modelled entities. [*Meerut, MSc* (*OR*), *2000, AMIE, 2008*]
5. What is a model? Discuss various classification schemes of models. [*CA*, *May*, *2000*]
6. (a) Discuss, in brief, the role of OR models in decision-making. What areas of OR have made a significant impact on the decision-making process?
   (b) Discuss the importance of operations research in the decision-making process.
   (c) Give some examples of various types of models. What is a mathematical model? Develop two examples of mathematical models.
7. Explain how and why operations research methods have been valuable in aiding executive decisions. [*Meerut, MSc* (*Stat.*), *2001*]
8. State the different types of models used in OR. Explain briefly the general methods for solving these OR models. [*Indore, MSc* (*Maths*), *2002; Punjab BSc* (*Mech. Engg.*), *2003*]
9. What reasons can you think of to explain the fact that many models are built and never implemented? Does the absence of implementation mean that the entire activity of model development was a waste.
10. 'Quantifying the elements of a decision problem is the easy part; the hard part is solving the model.' Do you agree? Why or why not?
11. In constructing any OR model, it is essential to realize that the most important purpose of the modelling process is 'to help manager better'. Keeping this purpose in mind, state any four OR models that can be of help chartered accountants in advising their clients. [*CA, May*, *2001*]
12. (a) Explain the scope and methodology of OR, the main phases of OR and techniques used in solving an OR problem.
    (b) Explain the steps involved in the solution of an operations research problem. [*Allahabad, MBA, 2002*]
13. How can OR models be classified? What is the best classification in terms of learning and understanding the fundamentals of OR?
14. Explain the essential characteristics of the following types of models:
    (a) Allocation models (b) Competitive games
    (c) Inventory (d) Waiting line.
15. (a) Describe briefly the applications of OR in managerial decision-making. Give suitable examples from different areas.
    (b) Explain the concept, scope and tools of OR as applicable to business and industry. [*Meerut, MSc* (*Stat.*), *2003*]
16. Define an OR model and give four examples. State their properties, advantages and limitations.
17. Assume that you have to make a decision. Describe your decision with specific reference to:
    (a) The alternative courses of action available.
    (b) The criteria you have set for the evaluation of the available alternatives.
    (c) The variables with which you propose to construct a model to assist you in making the decision.
    (d) The extent to which the variables and the relationships between them are quantifiable.
18. What are the advantages and disadvantages of operations research models? Why is it necessary to test models and how would you go about testing a model?
19. State any four areas for the application of OR techniques in financial management. How do they improve the performance of an organization?
20. It is common in business to insure against the occurrence of events which are subject to varying degrees of uncertainty, for example, ill health of senior executives. At the same time, the use of formal analytical models to assist in the process of making decisions on business problems, which are generally subject to uncertainty, does not appear to be very widespread.
    Describe the model building approach to the analysis of business problems under conditions of uncertainty. Discuss the apparent inconsistency in a company's willingness to insure when formal analytical models of an operations research nature, which allow for uncertainty, are relatively employed.
21. Recent developments in the field of computer technology have enabled operations research to integrate its models into information systems and thus make itself a part of decision-making procedures of many organizations. Comment.
22. Explain various types of OR models and indicate their applications to production, inventory and distribution systems.
23. Explain the distinction between
    (a) Static and dynamic models.
    (b) Analytical and simulation models.
    (c) Descriptive and prescriptive models.
24. Write briefly about the following:
    (a) Iconic models. (b) Analogue models.
    (c) Mathematical models (or symbolic models).
25. 'Whether in a private, non-profit, or public organization, the most important and the distinguishing function of a manager is problem solving. The field of quantitative methods is dedicated to aiding managers in their problem-solving efforts. This is accomplished through the use of mathematical models to analyse the problems.' Discuss. [*Delhi Univ., MBA, 2004*]
26. 'Executives at all levels in business and industry come across the problems of making decisions at every stage in their day-to-day activities. Operations research provides them with various quantitative techniques for decision-making and enhances their ability to make long-range plans and solve everyday problems of running a business and industry with greater efficiency, competence and confidence.' Elaborate the statement with examples. [*Delhi Univ., MBA, 2005; IGNOU, Diploma in Mgt*, *2003*]
27. Comment on the following statements;
    (a) Operations research advocates a system's approach and is concerned with optimization. [*Delhi Univ., MBA* (*HCA*), *2004*]
    (b) Operations research replaces management by personality. [*Delhi Univ, MBA* (*HCA*), *2005*]
28. Suppose you are being interviewed by the manager of a commercial firm for a job in the research department which deals with the applications of quantitative techniques. Explain the scope and purpose of quantitative techniques and its usefulness to the firm. Give some examples of the applications of quantitative techniques in industry. [*Ajmer Univ., MBA, 2002*]
29. Operations research is an ongoing process. Explain this statement with examples.

30. Distinguish between model results that recommend a decision and model results that are descriptive.
31. 'Mathematics . . . tends to lull the unsuspecting into believing that he who thinks elaborately thinks well'. Do you think that the best OR models are the ones that are most elaborate and mathematically complex? Why?
32. 'The hard problems are those for which models do not exist' Interpret this statement. Give some examples.

CHAPTER SUMMARY

Operations research is the scientific approach to decision making. This approach includes defining the problem, developing a model, solving the model, testing the solution, analyzing the results, and implementing the results. In using operations research approach, however, there can be potential problems, including conflicting viewpoints, the impact of quantitative analysis on various sections of the organization, beginning assumptions, outdated solutions, understanding the model, obtaining relevant input data, obtaining optimal solution, testing the solution, and analyzing the results. In using the operations research approach, implementation is not the final step because there can be a lack of commitment to the approach and resistance to change.

CHAPTER CONCEPTS QUIZ

True or False

1. Operations research is that art of winning wars without actually fighting them.
2. Operations research approach does not adequately meet the needs of business and industry.
3. Static model represents a system at some specified time without doing any change over a period of time.
4. Operations research is a scientific approach to problem solving for executives.
5. Simulation is one of the most widely used quantitative techniques.
6. The mathematical modeling of inventory control problem are among the best operations research techniques.
7. Simulation models are comparatively less flexible than any mathematical models.
8. All decisions are made either on the basis of intuition or by using more formal approaches.
9. The quantitative approach to decision modeling is a rational approach.
10. Operations research practioners can predict about the future events.

Fill in the Blanks

11. Dynamic programming involves the optimization of the ..... decision processes.
12. Media planning is one of the ..... of operations research.
13. There are many real world problems that don't possess any ..... solution.
14. Operations research is a ..... approach to problem solving for executives.
15. Operations research increases the ..... capabilities of a decision-maker.
16. Model in which at least one decision variable is random is known as ..... model.
17. ..... is one of the situations which could cause decision-maker to move towards a quantitative approach.
18. The manager should rely on ....., ... and ..... while using a quantitative approach for problem solving.
19. The constraints may be in the form of ..... or ..... .
20. A ..... is sometimes described as a rule of thumb that usually works.

Multiple Choice

21. Operations research approach is
    (a) multi-disciplinary (b) scientific
    (c) intuitive (d) all of the above
22. Operations research practioners do not
    (a) predict future operations
    (b) build more than one model
    (c) collect relevant data
    (d) recommend decision and accept
23. For analyzing a problem, decision-makers should study
    (a) its qualitative aspects (b) its quantitative aspects
    (c) both (a) and (b) (d) neither (a) nor (b)
24. Decision variables are
    (a) controllable (b) uncontrollable
    (c) parameters (d) none of the above
25. A model is
    (a) an essence of reality (b) an approximation
    (c) an idealization (d) all of the above
26. Managerial decisions are based on
    (a) an evaluation of quantitative data
    (b) the use of qualitative factors
    (c) results generated by formal models
    (d) all of the above
27. The use of decision models
    (a) is possible when the variable's value is known
    (b) reduces the scope of judgement and intuition known with certainty in decision-making
    (c) require the use of computer software
    (d) none of the above
28. Every mathematical model
    (a) must be deterministic
    (b) requires computer aid for its solution
    (c) represents data in numerical form
    (d) all of the above
29. A physical model is example of
    (a) an iconic model (b) an analogue model
    (c) a verbal model (d) a mathematical model
30. An optimization model
    (a) provides the best decision
    (b) provides decision within its limited context
    (c) helps in evaluating various alternatives
    (d) all of the above
31. The quantitative approach to decision analysis is a
    (a) logical approach (b) rational approach
    (c) scientific approach (d) all of the above

32. The process of modifying an OR model to observe the effect upon its outputs is called
    (a) sensitivity analysis (b) cost/benefit analysis
    (c) model validation (d) input variation
33. Operations research practitioners do not
    (a) take responsibility for solution implementation
    (b) collect essential data
    (c) predict future actions/operations
    (d) build more than one model
34. OR approach is typically based on the use of
    (a) Physical model (b) Mathematical model
    (c) Iconic model (d) Descriptive model
35. The qualitative approach to decision analysis relies on
    (a) Experience (b) Judgement
    (c) Intuition (d) All of the above

Answers to Quiz

1. T	2. T	3. T	4. T	5. T	6. T	7. F	8. F	9. T	10. F
11. Multistage	12. application		13. optimal		14. scientific		15. creativity		
16. stochastic	17. problem is complex		18. experience, intuitions, judgement			19. equalities, inequalities			
20. heuristic	21. (a)	22. (a)	23. (c)	24. (a)	25. (d)	26. (d)	27. (d)	28. (c)	29. (a)
30. (d)	31. (c)	32. (a)	33. (c)	34. (b)	35. (d)				

CASE STUDY

Case 1.1: Daily Newspaper

A meeting of 'Board of Governers' of a daily newspaper was called by its Managing Director to consider business proposals from the promotion department.

The promotion department submitted two alternatives proposals to increase daily circulation of the paper. The proposals, indicating the increased circulation as well as the cost of obtaining this increase is shown in Table 1. The cost is incurred in the first month, and the increase in circulation generally lasts about three months.

For each proposal, the advertising volume (in thousands of column cm) at the current rate of Rs 1,000 per column cm, and at the increased rate of Rs 1,100 per column cm is also shown in Table 1. Even advertising also has seasonal variations, above or below the normal level. In particular, next month should be at normal level, the succeeding month at 10 per cent above the normal, and the third month at 8 per cent below normal.

Table 1 Promotion Alternatives

Proposal	*Daily Average Circulation*	*Cost of Additional Circulation (Rs 1,000s)*	*Advertising Volume (in thousand column cm)*	
			Current Rate (Rs 1,000 per column cm)	*Increased Rate (Rs 1,100 per column cm)*
A (Existing)	2,00,000	0	335	300
B	2,30,000	500	345	325
C	2,40,000	900	350	335

Since various parts of the Sunday edition are printed during the entire week, the promotion department also proposed to sub-contract the printing of various parts of the Sunday edition. It is believed that such an act will be able to reduce (or eliminate) overtime in the press room. A local printer has consented to do the printing and supply the newsprint for Rs 1,04,000 per 10 lakh sheets printed, with the condition that he will be given three months' contract with the same amount of work each month.

Table 2 Cost of Typesetting

Type of Sheet (per column cm)	*Variable Cost (Rs)*
Advertising sheet	180
Editorial, news and other features	220

The main sources of revenue generation for the newspaper are: (i) daily sales, and (ii) sales of advertising space. Operational costs incurred are broadly classified as costs related to: (i) distribution, (ii) editorial and news coverage, (iii) typesetting, photo-engraving, layout and artwork, (iv) running of presses to produce the paper, and (v) newsprint.

It is understood that a larger circulation should generate more advertising revenue because the paper would be read by more readers and therefore, advertisers would like to release more advertisements in the paper. Also, readers buy the paper partly for the advertisements.

	Variable Cost Per Lakh Sheets Printed (Rs)	*Capacity Per Lakh Sheets Per Day*
Regular time	3,000	24
Overtime	4,600	Unlimited

Table 3
Press Room Costs and Capacity

For planning purposes, a month is considered as one unit and average values for circulation, advertising and news are used. The current average daily circulation of the paper is 2 lakh and average net profit is Re 0.50 on each paper, per day.

The paper is currently using 4,000 column cm space for advertising per day and this generates revenue at the rate of Rs 1,000 per column cm. The remaining, but fixed space of 800 column cm, is used for editorial features and news makeup.

The variable costs of typesetting and press room are shown in Tables 2 and 3, respectively. A sheet is one copy of the page of the newspaper. The regular capacity of the press room is 24 lakh sheets per day.

Department	*Fixed Cost per Month (Rs 100s)*
Typesetting	2,500
Editorial, news and features	3,200
Press room	9,000
Distribution	26,000

Table 4
Fixed Costs

The variable distribution costs depend on the circulation and size. The average distribution cost is Rs 1,600 per tonne of newspaper. The fixed costs for other activities are shown in Table 4. The variable cost of newsprint is Rs 4,800 per tonne. The press room can effectively print 50,000 sheets from one tonne of newsprint. The quota of the newspaper is 48 tonnes of newsprint per month. However, additional newsprint can be obtained at an extra cost of Rs 6,400 per tonne.

Questions for Discussion

1. Identify decision variables and constraints to build a model. Write down the relationships in the form of equations/ inequalities.
2. Consider the following initial values for the decision variables.
   Initial circulation : 2 lakh/day
   Price of advertising : Rs 1,000/cm
   Initial space used for advertising : 4,000 column cm
   Sheets printed outside : Nil
   Examine (a) the financial viability of various proposals.
   (b) the possibility of increasing circulation by one per cent other conditions remaining same, as well as the effect of the increase on total profit per year.

Case 1.2: Supply Chain Management

A global consumer-electronics firm wants to automate its existing fulfillment process using ERP, APS, and the Internet. Currently, its regional offices around the globe group, orders from customers, typically consumer electronics chains, in their regions once a week for 26 weeks of a rolling horizon. They then send the grouped orders to the headquarters. The headquarters assign current and planned inventory to these orders and then spends them directly from warehouses at the plant locations to customer warehouses around the globe. In the future, customers may order from headquarters directly, and external suppliers may fill orders. For the first four weeks of the planning horizon, the orders are firm, while later orders may change in time and therefore serve as forecasts. The headquarters, regional offices (or customers), and plants (or suppliers) can use the Internet, ERP, and APS systems (as shown in Fig. 1.4) as follows, while maintaining the existing fulfillment process, with all steps except (f) and (g) to be run once a week in the following sequence:

(a) Regional offices (or customers) send orders through the Internet to a database at headquarters and overwrite the orders for corresponding weeks placed a week earlier. Unfilled orders from past weeks in the ERP system also go to this database.
(b) Plants (or suppliers) use the Internet to enter the planned replenishments by date and ship-from location in the same database. The ERP system maintains records of current inventory at each plant (supplier) location as stocks are added or taken out.

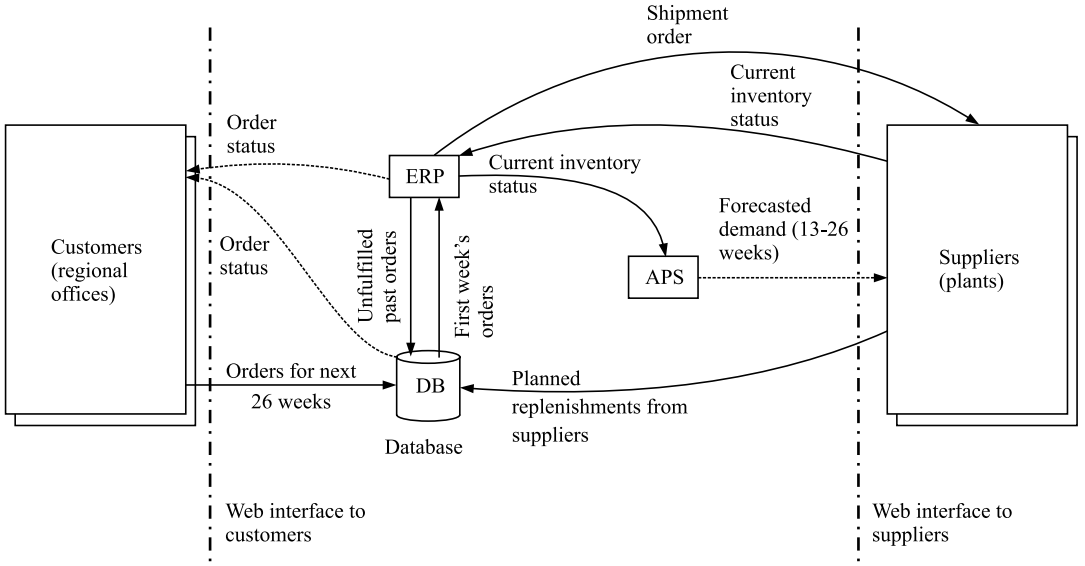


Fig. 1.4 Illustration of using Web in Conjunction with its ERP and APS Systems to Improve Supply Chain Management

(c) The APS system takes all the orders and the planned replenishments for the entire time horizon of 26 weeks from the database and takes the current inventory from the ERP system. It runs either a heuristic or a linear-programming-based model to allocate current and planned inventory among these orders, balancing on the way delays and customers according to their importance. With either model, the system gives each order an allocated quantity, target shipping date, and a target shipper location. The firm considers planned replenishments fixed for the first 12 weeks and flexible for weeks 13 through 26 weeks, so the APS system take the first 12 weeks' replenishment from suppliers as a constraint and the next 14 weeks' replenishment as a decision variable. (The heuristic for a single shipping location works as follows: taking the time periods in sequence, for each time period, it sorts unfilled orders by due date and customer importance and then allocates the maximum possible inventory to orders starting from the top of the list. The linear-programming-based solution is a multi-time-period network model with current and planned inventory as source nodes, orders as demand nodes, and delay penalties per unit, time-based on customer importance.)

(d) Through Internet-enabled collaboration, the APS system makes the forecast for weeks 13 through to 26 available to the suppliers.

(e) The ERP system imports orders for the current week from the database and updates unfilled orders from previous weeks. All of these orders now have a shipping quantity and date as determined by the APS module in step (c).

(f) The ERP system on a daily basis executes orders based on the shipping date and the inventory at the shipping location, issuing transportation orders and updating order statuses accordingly.

(g) The regional offices (or customers) can use the Internet any time after the APS run to view the shipping quantities and shipping dates for their planned orders. The ERP system provides information on later orders.

Questions for Discussion

1. If this firm were to join an electronic marketplace, it might have to radically change and use OR to make improvements even before joining such a marketplace. Suggest the kind of improvement it should adopt in order to make the process more responsive to changes in demand.
2. The firm could make the supply chain even more responsive if it allowed changes to the suppliers' replenishment schedule within the first 12 weeks. Then, it could tie the suppliers' production scheduling system(s) to its APS system. This would let a customer enter a tentative order through the Web to get an immediate response as to whether or not the firm could fill the order by the specified date. Suggest the method which could provide this functionality in the present context.

PUZZLES IN OPERATIONS RESEARCH*

Students are advised to read the chapter on 'Integer Programming' and 'Assignment Problems' before attempting to solve the following puzzles.

Less and Less # 1 The puzzle given below can initially be formulated as the integer programming problem.

$$\text{Maximize } Z = \sum_{j=1}^{47} (c_{1j} + c_{2j} + c_{3j}) x_j$$

* Based on Glenn Weber, *Puzzle Contests in MS/OR Education,* Interface 20, March-April 1990, pp. 72–76.

subject to the constraints

$$\sum_{j=1}^{47} x_j = 8; \quad \sum_{j=1}^{47} (c_{1j} - c_{2j})\, x_j \leq -1; \quad \sum_{j=1}^{47} (c_{1j} - c_{3j})\, x_j \leq -1$$

and $x_j = 0, 1$ for $j = 1, 2, \ldots, 47$.

where c_{ij} is the numerical value of letter i in word j and $x_j = 1$ if word j is chosen.

Less and Less # 1

Place a different word from the word list on each of the 8 lines below. All letters are assigned values. Use the letter values to score the columns of the words. The total of column 1 (extreme left column) must be less than the total of column 2 . . . and the total of column 1 must also be less than the total of column 3. Add the column totals to determine your score. *High score wins*. Ties, if any, will be broken by the highest total for column 1. If there are still ties, column 2 will be judged for the highest total.

Letter Values

A - 1	H - 8	O - 15	V - 22
B - 2	I - 9	P - 16	W - 23
C - 3	J - 10	Q - 17	X - 24
D - 4	K - 11	R - 18	Y - 25
E - 5	L - 12	S - 19	Z - 26
F - 6	M - 13	T - 20	
G - 7	N - 14	U - 21	

1.	____	____	____
2.	____	____	____
3.	____	____	____
4.	____	____	____
5.	____	____	____
6.	____	____	____
7.	____	____	____
8.	____	____	____

Total = _____

Word List

ADV	AFT	BET	BKS	CCW	CIR	DER	DIP	EAT	EGO	FAR
FIN	GHQ	GOO	HAT	HOI	HUG	ION	IVE	JCS	JOE	KEN
KKK	LIP	LYE	MOL	MTG	NES	NTH	OIL	OSF	PIP	PRF
QMG	QUE	ROE	RUG	STG	SIP	TUE	TVA	UTE	VIP	WHO
XIN	YES	ZIP								

Hint The solution of this IP problem has an objective function value of $Z = 330$. The alternative optimal solutions must be considered in order to handle ties, as stipulated in the rules. The easiest way to handle this is to include in the constraint set the objective function.

$$\sum_{j=1}^{47} (c_{1j} + c_{2j} + c_{3j})\, x_j = 330 \text{ and to use Max } \sum_{j=1}^{47} c_{1j} x_j$$

as the new objective function. The solution to this revised formulation includes the words BKS, GOD, ION, LYE, NTH, PIP, VIP, ZIP with column totals of 108, 113, and 109 and the overall total of 330. There was no need to consider maximizing the second column to handle additional ties since, if 113 could be increased, 109 would have to decrease and this world violate the condition that the total of column 1 must be less than the total of column 3.

Assign the Values # 2 This puzzle is a variation of assignment problem in which an even letter is assigned a number, and to guarantee that they are different, every number is assigned a letter. Assuming that the group totals can be made equal, the objective is to solve for ties by maximizing the total of group 1.

$$\text{Maximize } Z = \sum_{j=1}^{9} \left\{ c_{i1} \left(\sum_{j=1}^{9} [j]\, x_{ij} \right) \right\}$$

subject to the constraints

$$\sum_{j=1}^{9} \left\{ (c_{i1} - c_{i2}) \left(\sum_{j=1}^{9} [j] x_{ij} \right) \right\} = 0; \quad \sum_{j=1}^{9} x_{ij} = 1,\ i = 1, 2, \ldots, 9$$

$$\sum_{j=1}^{9} x_{ij} = 1,\ j = 1, 2, \ldots, 9; \quad \text{and } x_{ij} = 0, 1 \text{ for all } i \text{ and } j$$

where c_{ik} is the number of times letters i appears in group k and $x_{ij} = 1$ if letter i is assigned value j.

Assign the Values # 2

Assign values 1 to 9 to the letters in the letter-value chart below. Each letter must have a different value. Determine totals for both groups by using the letter values you assigned. You must now find the difference between the totals of Groups 1 and 2 to determine your score. *Low score wins.* A score of 0 (equal totals) would be perfect. Ties, if any, will be broken by judging the total of Group 1 for the highest score.

Group 1		**Group 2**	
1. AREA -		1. ERST -	
2. FORT -		2. FOOT -	
3. HOPE -		3. HEAT -	
4. SPAR -		4. PAST -	
5. THAT -		5. PROG -	
6. TREE -	_____	6. STOP -	_____

Letter Values

A -	P -	E -	Total — Group 1 = _____
R -	F -	S -	Total — Group 2 = _____
H -	T -	O -	Score = _____

Hint The solution of this equal-group-total-assumption formulation has the letter values A – 6, E – 7, F – 2, H – 1, O – 8, P – 4, R – 5, S – 3, T – 9 with equal-group totals of 139. If this formulation had no feasible solution, then it would be necessary to solve a sequence of problems in which the right-hand side of the first constraint is varied with increasing positive and negative integer values until a feasible solution is found.

Chapter 2

Linear Programming: Applications and Model Formulation

"Whenever there is a hard job to be done I assign it to a lazy man; he is sure to find an easy way of doing it."

– Walter Chrysler

PREVIEW

Linear programming (LP) is a widely used mathematical modelling technique developed to help decision makers in planning and decision-making regarding optimal use of scarce resources. This chapter is devoted to illustrate the applications of LP programming in different functional areas of management and how LP models are formulated.

LEARNING OBJECTIVES

After reading this chapter you should be able to

- identify situations in which linear programming technique can be applied.
- understand fundamental concepts and general mathematical structure of a linear programming model.
- express objective function and resource constraints in LP model in terms of decision variables and parameters.
- appreciate the limitations and assumptions of linear programming technique with a view to interpret the solution.

CHAPTER OUTLINE

2.1 Introduction
2.2 Structure of Linear Programming Model
2.3 Advantages of Using Linear Programming
2.4 Limitations of Linear Programming
2.5 Application Areas of Linear Programming
2.6 General Mathematical Model of Linear Programming Problem
2.7 Guidelines on Linear Programming Model Formulation
2.8 Examples of LP Model Formulation
- Conceptual Questions
- Self Practice Problems
- Hints and Answers

❑ Chapter Summary
❑ Chapter Concepts Quiz
❑ Case Study

2.1 INTRODUCTION

The application of specific operations research techniques to determine the choice among several courses of action, so as to get an optimal value of the measures of effectiveness (objective or goal), requires to formulate (or construct) a mathematical model. Such a model helps to represent the essence of a system that is required for decision-analysis. The term *formulation* refers to the process of converting the verbal description and numerical data into mathematical expressions, which represents the relationship among relevant decision variables (or factors), objective and restrictions (constraints) on the use of scarce resources (such as labour, material, machine, time, warehouse space, capital, energy, etc.) to several competing activities (such as products, services, jobs, new equipment, projects, etc.) on the basis of a given criterion of optimality. The term *scarce resources* refers to resources that are not available in infinite quantity during the planning period. The criterion of optimality is generally either performance, return on investment, profit, cost, utility, time, distance and the like.

Linear Programming is a mathematical technique useful for allocation of 'scarce' or 'limited' resources, to several competing activities on the basis of a given criterion of optimality.

In 1947, during World War II, George B Dantzing while working with the US Air Force, developed LP model, primarily for solving military logistics problems. But now, it is extensively being used in all functional areas of management, airlines, agriculture, military operations, education, energy planning, pollution control, transportation planning and scheduling, research and development, health care systems, etc. Though these applications are diverse, all LP models have certain common properties and assumptions – that are essential for decision-makers to understand before their use.

Before discussing the basic concepts and applications of linear programming, it is important to understand the meaning of the words – *linear* and *programming.* The word *linear* refers to linear relationship among variables in a model. That is, a given change in one variable causes a proportional change in another variable. For example, doubling the investment on a certain project will also double the rate of return. The word *programming* refers to the mathematical modelling and solving of a problem that involves the use of limited resources, by choosing a particular *course of action* (or *strategy*) among the given courses of action (or strategies) in order to achieve the desired objective.

The usefulness of this technique is enhanced by the availability of several user-friendly computer software such as STORM, TORA, QSB_+, LINDO, etc. However, there is no computer software for building an LP model. Model building is an art that improves with practice. A variety of examples are given in this chapter to illustrate the formulation of an LP model.

2.2 STRUCTURE OF LINEAR PROGRAMMING MODEL

2.2.1 General Structure of an LP Model

The general structure of an LP model consists of following three basic components (or parts).

***Decision variables* (*activities*)** The evaluation of various courses of action (alternatives) and select the best to arrive at the optimal value of objective function, is guided by the nature of objective function and availability of resources. For this, certain activities (also called *decision variables*) usually denoted by $x_1, x_2, \ldots, x_n$ are conducted. The value of these variables (activities) represents the extent to which each of these is performed. For example, in a product-mix manufacturing problem, an LP model may be used to determine units of each of the products to be manufactured by using limited resources such as personnel, machinery, money, material, etc.

Four components of any LP model are: (i) decision variables, (ii) objective function, (iii) constraints, and (iv) non-negativity

The value of certain variables may or may not be under the decision-maker's control. If values are under the control of the decision-maker, then such variables are said to be *controllable*, otherwise they are said to be *uncontrollable*. These decision variables, usually interrelated in terms of consumption of resources, require simultaneous solutions. In an LP model all decision variables are continuous, controllable and non-negative. That is, $x_1 \geq 0, x_2 \geq 0, \ldots, x_n \geq 0$.

The objective function The objective function of each LP problem is expressed in terms of decision variables to optimize the criterion of optimality (also called *measure-of-performance*) such as profit, cost, revenue, distance etc. In its general form, it is represented as:

$$\text{Optimize (Maximize or Minimize) } Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n,$$

where Z is the measure-of-performance variable, which is a function of $x_1, x_2, \ldots, x_n$. Quantities $c_1, c_2, \ldots, c_n$ are parameters that represent the contribution of a unit of the respective variable $x_1, x_2, \ldots, x_n$ to the

measure-of-performance *Z*. The optimal value of the given objective function is obtained by the graphical method or simplex method.

The constraints There are always certain limitations (or constraints) on the use of resources, such as: labour, machine, raw material, space, money, etc., that limit the degree to which an objective can be achieved. Such constraints must be expressed as linear equalities or inequalities in terms of decision variables. The solution of an LP model must satisfy these constraints.

2.2.2 Assumptions of an LP Model

In all mathematical models, assumptions are made for reducing the complex real-world problems into a simplified form that can be more readily analyzed. The following are the major assumptions of an LP model:

Assumptions of an LP model are:
(i) certainty,
(ii) additivity,
(iii) proportionality, &
(iv) divisibility

1. ***Certainty:*** In LP models, it is assumed that all its parameters such as: availability of resources, profit (or cost) contribution per unit of decision variable and consumption of resources per unit of decision variable must be known and constant.
2. ***Additivity:*** The value of the objective function and the total amount of each resource used (or supplied), must be equal to the sum of the respective individual contribution (profit or cost) of the decision variables. For example, the total profit earned from the sale of two products A and B must be equal to the sum of the profits earned separately from A and B. Similarly, the amount of a resource consumed for producing A and B must be equal to the total sum of resources used for A and B individually.
3. ***Linearity (or proportionality):*** The amount of each resource used (or supplied) and its contribution to the profit (or cost) in objective function must be proportional to the value of each decision variable. For example, if production of one unit of a product uses 5 hours of a particular resource, then making 3 units of that product uses $3\times5 = 15$ hours of that resource.
4. ***Divisibility (or continuity):*** The solution values of decision variables are allowed to assume continuous values. For instance, it is possible to collect 6.254 thousand litres of milk by a milk dairy and such variables are divisible. But, it is not desirable to produce 2.5 machines and such variables are not divisible and therefore must be assigned integer values. Hence, if any of the variable can assume only integer values or are limited to discrete number of values, LP model is no longer applicable.

2.3 ADVANTAGES OF USING LINEAR PROGRAMMING

Following are certain advantages of using linear programming technique:

1. Linear programming technique helps decision-makers to use their productive resources effectively.
2. Linear programming technique improves the quality of decisions. The decision-making approach of the user of this technique becomes more objective and less subjective.
3. Linear programming technique helps to arrive at optimal solution of a decision problem by taking into account constraints on the use of resources. For example, saying that so many units of any product may be produced does not mean that all units can be sold.
4. Linear programming approach for solving decision problem highlight bottlenecks in the production processes. For example, when a bottleneck occurs, machine cannot produce sufficient number of units of a product to meet demand. Also, machines may remain idle.

2.4 LIMITATIONS OF LINEAR PROGRAMMING

In spite of having many advantages and wide areas of applications, there are some limitations associated with this technique. These are as follows:

1. Linear programming assumes linear relationships among decision variables. However, in real-life problems, decision variables, neither in the objective function nor in the constraints are linearly related.

2. While solving an LP model there is no guarantee that decision variables will get integer value. For example, how many men/machines would be required to perform a particular job, a non-integer valued solution will be meaningless. Rounding off the solution to the nearest integer will not yield an optimal solution.
3. The linear programming model does not take into consideration the effect of time and uncertainty.
4. Parameters in the model are assumed to be constant but in real-life situations, they are frequently neither known nor constant.
5. Linear programming deals with only single objective, whereas in real-life situations a decision problem may have conflicting and multiple objectives.

2.5 APPLICATION AREAS OF LINEAR PROGRAMMING

Linear programming is the most widely used technique of decision-making in business and industry and in various other fields. In this section, broad application areas of linear programming are discussed:

Applications in Agriculture

These applications fall into categories of farm economics and farm management. The former deals with inter-regional competition, optimum allocation of crop production, efficient production patterns under regional land resources and national demand constraints, while the latter is concerned with the problems of the individual farm such as allocation of limited resources such as acreage, labour, water supply, working capital, etc., so as to maximize the net revenue.

Applications in Military

Military applications include (i) selection of an air weapon system against the enemy, (ii) ensuring minimum use of aviation gasoline (iii) updating supply-chain to maximize the total tonnage of bombs dropped on a set of targets and takes care of the problem of community defence against disaster at the lowest possible cost.

Production Management

Product Mix To determine the quantity of several different products to be produced, knowing their per unit profit (cost) contribution and amount of limited production resources used. The objective is to maximize the total profit subject to all constraints.

- *Production Planning* This deals with the determination of minimum cost production plan over the planning period, of an item with a fluctuating demand, while considering the initial number of units in inventory, production capacity, constraints on production, manpower and all relevant cost factors. The objective is to minimize total operation costs.
- *Assembly-line Balancing* This problem is likely to arise when an item can be made by assembling different components. The process of assembling requires some specified sequence(s). The objective is to minimize the total elapse time.
- *Blending Problems* These problems arise when a product can be made from a variety of available raw materials, each of which has a particular composition and price. The objective here is to determine the minimum cost blend, subject to availability of the raw materials, and to minimum and maximum constraints on certain product constituents.
- *Trim Loss* When an item is made to a standard size (e.g. glass, paper sheet), the problem of determining which combination of requirements should be produced from standard materials in order to minimize the trim loss, arises.

Financial Management

- *Portfolio Selection* This deals with the selection of specific investment activity among several other activities. The objective here is to find the allocation which maximizes the total expected return or minimizes risk under certain limitations.
- *Profit Planning* This deals with the maximization of the profit margin from investment in plant facilities and equipment, cash in hand and inventory.

Marketing Management

- *Media Selection* The linear programming technique helps in determining the advertising media mix so as to maximize the effective exposure, subject to limitation of budget, specified exposure rates to different market segments, specified minimum and maximum number of advertisements in various media.
- *Travelling Salesman Problem* The salesman's problem is to find the shortest route from a given city to each of the specified cities and then returning to the original point of departure, provided no city would be visited twice during the tour. Such type of problems can be solved with the help of the modified assignment technique.
- *Physical Distribution* Linear programming determines the most economic and efficient manner of locating manufacturing plants and distribution centres for physical distribution.

Personnel Management

- *Staffing Problem* Linear programming is used to allocate optimum manpower to a particular job so as to minimize the total overtime cost or total manpower.
- *Determination of Equitable Salaries* Linear programming technique has been used in determining equitable salaries and sales incentives.
- *Job Evaluation and Selection* Selection of suitable person for a specified job and evaluation of job in organizations has been done with the help of the linear programming technique.

Other applications of linear programming lie in the area of administration, education, fleet utilization, awarding contracts, hospital administration, capital budgeting, etc.

2.6 GENERAL MATHEMATICAL MODEL OF LINEAR PROGRAMMING PROBLEM

The general linear programming problem (or model) with n decision variables and m constraints can be stated in the following form:

Optimize (Max. or Min.) $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$

subject to the linear constraints,

$$\begin{aligned} a_{11}x_1 &+ a_{12}x_2 + \ldots + a_{1n}x_n \ (\leq, =, \geq)\ b_1 \\ a_{21}x_1 &+ a_{22}x_2 + \ldots + a_{2n}x_n \ (\leq, =, \geq)\ b_2 \\ \vdots \quad & \quad \vdots \quad \quad \vdots \quad \quad \vdots \\ a_{m1}x_1 &+ a_{m2}x_2 + \ldots + a_{mn}x_n \ (\leq, =, \geq)\ b_m \end{aligned}$$

and $x_1, x_2, \ldots, x_n \geq 0$

The above formulation can also be expressed in a compact form as follows.

Optimize (Max. or Min.) $Z = \sum_{j=1}^{n} c_j x_j$ (Objective function) (1)

subject to the linear constraints

$$\sum_{j=1}^{n} a_{ij} x_j \ (\leq, =, \geq)\ b_i; \quad i = 1, 2, \ldots, m \qquad \text{(Constraints)} \qquad (2)$$

and $x_j \geq 0; \ j = 1, 2, \ldots, n$ (Non-negativity conditions) (3)

where, the c_j *s* are coefficients representing the per unit profit (or cost) of decision variable x_j to the value of objective function. The a_{ij}*'s* are referred as *technological coefficients* (*or input-output coefficients*). These represent the amount of resource, say i consumed per unit of variable (activity) x_j. These coefficients can be positive, negative or zero. The b_i represents the *total availability of the ith resource.* The term resource is used in a very general sense to include any numerical value associated with the right-hand side of a constraint. It is assumed that $b_i \geq 0$ for all i. However, if any $b_i < 0$, then both sides of constraint i is multiplied by –1 to make $b_i > 0$ and reverse the inequality of the constraint.

In the general LP problem, the expression (≤, =, ≥) means that in any specific problem each constraint may take only one of the three possible forms:

(i) less than or equal to (≤)
(ii) equal to (=)
(iii) greater than or equal to (≥)

2.7 GUIDELINES ON LINEAR PROGRAMMING MODEL FORMULATION

The effective use and application requires, as a first step, the mathematical formulation of an LP model. Steps of LP model formulation are summarized as follows:

Step 1: Identify the decision variables

(a) Express each constraint in words. For this you should first see whether the constraint is of the form $\geq$ (at least as large as), of the form $\leq$ (no larger than) or of the form = (exactly equal to).
(b) Express verbally the objective function.
(c) Verbally identify the decision variables with the help of Step (a) and (b). For this you need to ask yourself the question – *What decisions must be made in order to optimize the objective function?*

LP model formulation requires: (i) identification of decision variables and input data, (ii) formulation of constraints, and (iii) objective function

Having followed Step 1(a) to (c) decide the symbolic notation for the decision variables and specify their units of measurement. Such specification of units of measurement would help in interpreting the final solution of the LP problem.

Step 2: Identify the problem data

To formulate an LP model, identify the problem data in terms of constants, and parameters associated with decision variables. It may be noted that the decision-maker can control values of the variables but cannot control values in the data set.

Step 3: Formulate the constraints

Convert the verbal expression of the constraints in terms of resource requirement and availability of each resource. Then express each of them as linear equality or inequality, in terms of the decision variables defined in Step 1.

Values of these decision variables in the optimal LP problem solution must satisfy these constraints in order to constitute an acceptable (feasible) solution. Wrong formulation can either lead to a solution that is not feasible or to the exclusion of a solution that is actually feasible and possibly optimal.

Step 4: Formulate the objective function

Identify whether the objective function is to be maximized or minimized. Then express it in the form of linear mathematical expression in terms of decision variables along with profit (cost) contributions associated with them.

After gaining enough experience in model building, readers may skip verbal description. The following are certain examples of LP model formulation that may be used to strengthen the ability to translate a real-life problem into a mathematical model.

2.8 EXAMPLES OF LP MODEL FORMULATION

In this section a number of illustrations have been presented on LP model formulation with the hope that readers may gain enough experience in model building.

2.8.1 Examples on production

Example 2.1 A manufacturing company is engaged in producing three types of products: A, B and C. The production department produces, each day, components sufficient to make 50 units of A, 25 units of B and 30 units of C. The management is confronted with the problem of optimizing the daily production of the products in the assembly department, where only 100 man-hours are available daily for assembling the products. The following additional information is available:

Type of Product	*Profit Contribution per Unit of Product (Rs)*	*Assembly Time per Product (hrs)*
A	12	0.8
B	20	1.7
C	45	2.5

The company has a daily order commitment for 20 units of products A and a total of 15 units of products B and C. Formulate this problem as an LP model so as to maximize the total profit.

LP model formulation The data of the problem is summarized as follows:

Resources/Constraints	*Product Type*			*Total*
	A	*B*	*C*	
Production capacity (units)	50	25	30	
Man-hours per unit	0.8	1.7	2.5	100
Order commitment (units)	20	15 (both for B and C)		
Profit contribution (Rs/unit)	12	20	45	

Decision variables Let x_1, x_2 and x_3 = number of units of products A, B and C to be produced, respectively.

The LP model

Maximize (total profit) $Z = 12x_1 + 20x_2 + 45x_3$

subject to the constraints

(i) Labour and materials

(a) $0.8x_1 + 1.7x_2 + 2.5x_3 \le 100$, (b) $x_1 \le 50$, (c) $x_2 \le 25$, (d) $x_3 \le 30$

(ii) Order commitment

(a) $x_1 \ge 20$; (b) $x_2 + x_3 \ge 15$

and $x_1, x_2, x_3 \ge 0$.

Example 2.2 A company has two plants, each of which produces and supplies two products: A and B. The plants can each work up to 16 hours a day. In plant 1, it takes three hours to prepare and pack 1,000 gallons of A and one hour to prepare and pack one quintal of B. In plant 2, it takes two hours to prepare and pack 1,000 gallons of A and 1.5 hours to prepare and pack a quintal of B. In plant 1, it costs Rs 15,000 to prepare and pack 1,000 gallons of A and Rs 28,000 to prepare and pack a quintal of B, whereas in plant 2 these costs are Rs 18,000 and Rs 26,000, respectively. The company is obliged to produce daily at least 10 thousand gallons of A and 8 quintals of B.

Formulate this problem as an LP model to find out as to how the company should organize its production so that the required amounts of the two products be obtained at the minimum cost.

LP model formulation The data of the problem is summarized as follows:

Resources/Constraints		*Product*		*Total Availability (hrs)*
		A	*B*	
Preparation time (hrs)	Plant 1: 3 hrs/thousand gallons		1 hr/quintal	16
	Plant 2: 2 hrs/thousand gallons		1.5 hr/quintal	16
Minimum daily production	10 thousand gallons		8 quintals	
Cost of production (Rs)	Plant 1: 15,000/thousand gallons		28,000/quintals	
	Plant 2: 18,000/thousand gallons		26,000/quintals	

Decision variables Let

x_1, x_2 = quantity of product A (in '000 gallons) to be produced in plant 1 and 2, respectively.

x_3, x_4 = quantity of product B (in quintals) to be produced in plant 1 and 2, respectively.

The LP model

Minimize (total cost) $Z = 15{,}000x_1 + 18{,}000x_2 + 28{,}000x_3 + 26{,}000x_4$

subject to the constraints

(i) Preparation time

(a) $3x_1 + 2x_2 \le 16$, (b) $x_3 + 1.5x_4 \le 16$

(ii) Minimum daily production requirement

(a) $x_1 + x_2 \ge 10$, (b) $x_3 + x_4 \ge 8$

and $x_1, x_2, x_3, x_4 \ge 0$.

Example 2.3 An electronic company is engaged in the production of two components C_1 and C_2 that are used in radio sets. Each unit of C_1 costs the company Rs 5 in wages and Rs 5 in material, while each of C_2 costs the company Rs 25 in wages and Rs 15 in material. The company sells both products on one-period credit terms, but the company's labour and material expenses must be paid in cash. The selling price of C_1 is Rs 30 per unit and of C_2 it is Rs 70 per unit. Because of the company's strong monopoly in these components, it is assumed that the company can sell, at the prevailing prices, as many units as it produces. The company's production capacity is, however, limited by two considerations. First, at the beginning of period 1, the company has an initial balance of Rs 4,000 (cash plus bank credit plus collections from past credit sales). Second, the company has, in each period, 2,000 hours of machine time and 1,400 hours of assembly time. The production of each C_1 requires 3 hours of machine time and 2 hours of assembly time, whereas the production of each C_2 requires 2 hours of machine time and 3 hours of assembly time. Formulate this problem as an LP model so as to maximize the total profit to the company.

LP model formulation The data of the problem is summarized as follows:

Resources/Constraints	*Components*		*Total Availability*
	C_1	C_2	
Budget (Rs)	10/unit	40/unit	Rs 4,000
Machine time	3 hrs/unit	2 hrs/unit	2,000 hours
Assembly time	2 hrs/unit	3 hrs/unit	1,400 hours
Selling price	Rs 30	Rs 70	
Cost (wages + material) price	Rs 10	Rs 40	

Decision variables Let x_1 and x_2 = number of units of components C_1 and C_2 to be produced, respectively.

The LP model

Maximize (total profit) Z = Selling price – Cost price

$$= (30 - 10)\, x_1 + (70 - 40)\, x_2 = 20x_1 + 30x_2$$

subject to the constraints

(i) The total budget available

$$10x_1 + 40x_2 \le 4{,}000$$

(ii) Production time

(a) $3x_1 + 2x_2 \le 2{,}000$; (b) $2x_1 + 3x_2 \le 1{,}400$

and $x_1, x_2 \ge 0$.

Example 2.4 A company has two grades of inspectors 1 and 2, the members of which are to be assigned for a quality control inspection. It is required that at least 2,000 pieces be inspected per 8-hour day. Grade 1 inspectors can check pieces at the rate of 40 per hour, with an accuracy of 97 per cent. Grade 2 inspectors check at the rate of 30 pieces per hour with an accuracy of 95 per cent.

The wage rate of a Grade 1 inspector is Rs 5 per hour while that of a Grade 2 inspector is Rs 4 per hour. An error made by an inspector costs Rs 3 to the company. There are only nine Grade 1 inspectors and eleven Grade 2 inspectors available to the company. The company wishes to assign work to the available inspectors so as to minimize the total cost of the inspection. Formulate this problem as an LP model so as to minimize the daily inspection cost. [*Delhi Univ., MBA, 2004, 2006*]

LP model formulation The data of the problem is summarized as follows:

	Inspector	
	Grade 1	*Grade 2*
Number of inspectors	9	11
Rate of checking	40 pieces/hr	30 pieces/hr
Inaccuracy in checking	1 – 0.97 = 0.03	1 – 0.95 = 0.05
Cost of inaccuracy in checking	Rs 3/piece	Rs 3/piece
Wage rate/hour	Rs 5	Rs 4
Duration of inspection = 8 hrs per day		
Total pieces which must be inspected = 2,000		

Decision variables Let x_1 and x_2 = number of Grade 1 and 2 inspectors to be assigned for inspection, respectively.

The LP model

Hourly cost of each inspector of Grade 1 and 2 can be computed as follows:

Inspector Grade 1 : Rs $(5 + 3 \times 40 \times 0.03)$ = Rs 8.60

Inspector Grade 2 : Rs $(4 + 3 \times 30 \times 0.05)$ = Rs 8.50

Based on the given data, the LP model can be formulated as follows:

Minimize (daily inspection cost) $Z = 8\ (8.60x_1 + 8.50x_2) = 68.80x_1 + 68.00x_2$

subject to the constraints

(i) Total number of pieces that must be inspected in an 8-hour day

$$8 \times 40x_1 + 8 \times 30x_2 \geq 2000$$

(ii) Number of inspectors of Grade 1 and 2 available

(a) $x_1 \leq 9$, (b) $x_2 \leq 11$

and $x_1, x_2 \geq 0$.

Example 2.5 An electronic company produces three types of parts for automatic washing machines. It purchases casting of the parts from a local foundry and then finishes the part on drilling, shaping and polishing machines.

The selling prices of parts A, B and C are Rs 8, Rs 10 and Rs 14 respectively. All parts made can be sold. Castings for parts A, B and C, respectively cost Rs 5, Rs 6 and Rs 10.

The shop possesses only one of each type of casting machine. Costs per hour to run each of the three machines are Rs 20 for drilling, Rs 30 for shaping and Rs 30 for polishing. The capacities (parts per hour) for each part on each machine are shown in the table:

Machine	*Capacity per Hour*		
	Part A	*Part B*	*Part C*
Drilling	25	40	25
Shaping	25	20	20
Polishing	40	30	40

The management of the shop wants to know how many parts of each type it should produce per hour in order to maximize profit for an hour's run. Formulate this problem as an LP model so as to maximize total profit to the company. [*Delhi Univ., MBA, 2001, 2004, 2007*]

LP model formulation Let x_1, x_2 and x_3 = numbers of type A, B and C parts to be produced per hour, respectively.

Since 25 type A parts per hour can be run on the drilling machine at a cost of Rs 20, then Rs 20/25 = Re 0.80 is the drilling cost per type A part. Similar reasoning for shaping and polishing gives

$$\text{Profit per type A part} = (8 - 5) - \left(\frac{20}{25} + \frac{30}{25} + \frac{30}{40}\right) = 0.25$$

$$\text{Profit per type B part} = (10 - 6) - \left(\frac{20}{40} + \frac{30}{20} + \frac{30}{30}\right) = 1$$

$$\text{Profit per type C part} = (14 - 10) - \left(\frac{20}{25} + \frac{30}{20} + \frac{30}{40}\right) = 0.95$$

On the drilling machine, one type A part consumes 1/25th of the available hour, a type B part consumes 1/40th, and a type C part consumes 1/25th of an hour. Thus, the drilling machine constraint is

$$\frac{x_1}{25} + \frac{x_2}{40} + \frac{x_3}{25} \leq 1$$

Similarly, other constraints can be established.

The LP model

Maximize (total profit) $Z = 0.25\ x_1 + 1.00\ x_2 + 0.95\ x_3$

subject to the constraints

(i) Drilling machine: $\frac{x_1}{25} + \frac{x_2}{40} + \frac{x_3}{25} \leq 1$, (ii) Shaping machine: $\frac{x_1}{25} + \frac{x_2}{20} + \frac{x_3}{20} \leq 1$,

(iii) Polishing machine: $\frac{x_1}{40} + \frac{x_2}{30} + \frac{x_3}{40} \le 1,$

and $x_1, x_2, x_3 \ge 0.$

Example 2.6 A pharmaceutical company produces two pharmaceutical products: A and B. Production of both these products requires the same process – I and II. The production of B also results in a by-product C at no extra cost. The product A can be sold at a profit of Rs 3 per unit and B at a profit of Rs 8 per unit. Some quantity of this by-product can be sold at a unit profit of Rs 2, the remainder has to be destroyed and the destruction cost is Re 1 per unit. Forecasts show that only up to 5 units of C can be sold. The company gets 3 units of C for each unit of B produced. The manufacturing times are 3 hours per unit for A on process I and II, respectively, and 4 hours and 5 hours per unit for B on process I and II, respectively. Because the product C is a by product of B, no time is used in producing C. The available times are 18 and 21 hours of process I and II, respectively. Formulate this problem as an LP model to determine the quantity of A and B which should be produced, keeping C in mind, to make the highest total profit to the company. [*Delhi Univ., MBA* (*HCA*)*, 2001, 2008*]

LP model formulation The data of the problem is summarized as follows:

Constraints/Resources	*Time* (*hrs*) *Required by*			*Availability*
	A	*B*	*C*	
Process I	3	4	–	18 hrs
Process II	3	5	–	21 hrs
By-product ratio from B	–	1	3	5 units (max. units that
Profit per unit (Rs)	3	8	2	can be sold)

Decision variables Let

x_1, x_2 = units of product A and B to be produced, respectively

x_3, x_4 = units of product C to be produced and destroyed, respectively.

The LP model

Maximize (total profit) $Z = 3x_1 + 8x_2 + 2x_3 - x_4$

subject to the constraints

(i) Manufacturing constraints for product A and B

(a) $3x_1 + 4x_2 \le 18$, (b) $3x_1 + 5x_2 \le 21$

(ii) Manufacturing constraints for by-product C

(a) $x_3 \le 5$, (b) $-3x_2 + x_3 + x_4 = 0$

and $x_1, x_2, x_3, x_4 \ge 0.$

Example 2.7 A tape recorder company manufactures models A, B and C, which have profit contributions per unit of Rs 15, Rs 40 and Rs 60, respectively. The weekly minimum production requirements are 25 units for model A, 130 units for model B and 55 units for model C. Each type of recorder requires a certain amount of time for the manufacturing of the component parts for assembling and for packing. Specifically, a dozen units of model A require 4 hours for manufacturing, 3 hours for assembling and 1 hour for packaging. The corresponding figures for a dozen units of model B are 2.5, 4 and 2 and for a dozen units of model C are 6, 9 and 4. During the forthcoming week, the company has available 130 hours of manufacturing, 170 hours of assembling and 52 hours of packaging time. Formulate this problem as an LP model so as to maximize the total profit to the company.

LP model formulation The data of the problem is summarized as follows:

Resources/Constraints	*Models*			*Total Availability* (*hrs*)
	A	*B*	*C*	
Production requirement (units)	25	130	55	
Manufacturing time (per dozen)	4	2.5	6	130
Assembling time (per dozen)	3	4	9	170
Packaging time (per dozen)	1	2	4	52
Contribution per unit (Rs)	15	40	60	

Decision variables Let x_1, x_2 and x_3 = units of model A, B and C to be produced per week, respectively.

The LP model

Maximize (total profit) = $15x_1 + 40x_2 + 60x_3$

subject to the constraints

(i) Minimum production requirement:
(a) $x_1 \geq 25$, (b) $x_2 \geq 130$, (c) $x_3 \geq 55$

(ii) Manufacturing time : $\frac{4x_1}{12} + \frac{2.5x_2}{12} + \frac{6x_3}{12} \leq 130$

(iii) Assembling time : $\frac{3x_1}{12} + \frac{4x_2}{12} + \frac{9x_3}{12} \leq 170$

(iv) Packaging time : $\frac{x_1}{12} + \frac{2x_2}{12} + \frac{4x_3}{12} \leq 52$

and $x_1, x_2, x_3 \geq 0$.

Example 2.8 Consider the following problem faced by a production planner of a soft drink plant. He has two bottling machines A and B. A is designed for 8-ounce bottles and B for 16-ounce bottles. However, each can also be used for both types of bottles with some loss of efficiency. The manufacturing data is as follows:

Machine	*8-ounce Bottles*	*16-ounce Bottles*
A	100/minute	40/minute
B	60/minute	75/minute

The machines can be run for 8 hours per day, 5 days per week. The profit on an 8-ounce bottle is Rs 1.5 and on a 16-ounce bottle is Rs 2.5. Weekly production of the drink cannot exceed 3,00,000 bottles and the market can absorb 25,000, 8-ounce bottles and 7,000, 16-ounce bottles per week. The planner wishes to maximize his profit, subject of course, to all the production and marketing restrictions. Formulate this problem as an LP model to maximize total profit.

LP model formulation The data of the problem is summarized as follows:

Constraints	*Production*		*Availability*
	8-ounce Bottles	*16-ounce Bottles*	
Machine A time	100/minute	40/minute	8 × 5 × 60 = 2,400 minutes
Machine B time	60/minute	75/minute	8 × 5 × 60 = 2,400 minutes
Production	1	1	3,00,000 units/week
Marketing	1	–	25,000 units/week
	–	1	7,000 units/week
Profit/unit (Rs)	1.5	2.5	

Decision variables Let x_1 and x_2 = units of 8-ounce and 16-ounce bottles to be produced weekly, respectively

The LP model

Maximize (total profit) $Z = 1.5x_1 + 2.5x_2$

subject to the constraints

(i) Machine time : (a) $\frac{x_1}{100} + \frac{x_2}{40} \leq 2,400$ and (b) $\frac{x_1}{60} + \frac{x_2}{75} \leq 2,400$

(ii) Production : $x_1 + x_2 \leq 3,00,000$

(iii) Marketing : (a) $x_1 \leq 25,000$, (b) $x_2 \leq 7,000$

and $x_1, x_2 \geq 0$.

Example 2.9 A company engaged in producing tinned food has 300 trained employees on its rolls, each of whom can produce one can of food in a week. Due to the developing taste of public for this kind of food, the company plans to add to the existing labour force, by employing 150 people, in a phased manner, over the next five weeks. The newcomers would have to undergo a two-week training programme before being

put to work. The training is to be given by employees from among the existing ones and it is a known fact that one employee can train three trainees. Assume that there would be no production from the trainers and the trainees during training period, as the training is off-the-job. However, the trainees would be remunerated at the rate of Rs 300 per week, the same rate would apply as for the trainers.

The company has booked the following orders to supply during the next five weeks:

Week	:	1	2	3	4	5
No. of cans	:	280	298	305	360	400

Assume that the production in any week would not be more than the number of cans ordered for, so that every delivery of the food would be 'fresh'.

Formulate this problem as an LP model to develop a training schedule that minimizes the labour cost over the five-week period. [*Delhi Univ., MBA, 2003, 2005*]

LP model formulation The data of the problem is summarized as given below:

(i) Cans supplied

Week	:	1	2	3	4	5
Number	:	280	298	305	360	400

(ii) Each trainee has to undergo a two-week training.
(iii) One employee is required to train three trainees.
(iv) Every trained worker produces one can/week but there would be no production from trainers and trainees during training.
(v) Number of employees to be employed = 150
(vi) The production in any week is not to exceed the cans required.
(vii) Number of weeks for which newcomers would be employed: 5, 4, 3, 2, 1.

Observations based on given data are as follows:

(a) Workers employed at the beginning of the first week would get salary for all the five weeks; those employed at the beginning of the second week would get salary for four weeks and so on.
(b) The value of the objective function would be obtained by multiplying it by 300 because each person would get a salary of Rs 300 per week.
(c) Inequalities have been used in the constraints because some workers might remain idle in some week(s).

Decision variables Let x_1, x_2, x_3, x_4 and x_5 = number of trainees appointed in the beginning of week 1, 2, 3, 4 and 5, respectively.

The LP model

Minimize (total labour force) $Z = 5x_1 + 4x_2 + 3x_3 + 2x_4 + x_5$

subject to the constraints

(i) Capacity

(a) $300 - \frac{x_1}{3} \geq 280$,

(b) $300 - \frac{x_1}{3} - \frac{x_2}{3} \geq 298$

(c) $300 + x_1 - \frac{x_2}{3} - \frac{x_3}{3} \geq 305$,

(d) $300 + x_1 + x_2 - \frac{x_3}{3} - \frac{x_4}{3} \geq 360$

(e) $300 + x_1 + x_2 + x_3 - \frac{x_4}{3} - \frac{x_5}{3} \geq 400$

(ii) New recruitment

$$x_1 + x_2 + x_3 + x_4 + x_5 = 150$$

and $x_1, x_2, x_3, x_4, x_5 \geq 0$.

Example 2.10 XYZ company assembles and markets two types of mobiles – A and B. Presently 200 mobiles of each type are manufactured per week. You are advised to formulate the production schedule which will maximize the profits in the light of the following information:

Type	*Total Component Cost Per Mobile (Rs)*	*Man-Hours of Assembly Time Per Mobile*	*Average Man-Minutes of Inspection and Correction*	*Selling Price Per Mobile (Rs)*
A	2000	12	10	6000
B	1600	6	35	4800

The company employs 100 assemblers who are paid Rs 50 per hour actually worked and who will work up to a maximum of 48 hours per week. The inspectors, who are presently four, have agreed to a plan, whereby they average 40 hours of work per week each. However, the four inspectors have certain other administrative duties which have been found to take up an average of 8 hours per week between them. The inspectors are each paid a fixed wage of Rs 12000 per week.

Each mobile of either type requires one camera of same type. However, the company can obtain a maximum supply of 600 cameras per week. Their cost has been included in the component's cost, given for each mobile in the table above. The other cost incurred by the company are fixed overheads of Rs 20,000 per week.

LP model formulation Computation of contribution from radio types A and B is as follows:

		A	B
Component cost	:	2000	1600
Labour cost in assembly (Rs 50 per hour)	:	600	300
Labour cost for inspection			
$\frac{1200}{40}\times\frac{1}{60}$ = Re 0.50 per minute		5.00	17.5
Total variable cost		2605.00	1917.50
Selling price		6000	4800
Contribution (selling price – cost price)		3395	2882.50

Decision variables Let x_1 and x_2 = number of units of radio types A and B to be produced, respectively.

The LP model

Maximize (total contribution) $Z = 3395x_1 + 2882.5x_2 - 20{,}000$

subject to the constraints

(i) $12x_1 + 6x_2 \le 48 \times 100$, (ii) $10x_1 + 35x_2 \le 4 \times \left(40 - \frac{25}{3}\right) \times 60 = 7{,}600$

(iii) $x_1 + x_2 \le 600$

and $x_1, x_2 \ge 0$.

Example 2.11 A plastic products manufacturer has 1,200 boxes of transparent wrap in stock at one factory and another 1,200 boxes at its second factory. The manufacturer has orders for this product from three different retailers, in quantities of 1,000, 700 and 500 boxes, respectively. The unit shipping costs (in rupees per box) from the factories to the retailers are as follows:

	Retailer I	*Retailer II*	*Retailer III*
Factory A	14	11	13
Factory B	13	13	12

Determine a minimum cost shipping schedule for satisfying all demands from current inventory. Formulate this problem as an LP model.

LP model formulation Given that the total number of boxes available at factory A and B = total number of boxes required by retailers 1, 2 and 3.

Decision variables Let x_1, x_2 and x_3 = number of boxes to be sent from factory A to retailer 1; factory B to retailer 2 and factory C to retailer 3, respectively.

	Number of Boxes to be Sent		
	Retailer 1	*Retailer 2*	*Retailer 3*
Factory A	x_1	x_2	$1{,}200 - (x_1 + x_2)$
Factory B	$1{,}000 - x_1$	$700 - x_2$	$500 - [(1{,}000 - x_1) + (700 - x_2)]$

The LP model

Minimize (total distance) $Z = 14x_1 + 13x_2 + 11\ (1{,}200 - x_1 - x_2) + 13\ (1{,}000 - x_1) +$
$13\ (700 - x_2) + 12\ (x_1 - x_2 - 700) = 2x_1 + x_2 + 26{,}900$

subject to the constraints

(i) $x_1 + x_2 \le 1{,}200$, (ii) $x_1 \le 1{,}000$, (iii) $x_2 \le 700$;

and $x_1, x_2 \ge 0$.

Example 2.12 A company produces two types of sauces: A and B. Both these sauces are made by blending two ingredients – X and Y. A certain level of flexibility is permitted in the formulae of these products. Indeed, the restrictions are that (i) B must contain no more than 75 per cent of X, and (ii) A must contain no less than 25 per cent of X, and no less than 50 per cent of Y. Up to 400 kg of X and 300 kg of Y could be purchased. The company can sell as much of these sauces as it produces at a price of Rs 18 for A and Rs 17 for B. The X and Y cost Rs 1.60 and 2.05 per kg, respectively.

The company wishes to maximize its net revenue from the sale of these sauces. Formulate this problem as an LP model.

LP model formulation Let x_1, x_2 = kg of sauces A and B to be produced, respectively.
y_1, y_2 = kg of ingredient X used to make sauces A and B, respectively.
y_3, y_4 = kg of ingredient Y used to make sauces A and B, respectively.

The LP model

Maximize $Z = 18x_1 + 17x_2 - 1.60\,(y_1 + y_2) - 2.05\,(y_3 + y_4)$

subject to the constraints

(i) $y_1 + y_3 - x_1 = 0$,

(ii) $y_2 + y_4 - x_2 = 0$

(iii) $\left.\begin{aligned} y_1 + y_2 &\le 400 \\ y_3 + y_4 &\le 300 \end{aligned}\right\}$ (Purchase)

(iv) $\left.\begin{aligned} y_1 - 0.25\,x_1 &\ge 0 \\ y_2 - 0.50\,x_2 &\ge 0 \end{aligned}\right\}$ (Sauce A)

(v) $y_2 - 0.75x_2 \ge 0$ (Sauce B)

and $x_1, x_2, y_1, y_2, y_3, y_4 \ge 0$.

Example 2.13 A complete unit of a certain product consists of four units of component A and three units of component B. The two components (A and B) are manufactured from two different raw materials of which 100 units and 200 units, respectively, are available. Three departments are engaged in the production process with each department using a different method for manufacturing the components per production run and the resulting units of each component are given below:

Department	*Input of Raw Materials per Run (units)*		*Output of Components per Run (units)*	
	I	*II*	*A*	*B*
1	7	5	6	4
2	4	8	5	8
3	2	7	7	3

Formulate this problem as an LP model to determine the number of production runs for each department which will maximize the total number of complete units of the final product.

LP model formulation Let x_1, x_2 and x_3 = number of production runs for departments 1, 2 and 3, respectively.

Since each unit of the final product requires 4 units of component A and 3 units of component B, therefore maximum number of units of the final product cannot exceed the smaller value of

$$\left\{\frac{\text{Total number of units of A produced}}{4};\ \frac{\text{Total number of units of B produced}}{3}\right\}$$

or $$\left\{\frac{6x_1 + 5x_2 + 7x_3}{4} \text{ and } \frac{4x_1 + 8x_2 + 3x_3}{3}\right\}$$

Also if y is the number of component units of final product, then we obviously have

$$\frac{6x_1 + 5x_2 + 7x_3}{4} \ge y \text{ and } \frac{4x_1 + 8x_2 + 3x_3}{3} \ge y$$

The LP model

$$\text{Maximize } Z = \text{Min}\left\{\frac{6x_1 + 5x_2 + 7x_3}{4};\ \frac{4x_1 + 8x_2 + 3x_3}{3}\right\}$$

subject to the constraints

(i) Raw material

(a) $7x_1 + 4x_2 + 2x_3 \le 100$ (Material I), (b) $5x_1 + 8x_2 + 7x_3 \le 200$ (Material II)

(ii) Number of component units of final product

(a) $6x_1 + 5x_2 + 7x_3 - 4y \ge 0$, (b) $4x_1 + 8x_2 + 3x_3 - 4y \ge 0$

and $x_1, x_2, x_3 \ge 0$.

Example 2.14 ABC company manufactures three grades of paint: Venus, Diana and Aurora. The plant operates on a three-shift basis and the following data is available from the production records:

Requirement of Resource	*Grade*			*Availability* (*capacity/month*)
	Venus	*Diana*	*Aurora*	
Special additive (kg/litre)	0.30	0.15	0.75	600 tonnes
Milling (kilolitres per machine shift)	2.00	3.00	5.00	100 machine shifts
Packing (kilolitres per shift)	12.00	12.00	12.00	80 shifts

There are no limitations on the other resources. The particulars of sales forecasts and the estimated contribution to overheads and profits are given below:

	Venus	*Diana*	*Aurora*
Maximum possible sales per month (kilolitres)	100	400	600
Contribution (Rs/kilolitre)	4,000	3,500	2,000

Due to the commitments already made, a minimum of 200 kilolitres per month, of Aurora, must be supplied the next year.

Just when the company was able to finalize the monthly production programme for the next 12 months, it received an offer from a nearby competitor for hiring 40 machine shifts per month of milling capacity for grinding Diana paint that could be spared for at least a year. However, due to additional handling at the competitor's facility, the contribution from Diana would be reduced by Re 1 per litre.

Formulate this problem as an LP model for determining the monthly production programme to maximize contribution. [*Delhi Univ., MBA, 2006*]

LP model formulation Let

x_1 = quantity of Venus (kilolitres) produced in the company

x_2 = quantity of Diana (kilolitres) produced in the company

x_3 = quantity of Diana (kilolitres) produced by hired facilities

x_4 = quantity of Aurora (kilolitres) produced in the company

The LP model

Maximize (total profit) $Z = 4{,}000x_1 + 3{,}500x_2 + (3{,}500 - 1{,}000)x_3 + 2{,}000x_4$

subject to the constraints

(i) Special additive : $0.30x_1 + 0.15x_2 + 0.15x_3 + 0.75x_4 \le 600$

(ii) Own milling facility : $\frac{x_1}{2} + \frac{x_2}{3} + \frac{x_4}{5} \le 100$

(iii) Hired milling facility : $\frac{x_3}{3} \le 40$

(iv) Packing : $\frac{x_1}{12} + \frac{x_2 + x_3}{12} + \frac{x_4}{12} \le 80$

(v) Marketing:

(i) $x_1 \le 100$ (Venus); (ii) $x_2 + x_3 \le 400$ (Diana); (iii) $200 \le x_4 \le 600$ (Aurora)

and $x_1, x_2, x_3, x_4 \ge 0$.

Example 2.15 Four products have to be processed through a particular plant, the quantities required for the next production period are:

Product 1 : 2,000 units Product 2 : 3,000 units

Product 3 : 3,000 units Product 4 : 6,000 units

There are three production lines on which the products could be processed. The rates of production in units per day and the total available capacity in days are given in the following table. The corresponding cost of using the lines is Rs 600, Rs 500 and Rs 400 per day, respectively.

Production Line (*days*)	*Product*				*Maximum Line*
	1	*2*	*3*	*4*	
1	150	100	500	400	20
2	200	100	760	400	20
3	160	80	890	600	18
Total	2,000	3,000	3,000	6,000	

Formulate this problem as an LP model to minimize the cost of operation.

LP model formulation Let x_{ij} = number of units of product i (i = 1, 2, 3, 4) produced on production line j (j = 1, 2, 3)

The LP model

Minimize (total cost) $Z = 600\sum_{i=1}^{4} x_{i1} + 500\sum_{i=1}^{4} x_{i2} + 400\sum_{i=1}^{4} x_{i3}$

subject to the constraints

(i) Production: (a) $\sum_{i=1}^{3} x_{i1} = 2{,}000,$ (b) $\sum_{i=1}^{3} x_{i2} = 3{,}000$

(c) $\sum_{i=1}^{3} x_{i3} = 3{,}000,$ (d) $\sum_{i=1}^{3} x_{i4} = 6{,}000$

(ii) Line capacity

(a) $\frac{x_{11}}{150} + \frac{x_{12}}{100} + \frac{x_{13}}{500} + \frac{x_{14}}{400} \le 20,$ (b) $\frac{x_{21}}{200} + \frac{x_{22}}{100} + \frac{x_{23}}{760} + \frac{x_{24}}{400} \le 20$

(c) $\frac{x_{31}}{160} + \frac{x_{32}}{80} + \frac{x_{33}}{890} + \frac{x_{34}}{600} \le 18$

and $x_{ij} \ge 0$ for all i and j.

Example 2.16 XYZ company produces a specific automobile spare part. A contract that the company has signed with a large truck manufacturer calls for the following 4-month shipping schedule.

Month	*Number of Parts to be Shipped*
January	3,000
February	4,000
March	5,000
April	5,000

The company can manufacture 3,000 parts per month on a regular time basis and 2,000 parts per month on an overtime basis. Its production cost is Rs 15,000 for a part produced during regular time and 25,000 for a part produced during overtime. Its monthly inventory holding cost is Rs 500. Formulate this problem as an LP model to minimize the overall cost.

LP model formulation Let x_{ijk} = number of units of automobile spare part manufactured in month i (i = 1, 2, 3, 4) using shift j (j = 1, 2) and shipped in month k (k = 1, 2, 3, 4)

The LP model

Minimize (total cost) Z = Regular time production cost + Overtime production cost + One-month inventory cost + Two-month inventory cost + Three-month inventory cost

$= 15{,}000(x_{111} + x_{112} + x_{113} + x_{114} + x_{212} + x_{213} + x_{214} + x_{313} + x_{314} + x_{414})$
$+ 25{,}000(x_{121} + x_{122} + x_{123} + x_{124} + x_{222} + x_{223} + x_{224} + x_{323} + x_{324} + x_{424})$
$+ 500(x_{112} + x_{122} + x_{213} + x_{223} + x_{314} + x_{324}) + 1{,}000(x_{113} + x_{123} + x_{214} + x_{224}) + 1{,}500(x_{114} + x_{124})$

subject to the constraints

(i) Monthly regular time production

(a) $x_{111} + x_{112} + x_{113} + x_{114} \le 3{,}000$, (b) $x_{212} + x_{213} + x_{214} \le 3{,}000$

(c) $x_{313} + x_{314} \le 3{,}000$, (d) $x_{414} \le 3{,}000$

(ii) Monthly overtime production constraints

(a) $x_{121} + x_{122} + x_{123} + x_{124} \le 2{,}000$, (b) $x_{222} + x_{223} + x_{224} \le 2{,}000$

(c) $x_{323} + x_{324} \le 2{,}000$, (d) $x_{424} \le 2{,}000$

(iii) Monthly demand constraints

(a) $x_{111} + x_{121} = 3{,}000$, (b) $x_{112} + x_{122} + x_{212} + x_{222} = 4{,}000$

(c) $x_{113} + x_{123} + x_{213} + x_{223} + x_{313} + x_{323} = 5{,}000$

(d) $x_{114} + x_{124} + x_{214} + x_{224} + x_{314} + x_{324} + x_{414} + x_{424} = 5{,}000$

and $x_{ijk} \ge 0$ for all i, j and k.

2.8.2 Examples on Marketing

Example 2.17 An advertising company wishes to plan an advertising campaign for three different media: television, radio and a magazine. The purpose of the advertising is to reach as many potential customers as possible. The following are the results of a market study:

	Television			
	Prime Day (*Rs*)	*Prime Time* (*Rs*)	*Radio* (*Rs*)	*Magazine* (*Rs*)
Cost of an advertising unit	40,000	75,000	30,000	15,000
Number of potential customers reached per unit	4,00,000	9,00,000	5,00,000	2,00,000
Number of women customers reached per unit	3,00,000	4,00,000	2,00,000	1,00,000

The company does not want to spend more than Rs 8,00,000 on advertising. It is further required that

(i) at least 2 million exposures take place amongst women,

(ii) the cost of advertising on television be limited to Rs 5,00,000,

(iii) at least 3 advertising units be bought on prime day and two units during prime time; and

(iv) the number of advertising units on the radio and the magazine should each be between 5 and 10.

Formulate this problem as an LP model to maximize potential customer reach.

LP model formulation Let x_1, x_2, x_3 and x_4 = number of advertising units bought in prime day and time on television, radio and magazine, respectively.

The LP model

Maximize (total potential customer reach) $Z = 4{,}00{,}000x_1 + 9{,}00{,}000x_2 + 5{,}00{,}000x_3 + 2{,}00{,}000x_4$

subject to the constraints

(i) Advertising budget: $40{,}000x_1 + 75{,}000x_2 + 30{,}000x_3 + 15{,}000x_4 \le 8{,}00{,}000$

(ii) Number of women customers reached by the advertising campaign

$$3{,}00{,}000x_1 + 4{,}00{,}000x_2 + 2{,}00{,}000x_3 + 1{,}00{,}000x_4 \ge 20{,}00{,}000$$

(iii) Television advertising : (a) $40{,}000x_1 + 75{,}000x_2 \le 5{,}00{,}000$; (b) $x_1 \ge 3$; (c) $x_2 \ge 2$

(iv) Radio and magazine advertising : (a) $5 \le x_3 \le 10$; (b) $5 \le x_4 \le 10$

and $x_1, x_2, x_3, x_4 \ge 0$.

Example 2.18 A businessman is opening a new restaurant and has budgeted Rs 8,00,000 for advertisement, for the coming month. He is considering four types of advertising:

(i) 30 second television commercials

(ii) 30 second radio commercials

(iii) Half-page advertisement in a newspaper

(iv) Full-page advertisement in a weekly magazine which will appear four times during the coming month.

The owner wishes to reach families (a) with income over Rs 50,000 and (b) with income under Rs 50,000. The amount of exposure of each media to families of type (a) and (b) and the cost of each media is shown below:

Media	*Cost of Advertisement (Rs) Rs 50,000 (a)*	*Exposure to Families with Annual Income Over Rs 50,000 (b)*	*Exposure to Families with Annual Income Under*
Television	40,000	2,00,000	3,00,000
Radio	20,000	5,00,000	7,00,000
Newspaper	15,000	3,00,000	1,50,000
Magazine	5,000	1,00,000	1,00.000

To have a balanced campaign, the owner has determined the following four restrictions:

(i) there should be no more than four television advertisements
(ii) there should be no more than four advertisements in the magazine
(iii) there should not be more than 60 per cent of all advertisements in newspaper and magazine put together
(iv) there must be at least 45,00,000 exposures to families with annual income of over Rs 50,000.

Formulate this problem as an LP model to determine the number of each type of advertisement to be given so as to maximize the total number of exposures.

LP model formulation Let x_1, x_2, x_3 and x_4 = number of television, radio, newspaper, magazine advertisements to be pursued, respectively.

The LP model

Maximize (total number of exposures of both groups) Z

$$= (2,00,000 + 3,00,000)\, x_1 + (5,00,000 + 7,00,000)\, x_2 + (3,00,000 + 1,50,000)\, x_3 + (1,00,000 + 1,00,000)\, x_4$$

$$= 5,00,000\, x_1 + 12,00,000\, x_2 + 4,50,000\, x_3 + 2,00,000\, x_4$$

subject to the constraints

(i) Available budget : $40{,}000x_1 + 20{,}000x_2 + 15{,}000x_3 + 5{,}000x_4 \le 8{,}00{,}000$

(ii) Maximum television advertisement : $x_1 \le 4$

(iii) Maximum magazine advertisement

$x_4 \le 4$ (because magazine will appear only four times in the next month)

(iv) Maximum newspaper and magazine advertisement

$$\frac{x_3 + x_4}{x_1 + x_2 + x_3 + x_4} \le 0.60 \quad \text{or} \quad -0.6x_1 - 0.6x_2 + 0.4x_3 + 0.4x_4 \le 0$$

(v) Exposure to families with income over Rs 50,000

$$2{,}00{,}000x_1 + 5{,}00{,}000x_2 + 3{,}00{,}000x_3 + 1{,}00{,}000x_4 \ge 45{,}00{,}000$$

and $x_1, x_2, x_3, x_4 \ge 0$.

Example 2.19 An advertising agency is preparing an advertising campaign for a group of agencies. These agencies have decided that different characteristics of their target customers should be given different importance (weightage). The following table gives the characteristics with their corresponding importance (weightage).

	Characteristics	*Weightage (%)*
Age	25–40 years	20
Annual income	Above Rs 60,000	30
Female	Married	50

The agency has carefully analyzed three media and has compiled the following data:

Data Item	*Media*		
	Women's Magazine (%)	*Radio (%)*	*Television (%)*
Reader characteristics			
(i) Age: 25–40 years	80	70	60
(ii) Annual income: Above Rs 60,000	60	50	45
(iii) Females/Married	40	35	25
Cost per advertisement (Rs)	9,500	25,000	1,00,000
Minimum number of advertisement allowed	10	5	5
Maximum number of advertisement allowed	20	10	10
Audience size (1000s)	750	1,000	1,500

The budget for launching the advertising campaign is Rs 5,00,000. Formulate this problem as an LP model for the agency to maximize the total expected effective exposure.

LP model formulation Let x_1, x_2 and x_3 = number of advertisements made using advertising media: women's magazines, radio and television, respectively.

The effectiveness coefficient corresponding to each of the advertising media is calculated as follows:

Media	*Effectiveness Coefficient*
Women's magazine	0.80 (0.20) + 0.60 (0.30) + 0.40 (0.50) = 0.54
Radio	0.70 (0.20) + 0.50 (0.30) + 0.35 (0.50) = 0.46
Television	0.60 (0.20) + 0.45 (0.30) + 0.25 (0.50) = 0.38

The coefficient of the objective function, i.e. effective exposure for all the three media employed, can be computed as follows:

Effective exposure = Effectiveness coefficient × Audience size

where effectiveness coefficient is a weighted average of audience characteristics. Thus, the effective exposure of each media is as follows:

$$\text{Women's magazine} = 0.54 \times 7{,}50{,}000 = 4{,}05{,}000$$
$$\text{Radio} = 0.46 \times 10{,}00{,}000 = 4{,}60{,}000$$
$$\text{Television} = 0.38 \times 15{,}00{,}000 = 5{,}70{,}000$$

The LP model

Maximize (effective exposure) $Z = 4{,}05{,}000x_1 + 4{,}60{,}000x_2 + 5{,}70{,}000x_3$

subject to the constraints

(i) Budget: $9{,}500x_1 + 25{,}000x_2 + 1{,}00{,}000x_3 \leq 5{,}00{,}000$

(ii) Minimum number of advertisements allowed

(a) $x_1 \geq 10$; (b) $x_2 \geq 5$; and (c) $x_3 \geq 5$

(iii) Maximum number of advertisements allowed constraints

(a) $x_1 \leq 20$; (b) $x_2 \leq 10$; and (c) $x_3 \leq 10$

and $x_1, x_2, x_3 \geq 0$.

2.8.3 Examples on Finance

Example 2.20 An engineering company planned to diversify its operations during the year 2005-06. The company allocated capital expenditure budget equal to Rs 5.15 crore in the year 2005 and Rs 6.50 crore in the year 2006. The company had to take five investment projects under consideration. The estimated net returns at that present value and the expected cash expenditures on each project in those two years are as follows.

Assume that the return from a particular project would be in direct proportion to the investment in it, so that, for example, if in a project, say A, 20% (of 120 in 2005 and of 320 in 2006) was invested, then the resulting net return in it would be 20% (of 240). This assumption also implies that individuality of the project should be ignored. Formulate this capital budgeting problem as an LP model to maximize the net return.

Project	*Estimated Net Returns (in '000 Rs)*	*Cash Expenditure (in '000 Rs)*	
		Year 2005	*Year 2006*
A	240	120	320
B	390	550	594
C	80	118	202
D	150	250	340
E	182	324	474

LP model formulation Let x_1, x_2, x_3, x_4 and x_5 = proportion of investment in projects A, B, C, D and E, respectively.

The LP model

Maximize (net return) = $240x_1 + 390x_2 + 80x_3 + 150x_4 + 182x_5$

subject to the constraints

(i) Capital expenditure budget

(a) $120x_1 + 550x_2 + 118x_3 + 250x_4 + 324x_5 \leq 515$ [For year 2005]

(b) $320x_1 + 594x_2 + 202x_3 + 340x_4 + 474x_5 \leq 650$ [For year 2006]

(ii) 0-1 integer requirement constraints

(a) $x_1 \leq 1$, (b) $x_2 \leq 1$, (c) $x_3 \leq 1$, (d) $x_4 \leq 1$, (e) $x_5 \leq 1$

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

Example 2.21 XYZ is an investment company. To aid in its investment decision, the company has developed the investment alternatives for a 10-year period, as given in the following table. The return on investment is expressed as an annual rate of return on the invested capital. The risk coefficient and growth potential are subjective estimates made by the portfolio manager of the company. The terms of investment is the average length of time period required to realize the return on investment as indicated.

Investment Alternative	*Length of Investment*	*Annual Rate of Return (Year)*	*Risk Coefficient*	*Growth Potential Return (%)*
A	4	3	1	0
B	7	12	5	18
C	8	9	4	10
D	6	20	8	32
E	10	15	6	20
F	3	6	3	7
Cash	0	0	0	0

The objective of the company is to maximize the return on its investments. The guidelines for selecting the portfolio are:

(i) The average length of the investment for the portfolio should not exceed 7 years.

(ii) The average risk for the portfolio should not exceed 5.

(iii) The average growth potential for the portfolio should be at least 10%.

(iv) At least 10% of all available funds must be retained in the form of cash, at all times.

Formulate this problem as an LP model to maximize total return.

LP model formulation Let x_j = proportion of funds to be invested in the jth investment alternative ($j = 1, 2, \ldots, 7$)

The LP model

Maximize (total return) $Z = 0.03x_1 + 0.12x_2 + 0.09x_3 + 0.20x_4 + 0.15x_5 + 0.06x_6 + 0.00x_7$

subject to the constraints

(i) Length of investment : $4x_1 + 7x_2 + 8x_3 + 6x_4 + 10x_5 + 3x_6 + 0x_7 \leq 7$

(ii) Risk level : $x_1 + 5x_2 + 4x_3 + 8x_4 + 6x_5 + 3x_6 + 0x_7 \leq 5$

(iii) Growth potential : $0x_1 + 0.18x_2 + 0.10x_3 + 0.32x_4 + 0.20x_5 + 0.07x_6 + 0x_7 \geq 0.10$

(iv) Cash requirement : $x_7 \geq 0.10$

(v) Proportion of funds : $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 1$

and $x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$.

Example 2.22 An investor has three investment opportunities available to him at the beginning of each years, for the next 5 years. He has a total of Rs 5,00,000 available for investment at the beginning of the first year. A summary of the financial characteristics of the three investment alternatives is presented in the following table:

Investment Alternative	*Allowable Size of Initial Investment (Rs)*	*Return (%)*	*Timing of Return*	*Immediate Reinvestment Possible?*
1	1,00,000	19	1 year later	yes
2	unlimited	16	2 years later	yes
3	50,000	20	3 years later	yes

The investor wishes to determine the investment plan that will maximize the amount of money which can be accumulated by the beginning of the 6th year in the future. Formulate this problem as an LP model to maximize total return. [*Delhi Univ., MBA, 2002, 2008*]

LP model formulation Let

x_{ij} = amount to be invested in investment alternative, i (i = 1, 2, 3) at the beginning of the year j (j =1, 2, . . ., 5)

y_j = amount not invested in any of the investment alternatives in period j

The LP model

Minimize (total return) $Z = 1.19x_{15} + 1.16x_{24} + 1.20x_{33} + y_5$

subject to the constraints

(i) Yearly cash flow

(a) $x_{11} + x_{21} + x_{31} + y_1 = 5{,}00{,}000$ (year 1)

(b) $-y_1 - 1.19x_{11} + x_{12} + x_{22} + x_{32} + y_2 = 0$ (year 2)

(c) $-y_2 - 1.16x_{21} - 1.19x_{12} + x_{23} + x_{23} + x_{33} + y_3 = 0$

(d) $-y_3 - 1.20x_{31} - 1.16x_{22} - 1.19x_{13} + x_{14} + x_{24} + x_{34} + y_4 = 0$ (year 4)

(e) $-y_4 - 1.20x_{32} - 1.16x_{23} - 1.19x_{14} + x_{15} + x_{25} + x_{35} + y_5 = 0$ (year 5)

(ii) Size of investment

$x_{11} \le 1{,}00{,}000$, $x_{12} \le 1{,}00{,}000$, $x_{13} \le 1{,}00{,}000$, $x_{14} \le 1{,}00{,}000$, $x_{15} \le 1{,}00{,}000$

$x_{31} \le 50{,}000$, $x_{32} \le 50{,}000$, $x_{33} \le 50{,}000$, $x_{34} \le 50{,}000$, $x_{35} \le 50{,}000$

and $x_{ij}, y_j \ge 0$ for all i and j.

Remark To formulate the first set of constraints of yearly cash flow, the following situation is adopted:

$$\frac{\text{Investment alternatives}}{x_{12} + x_{22} + x_{32} + y_2} = \frac{\text{Investment alternatives}}{y_1 + 1.19x_{11}}$$

or $$-y_1 - 1.19x_{11} + x_{12} + x_{22} + x_{32} + y_2 = 0.$$

Example 2.23 A leading CA is attempting to determine the 'best' investment portfolio and is considering six alternative investment proposals. The following table indicates point estimates for the price per share, the annual growth rate in the price per share, the annual dividend per share and a measure of the risk associated with each investment.

Portfolio Data

Shares Under Consideration	*A*	*B*	*C*	*D*	*E*	*F*
Current price per share (Rs)	80.00	100.00	160.00	120.00	150.00	200.00
Projected annual growth rate	0.08	0.07	0.10	0.12	0.09	0.15
Projected annual dividend per share (Rs)	4.00	4.50	7.50	5.50	5.75	0.00
Projected risk return	0.05	0.03	0.10	0.20	0.06	0.08

The total amount available for investment is Rs 25 lakh and the following conditions are required to be satisfied:

(i) The maximum rupee amount to be invested in alternative F is Rs 2,50,000.

(ii) No more than Rs 5,00,000 should be invested in alternatives A and B combined.

(iii) Total weighted risk should not be greater than 0.10, where

$$\text{Total weighted risk} = \frac{(\text{Amount invested in alternative } j)\,(\text{Risk of alternative } j)}{\text{Total amount invested in all the alternatives}}$$

(iv) For the sake of diversity, at least 100 shares of each stock should be purchased.

(v) At least 10 per cent of the total investment should be in alternatives A and B combined.

(vi) Dividends for the year should be at least 10,000.

Rupee return per share of stock is defined as the price per share one year hence, less current price per share plus dividend per share. If the objective is to maximize total rupee return, formulate this problem as an LP model for determining the optimal number of shares to be purchased in each of the shares under consideration. You may assume that the time horizon for the investment is one year.

LP model formulation Let x_1, x_2, x_3, x_4, x_5 and x_6 = number of shares to be purchased in each of the six investment proposals A, B, C, D, E and F, respectively.

Rupee return per share = Price per share one year hence – Current price per share + Dividend per share

= Current price per share × Projected annual growth rate (i.e. Projected growth each year + Dividend per share).

Thus, we compute the following data:

Investment Alternatives	:	*A*	*B*	*C*	*D*	*E*	*F*
No. of shares purchased	:	x_1	x_2	x_3	x_4	x_5	x_6
Projected growth for each share (Rs)	:	6.40	7.00	16.00	14.40	13.50	30.00
Projected annual dividend per share (Rs)	:	4.00	4.50	7.50	5.50	5.75	0.00
Return per share (Rs)	:	10.40	11.50	23.50	19.90	19.25	30.00

The LP model

Maximize (total return) $R = 10.40x_1 + 11.50x_2 + 23.50x_3 + 19.90x_4 + 19.25x_5 + 30.00x_6$

subject to the constraints

(i) $80x_1 + 100x_2 + 160x_3 + 120x_4 + 150x_5 + 200x_6 \le 25{,}00{,}000$ (total fund available)

(ii) $200x_6 \le 2{,}50{,}000$ [from condition (i)]

(iii) $80x_1 + 100x_2 \le 5{,}00{,}000$ [from condition (ii)]

(iv) $$\frac{80x_1(0.05) + 100x_2(0.03) + 160x_3(0.10) + 120x_4(0.02) + 150x_5(0.06) + 200x_6(0.08)}{80x_1 + 100x_2 + 160x_3 + 120x_4 + 150x_5 + 200x_6} \le 1$$

$$4x_1 + 3x_2 + 16x_3 + 24x_4 + 9x_5 + 16x_6 \le 8x_1 + 10x_2 + 16x_3 + 12x_4 + 15x_5 + 20x_6$$

$$-4x_1 - 7x_1 + 0x_3 + 12x_4 - 6x_5 - 4x_6 \le 0$$

(v) $x_1 \ge 100,\ x_2 \ge 100,\ x_3 \ge 100,\ x_4 \ge 100,\ x_5 \ge 100,\ x_6 \ge 100$ [from condition (iv)]

(vi) $80x_1 + 100x_2 \ge 0.10\,(80x_1 + 100x_2 + 160x_3 + 120x_4 + 150x_5 + 200x_6)$ [from condition (v)]

$$80x_1 + 100x_2 \ge 8x_1 + 10x_2 + 16x_3 + 12x_4 + 15x_5 + 20x_6$$

$$72x_1 + 90x_2 - 16x_3 - 12x_4 - 15x_5 - 20x_6 \ge 0$$

(vii) $4x_1 + 4.5x_2 + 7.5x_3 + 5.5x_4 + 5.75x_5 \ge 10{,}000$ [from condition (vi)]

and $x_j \ge 0;\ j = 1, 2, 3, 4, 5$ and 6.

Example 2.24 A company must produce two products over a period of three months. The company can pay for materials and labour from two sources: company funds and borrowed funds.

The firm has to take three decisions:

(a) How many units of product 1 should it produce?

(b) How many units of product 2 should it produce?

(c) How much money should it borrow to support the production of the products?

The firm must take these decisions in order to maximize the profit contribution, subject to the conditions stated below:

(i) Since the company's products enjoy a seller's market, the company can sell as many units as it can produce. The company would therefore like to produce as many units as possible, subject to its production capacity and financial constraints. The capacity constraints, together with cost and price data, are shown in the following table:

Capacity, Price and Cost Data

Product	*Selling Price (Rs per Unit)*	*Cost of Production (Rs per Unit)*	*Required Hours per Unit in Department*		
			A	*B*	*C*
1	14	10	0.5	0.3	0.2
2	11	8	0.3	0.4	0.1
Available hours per production period of three months :			500.00	400.00	200.00

(ii) The available company funds during the production period will be Rs 3 lakh.

(iii) A bank will give loans up to Rs 2 lakh per production period at an interest rate of 20 per cent per annum provided that company's acid (quick) test ratio is at 1 to 1 while the loan is outstanding. Take a simplified acid-test ratio given by

$$\frac{\text{Surplus cash on hand after production + Accounts receivable}}{\text{Bank borrowings + Interest occurred thereon}}$$

(iv) Also make sure that the needed funds are made available for meeting production costs. Formulate this problem as an LP model.

LP model formulation Let x_1, x_2 = number of units of products 1 and 2 produced, respectively.
x_3 = amount of money borrowed.

The LP model

Profit contribution per unit of each product = (Selling price – Variable cost of production)

Maximize Z = Total profit by producing two products – Cost of borrowed money
$= (14 - 10)x_1 + (11 - 8)x_2 - 0.05x_3 = 4x_1 + 3x_2 - 0.05x_3$
(since the interest rate is 20 per cent per annum, it will be 5 per cent for a period of three months)

subject to the constraints

(i) The production capacity constraints for each department

(a) $0.5x_1 + 0.3x_2 \leq 500$, (b) $0.3x_1 + 0.4x_2 \leq 400$, (c) $0.2x_1 + 0.1x_2 \leq 200$

(ii) The funds available for production are the sum of Rs 3,00,000 in cash that the firm has and borrowed funds maximum up to Rs 2,00,000. Consequently, production is limited to the extent that the funds are available to pay for production costs. Thus, we write the constraint as:

Funds required for production ≤ Funds available

$$10x_1 + 8x_2 \leq 3,00,000 + x_3$$

$$10x_1 + 8x_2 - x_3 \leq 3,00,000$$

(iii) Borrowed funds constraint [from condition (iii) of the problem]

$$x_1 \leq 2,00,000$$

(iv) Acid-test condition constraint

$$\frac{\text{Surplus cash on hand after production + Accounts receivable}}{\text{Bank borrowings + Interest accrued thereon}} \geq 1$$

$$\frac{(3,00,000 + x_3 - 10x_1 - 8x_2) + 14x_1 + 11x_2}{x_3 + 0.2x_3} \geq 1$$

$$3,00,000 + x_3 + 4x_1 + 3x_2 \geq x_3 + 0.2x_3$$

or $$-4x_1 - 3x_2 + 0.2x_3 \leq 3,00,000$$

and $$x_1, x_2, x_3 \geq 0.$$

Example 2.25 The most recent audited summarized balance sheet of Shop Financial Service is given below:

The company intends to enhance its investment in the lease portfolio by another Rs 1,000 lakh. For this purpose, it would like to raise a mix of debt and equity in such a way that the overall cost of raising additional funds is minimized. The following constraints apply to the way the funds can be mobilized:

(i) Total debt divided by net owned funds, cannot exceed 10.

(ii) Amount borrowed from financial institutions cannot exceed 25 per cent of the net worth.

(iii) Maximum amount of bank borrowings cannot exceed three times the net owned funds.

Balance Sheet as on 31 March 2008

Liabilities	(*Rs lakh*)	*Assets*	(*Rs lakh*)
Equity Share Capital	65	*Fixed Assets:*	
Reserves & Surplus	110	Assets on Lease	
		(Original Cost: Rs 550 lakhs)	375
Term Loan from IFCI	80	Other Fixed Assets	50
Public Deposits	150	Investments (on wholly owned subsidiaries)	20
Bank Borrowings	147	*Current Assets:*	
Other Current Liabilities	50	Stock on Hire	80
	602	Receivables	30
		Other Current Assets	35
		Miscellaneous Expenditure (not written off)	12
			602

(iv) The company would like to keep the total public deposit limited to 40 per cent of the total debt. The post-tax costs of the different sources of finance are as follows:

Equity	Term Loans	Public Deposits	Bank Borrowings
2.5%	8.5%	7%	10%

Formulate this problem as an LP model to minimize cost of funds raised.

***Note*:** (a) Total Debt = Term loans from Financial Institutions + Public deposits + Bank borrowings
(b) Net worth = Equity share capital + Reserves and surplus
(c) Net owned funds = Net worth – Miscellaneous expenditures

LP model formulation Let x_1, x_2, x_3 and x_4 = quantity of additional funds (in lakh) raised on account of additional equity, term loans, public deposits, bank borrowings, respectively.

The LP model

Minimize (cost of additional funds raised) $Z = 0.025x_1 + 0.085x_2 + 0.07x_3 + 0.1x_4$

subject to the constraints

(i) $\dfrac{\text{Total Debt}}{\text{Net owned funds}} \le 10$ or $\dfrac{\text{Existing debt+Additional total debt}}{\text{(Equity share capital+Reserve \& surplus + Additional equity – Misc. exp.)}} \le 10$

$$\frac{80+150+147+x_2+x_3+x_4}{(65+110+x_1)-12} \le 10 \quad \text{or} \quad \frac{x_2+x_3+x_4+377}{x_1+163} \le 10$$

$$x_2 + x_3 + x_4 + 377 \le 10\,x_1 + 1{,}630 \quad \text{or} \quad -10x_1 + x_2 + x_3 + x_4 \le 1{,}253.$$

(ii) Amount borrowed (from financial institutions) ≤ 25% of net worth

or (Existing long-term loan from financial institutions + Additional loan) ≤ 25% (Existing equity capital + Reserve & surplus + Addl. equity capital)

$$80 + x_2 \le 0.25\,(175 + x_1)$$
$$320 + 4x_1 \le 175 + x_1$$
$$-x_1 + 4x_2 \le -145 \text{ or } x_1 - 4x_2 \ge 145.$$

(iii) Maximum bank borrowings ≤ 3 (Net owned funds)

or (Existing bank borrowings + Addl. bank borrowings ≤ 3 (Existing equity capital + Reserves & surplus + Addl. equity capital – Misc. exp.)

$$(147 + x_4) \le 3\,(65 + 110 + x_1 - 12)$$
$$x_4 - 3x_1 \le 525 - 36 - 147$$
$$-3x_1 + x_4 \le 342.$$

(iv) Total public deposit ≤ 40% of total debt.

or (Existing public deposits + Addl. public deposits) ≤ 0.40 (Existing total debt + Addl. total debt)

or $150 + x_3 \le 0.40\,(80 + 150 + 147 + x_2 + x_3 + x_4)$ or $150 + x_3 \le 0.40\,(x_2 + x_3 + x_4 + 377)$

$1{,}500 + 10x_3 \le 4x_2 + 4x_3 + 4x_4 + 1{,}508$ or $-4x_2 + 6x_3 - 4x_4 \le 8.$

(v) Addl. equity capital + Addl. term loan + Addl. public deposits + Addl. bank borrowings = 1,000 (since the company wants to enhance the investment by Rs 1,000 lakh)

or $x_1 + x_2 + x_3 + x_4 = 1{,}000$

and $x_1, x_2, x_3, x_4 \ge 0.$

Example 2.26 Renco-Foundries is in the process of drawing up a Capital Budget for the next three years. It has funds to the tune of Rs 1,00,000 that can be allocated among projects A, B, C, D and E. The net cash flows associated with an investment of Re 1 in each project are provided in the following table.

Cash Flow at Time

Investment in	*0*	*1*	*2*	*3*
A	– Re 1	+ Re 0.5	+ Re 1	Re 0
B	Re 0	– Re 1	+ Re 0.5	+ Re 1
C	– Re 1	+ Rs 1.2	Re 0	Re 0
D	– Re 1	Re 0	Re 0	Rs 1.9
E	Re 0	Re 0	– Re 1	Rs 1.5

***Note*:** Time 0 = present, Time 1 = 1 year from now. Time 2 = 2 years from now. Time 3 = 3 years from now.

For example, Re 1 invested in investment B requires a Re 1 cash outflow at time 1 and returns Re 0.50 at time 2 and Re 1 at time 3.

To ensure that the firm remains reasonably diversified, the firm will not commit an investment exceeding Rs 75,000 for any project. The firm cannot borrow funds and therefore, the cash available for investment at any time is limited to the cash in hand. The firm will earn interest at 8 per cent per annum by parking the un-invested funds in money market investments. Assume that the returns from investments can be immediately re-invested. For example, the positive cash flow received from project C at time 1 can immediately be re-invested in project B. Formulate this problem as an LP model so as to maximize cash on hand at time 3. [*CA, 2000; Delhi Univ., MBA, 2007*]

LP model formulation Let x_1, x_2, x_3, x_4 and x_5 = Amount of rupees invested in investments A, B, C, D and E, respectively.

s_i = Money invested in money market instruments at time i (for $i = 0, 1, 2$).

Firm earns interest at 8 per cent per annum by parking the un-invested funds in money market instruments, hence Rs s_0, Rs s_1 and Rs s_2 which are invested in these instruments at times 0, 1 and 2 will become $1.08s_0$, $1.08s_1$ and $1.08s_2$ at times 1, 2 and 3, respectively.

Note: Cash available for investment in time t = cash on hand at time t.

From the given data, it can be computed that at time 3:

$$\text{Cash on hand} = x_1 \times 0 + x_2 \times 1 + x_3 \times 0 + 1.9x_4 + 1.5x_5 + 1.08s_2$$
$$= \text{Rs } (x_2 + 1.9x_4 + 1.5x_5 + 1.08s_2)$$

The LP model

Maximize (Cash on hand at time 3) $Z = x_2 + 1.9x_4 + 1.5x_5 + 1.08s_2$

subject to the constraints

At time 0: Total fund of Rs 1,00,000 is available for investing on projects A, C and D. That is

$$x_1 + x_2 + x_3 + s_0 = 1{,}00{,}000$$

At time 1: Rs $0.5x_1$, Rs $1.2x_3$, and Rs $1.08s_0$ will be available as a result of investment made at time 0. Since Rs x_2 and s_1 are invested in project B and money market instruments, respectively at time 1, therefore we write

$$0.5x_1 + 1.2x_3 + 1.08s_0 = x_2 + s_1$$

At time 2: Rs x_1; Rs $0.5x_2$ and Rs 1.08 s_1 will be available for investment. As Rs x_5 and Rs s_2 are invested at time 2. Thus

$$x_1 + 0.5x_2 + 1.08s_1 = x_5 + s_2$$

Also, since the company will not commit an investment exceeding Rs 75,000 in any project, therefore the constraint becomes: $x_i \leq 75{,}000$ for $i = 1, 2, 3, 4, 5$.

and $x_1, x_2, x_3, x_4, x_5, s_0, s_1, s_2 \geq 0$.

2.8.4 Examples on Agriculture

Example 2.27 A cooperative farm owns 100 acres of land and has Rs 25,000 in funds available for investment. The farm members can produce a total of 3,500 man-hours worth of labour during September – May and 4,000 man-hours during June – August. If any of these man-hours are not needed, some members of the firm would use them to work on a neighbouring farm for Rs 2 per hour during September – May and Rs 3 per hour during June – August. Cash income can be obtained from the three main crops and two types of livestock: dairy cows and laying hens. No investment funds are needed for the crops. However, each cow will require an investment outlay of Rs 3,200 and each hen will require Rs 15.

In additon each cow will also require 15 acres of land, 100 man-hours during the summer. Each cow will produce a net annual cash income of Rs 3,500 for the farm. The corresponding figures for each hen are: no acreage, 0.6 man-hours during September – May; 0.4 man-hours during June – August, and an annual net cash income of Rs 200. The chicken house can accommodate a maximum of 4,000 hens and the size of the cattle-shed limits the members to a maximum of 32 cows.

Estimated man-hours and income per acre planted in each of the three crops are:

	Paddy	*Bajra*	*Jowar*
Man-hours			
September-May	40	20	25
June-August	50	35	40
Net annual cash income (Rs)	1,200	800	850

The cooperative farm wishes to determine how much acreage should be planted in each of the crops and how many cows and hens should be kept in order to maximize its net cash income. Formulate this problem as an LP model to maximize net annual cash income.

LP model formulation The data of the problem is summarized as follows:

Constraints	*Cows*	*Hens*	*Crop*			*Extra Hours*		*Total Availability*
			Paddy	*Bajra*	*Jowar*	*Sept–May*	*June–Aug*	
Man-hours								
Sept–May	100	0.6	40	20	25	1	–	3,500
June–Aug	50	0.4	50	35	40	–	1	4,000
Land	1.5	–	1	1	1	–	–	100
Cow	1	–	–	–	–	–	–	32
Hens	–	1	–	–	–	–	–	4,000
Net annual cash income (Rs)	3,500	200	1,200	800	850	2	3	

Decision variables Let

x_1 and x_2 = number of dairy cows and laying hens, respectively.

x_3, x_4 and x_5 = average of paddy crop, *bajra* crop and *jowar* crop, respectively.

x_6 = extra man-hours utilized in Sept–May.

x_7 = extra man-hours utilized in June–Aug.

The LP model

Maximize (net cash income) $Z = 3{,}500x_1 + 200x_2 + 1{,}200x_3 + 800x_4 + 850x_5 + 2x_6 + 3x_7$

subject to the constraints

(i) Man-hours: $100x_1 + 0.6x_2 + 40x_3 + 20x_4 + 25x_5 + x_6 = 3{,}500$ (Sept-May duration)

$50x_1 + 0.4x_2 + 50x_3 + 35x_4 + 40x_5 + x_7 = 4{,}000$ (June-Aug duration)

(ii) Land availability: $1.5x_1 + x_3 + x_4 + x_5 \leq 100$

(iii) Livestock: (a) $x_1 \leq 32$ (dairy cows), (b) $x_2 \leq 4{,}000$ (laying hens)

and $x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$.

Example 2.28 A certain farming organization operates three farms of comparable productivity. The output of each farm is limited both by the usable acreage and by the amount of water available for irrigation. The data for the upcoming season is as shown below:

Farm	*Usable Acreage*	*Water Available (in cubic feet)*
1	400	1,500
2	600	2,000
3	300	900

The organization is considering planting crops which differ primarily in their expected profit per acre and in their consumption of water. Furthermore, the total acreage that can be devoted to each of the crops is limited by the amount of appropriate harvesting equipment available.

Crop	*Maximum Acreage*	*Water Consumption (in cubic feet)*	*Expected Profit per Acre (Rs)*
A	700	5	4,000
B	800	4	3,000
C	300	3	1,000

In order to maintain a uniform workload among the three farms, it is the policy of the organization that the percentage of the usable acreage planted be the same for each farm. However, any combination of the crops may be grown at any of the farms. The organization wishes to know how much of each crop should be planted at the respective farms in order to maximize expected profit.

Formulate this problem as an LP model in order to maximize the total expected profit.

LP model formulation The data of the problem is summarized below:

Crop	*Farm*			*Crop Requirement (in acres)*	*Expected Profit per Acre (Rs)*
	1	*2*	*3*		
A	x_{11}	x_{12}	x_{13}	700	4,000
B	x_{21}	x_{22}	x_{23}	800	3,000
C	x_{31}	x_{32}	x_{33}	300	1,000
Usable acreage	400	600	300		
Water available per acre	1,500	2,000	900		

Decision variables Let x_{ij} = number of acres to be allocated to crop i (i = 1, 2, 3) to farm j (j = 1, 2)

The LP model

Maximize (net profit) $Z = 4{,}000(x_{11} + x_{12} + x_{13}) + 3{,}000(x_{21} + x_{22} + x_{23}) + 1{,}000(x_{31} + x_{32} + x_{33})$ subject to the constraints

(i) Crop requirement

(a) $x_{11} + x_{12} + x_{13} \le 700$, (b) $x_{21} + x_{22} + x_{23} \le 800$, (c) $x_{31} + x_{32} + x_{33} \le 300$

(ii) Available acreage

(a) $x_{11} + x_{21} + x_{31} \le 400$, (b) $x_{12} + x_{22} + x_{32} \le 600$, (c) $x_{13} + x_{23} + x_{33} \le 300$

(iii) Water available (in acre feet)

(a) $5x_{11} + 4x_{21} + 3x_{31} \le 1{,}500$, (b) $5x_{12} + 4x_{22} + 3x_{32} \le 2{,}000$, (c) $5x_{13} + 4x_{23} + 3x_{33} \le 900$

(iv) Social equality

(a) $\frac{x_{11} + x_{21} + x_{31}}{400} = \frac{x_{12} + x_{22} + x_{32}}{600}$, (b) $\frac{x_{12} + x_{22} + x_{32}}{600} = \frac{x_{13} + x_{23} + x_{33}}{300}$,

(c) $\frac{x_{13} + x_{23} + x_{33}}{300} = \frac{x_{11} + x_{21} + x_{31}}{400}$

and $x_{ij} \ge 0$ for all i and j.

2.8.5 Examples on Transportation

Example 2.29 ABC manufacturing company wishes to develop its monthly production schedule for the next three months. Depending upon the sales commitments, the company can either keep the production constant, allowing fluctuation in inventory; or its inventories can be maintained at a constant level, with fluctuating production. Fluctuating production makes overtime work necessary, the cost of which is estimated to be double the normal production cost of Rs 12 per unit. Fluctuating inventories result in an inventory carrying cost of Rs 2 per unit/month. If the company fails to fulfil its sales commitment, it incurs a shortage cost of Rs 4 per unit/month. The production capacities for the next three months are in the table:

Month	*Production Capacity (units)*		*Sales (units)*
	Regular	*Overtime*	
1	50	30	60
2	50	0	120
3	60	50	40

Formulate this problem as an LP model to minimize the total production cost.

[*Delhi Univ., MBA, 2008*]

LP model formulation The data of the problem is summarized as follows:

Month	*Production Capacity*		*Sales*
	Regular	*Overtime*	
1	50	30	60
2	50	0	120
3	60	50	40

Normal production cost : Rs 12 per unit Overtime cost : Rs 24 per unit
Carrying cost : Rs 2 per unit per month Shortage cost : Rs 4 per unit per month

Assume five sources of supply: three regular and two overtime (because the second months overtime production is zero) production capacities. The demand for the three months will be the sales during these months.

All supplies against the order have to be made and can be made in the subsequent month if it is not possible to make them during the month of order, with additional cost equivalent to shortage cost, i.e. in month 2. The cumulative production of months 1 and 2 in regular and overtime is 130 units while the orders are for 180 units. This balance can be supplied during month 3 at an additional production cost of Rs 4.

The given information can now be presented in matrix form as follows:

	M_1	M_2	M_3	Production (supply)
M_1	12	14	16	50
M_2	16	12	14	50
M_3	20	16	12	60
M_1(OT)	24	26	28	30
M_2(OT)	32	28	24	50
Sales (demand)	60	120	40	

Decision variables Let x_{ij} = amount of commodity sent from source of supply i (i = 1, 2, . . ., 5) to destination j (j = 1, 2, 3)

The LP model

Minimize (total cost) $Z = 12x_{11} + 14x_{12} + 16x_{13} + 16x_{21} + 12x_{22} + 14x_{23} + 20x_{31} + 16x_{32} + 12x_{33} + 24x_{41} + 26x_{42} + 28x_{43} + 32x_{51} + 28x_{52} + 24x_{53}$

subject to the constraints

(i) Production (supply) constraints

(a) $x_{11} + x_{12} + x_{13} = 50$, (b) $x_{21} + x_{22} + x_{23} = 50$, (c) $x_{31} + x_{32} + x_{33} = 60$,
(d) $x_{41} + x_{42} + x_{43} = 30$, (e) $x_{51} + x_{52} + x_{53} = 30$

(ii) Sales (demand) constraints

(a) $x_{11} + x_{21} + x_{31} + x_{41} + x_{51} = 60$, (b) $x_{12} + x_{22} + x_{32} + x_{42} + x_{52} = 120$,
(c) $x_{13} + x_{23} + x_{33} + x_{43} + x_{53} = 40$

and $x_{ij} \geq 0$ for all i and j.

Example 2.30 A trucking firm has received an order to move 3,000 tonnes of industrial material to a destination 1,000 km away. The firm has available, at the moment, a fleet of 150 class-A 15-tonne trailer trucks and another fleet of 100 class-B 10-tonne trailer trucks. The operating costs of these trucks are Rs 3 and Rs 4 per tonne per km, respectively. Based on past experience, the firm has a policy of retaining at least one class-A truck with every two class-B trucks in reserve. It is desired to know how many of these two classes of vehicles should be despatched to move the material at minimal operating costs. Formulate this problem as an LP model.

LP model formulation Let x_1 and x_2 = number of class A and B trucks to be despatched, respectively.

The LP model

Minimize (total operating cost) $Z = 3x_1 + 4x_2$

subject to the constraints

$$15x_1 + 10x_2 \leq 3{,}000$$

$$\left.\begin{aligned} x_1 &\leq 149 \\ x_2 &\leq 98 \end{aligned}\right\}$$ (due to the policy of retaining at least one class-A truck with every two class-B truck in reserve)

and $x_1, x_2 \geq 0$.

Example 2.31 A ship has three cargo loads – forward, after and centre. Their capacity limits are:

	Weight (*kg*)	*Volume* (*cu cm*)
Forward	2,000	1,00,000
Centre	3,000	1,35,000
After	1,500	30,000

The following cargos are offered to be carried in the ship. The ship owner may accept all or any part of each commodity:

Commodity	*Weight (kg)*	*Volume (cu cm)*	*Profit (in Rs) per kg*
A	6,000	60	60
B	4,000	50	80
C	2,000	25	50

In order to preserve the trim of the ship, the weight in each cargo must be proportional to the capacity in kg. The cargo is to be distributed in a way so as to maximize profit. Formulate this problem as an LP model.

LP model formulation x_{iA}, x_{iB} and x_{iC} = weight (in kg) of commodities A, B and C to be accommodated in the direction i($i = 1, 2, 3$ – forward, centre and after), respectively.

The LP model

Maximize (total profit) $Z = 60\,(x_{1A} + x_{2A} + x_{3A}) + 80\,(x_{1B} + x_{2B} + x_{3B}) + 50\,(x_{1C} + x_{2C} + x_{3C})$

subject to the constraints

$$x_{1B} + x_{2B} + x_{3B} \le 4{,}000; \quad x_{1B} + x_{2B} + x_{3B} \le 4{,}000;$$
$$x_{1B} + x_{2B} + x_{3B} \le 4{,}000; \quad x_{1A} + x_{1B} + x_{1C} \le 2{,}000$$
$$x_{1A} + x_{2B} + x_{3C} \le 4{,}000; \quad x_{3A} + x_{3B} + x_{3C} \le 1{,}500$$
$$60x_{1A} + 50x_{1B} + 25x_{1C} \le 1{,}00{,}000$$
$$60x_{2A} + 50x_{2B} + 25x_{2C} \le 1{,}35{,}000$$
$$60x_{3A} + 50x_{3B} + 25x_{3C} \le 30{,}000$$

and $x_{iA}, x_{iB}, x_{iC} \ge 0$, for all i.

2.8.6 Examples on Personnel

Example 2.32 Evening shift resident doctors in a government hospital work five consecutive days and have two consecutive days off. Their five days of work can start on any day of the week and their schedule rotates indefinitely. The hospital requires the following minimum number of doctors to work on the given days:

Sun	Mon	Tues	Wed	Thus	Fri	Sat
35	55	60	50	60	50	45

No more than 40 doctors can start their five working days on the same day. Formulate this problem as an LP model to minimize the number of doctors employed by the hospital.

[*Delhi Univ., MBA (HCA), 2006*]

LP model formulation Let x_j = number of doctors who start their duty on day j ($j = 1, 2, \ldots, 7$) of the week.

The LP model

Minimize (total number of doctors) $Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$

subject to the constraints

(i) $x_1 + x_4 + x_5 + x_6 + x_7 \ge 35$, (ii) $x_2 + x_5 + x_6 + x_7 + x_1 \ge 55$

(iii) $x_3 + x_6 + x_7 + x_1 + x_2 \ge 60$, (iv) $x_4 + x_7 + x_1 + x_2 + x_3 \ge 50$

(v) $x_5 + x_1 + x_2 + x_3 + x_4 \ge 60$, (vi) $x_6 + x_2 + x_3 + x_4 + x_5 \ge 50$

(vii) $x_7 + x_3 + x_4 + x_5 + x_6 \ge 45$, (viii) $x_j \le 40$

and $x_j \ge 0$ for all j.

Example 2.33 A machine tool company conducts on-the-job training programme for machinists. Trained machinists are used as teachers for the programme, in the ratio of one for every ten trainees. The training programme lasts for one month. From past experience it has been found that out of the ten trainees hired, only seven complete the programme successfully and the rest are released.

Trained machinists are also needed for machining. The company's requirement for machining for the next three months is as follows: January 100, February 150 and March 200. In addition, the company requires 250 machinists by April. There are 130 trained machinists available at the beginning of the year. Pays per month are:

Each trainee : Rs 4,400
Each trained machinist (machining and teaching) : Rs 4,900
Each trained machinist idle : Rs 4,700

Formulate this problem as an LP model to minimize the cost of hiring and training schedule and the company's requirements.

LP model formulation Let

x_1, x_2 = trained machinist teaching and idle in January, respectively
x_3, x_4 = trained machinist teaching and idle in February, respectively
x_5, x_6 = trained machinist teaching and idle in March, respectively

The LP model

Minimize (total cost) Z = Cost of training programme (teachers and trainees) + Cost of idle machinists + Cost of machinists doing machine work (constant)

$$= 4{,}400\,(10x_1 + 10x_3 + 10x_5) + 4{,}900\,(x_1 + x_3 + x_5) + 4{,}700\,(x_2 + x_4 + x_6)$$

subject to the constraints

(i) Total trained machinists available at the beginning of January
= Number of machinists doing machining + Teaching + Idle

$$130 = 100 + x_1 + x_2 \quad \text{or} \quad x_1 + x_2 = 30$$

(ii) Total trained machinists available at the beginning of February
= Number of machinists in January + Joining after training programme

$$130 + 7x_1 = 150 + x_3 + x_4 \quad \text{or} \quad 7x_1 - x_3 - x_4 = 20$$

In January there are $10x_1$ trainees in the programme and out of those only $7x_1$ will become trained machinists.

(iii) Total trained machinists available at the beginning of March
= Number of machinists in January + Joining after training programme in January and February

$$130 + 7x_1 + 7x_3 = 200 + x_5 + x_6$$
$$7x_1 + 7x_3 - x_5 - x_6 = 70$$

(iv) Company requires 250 trained machinists by April

$$130 + 7x_1 + 7x_3 + 7x_5 = 250$$
$$7x_1 + 7x_3 + 7x_5 = 120$$

and $x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$.

Example 2.34 The super bazzar in a city daily needs anything between 22 to 30 workers in the bazzar depending on the time of day. The rush hours are between noon and 2 pm. The table indicates the number of workers needed at various hours when the bazzar is open.

The super bazzar now employs 24 full-time workers, but also needs a few part-time workers. A part-time worker must put in exactly 4 hours per day, but can start any time between 9 am and 1 pm. Full- time workers work from 9 am to 5 pm but are allowed an hour for lunch (half of the full-timers eat at 12 noon, the other half at 1 am). Full-timers thus provide 35 hours per week of productive labour time.

The management of the super bazzar limits part-time hours to a maximum of 50 per cent of the day's total requirement.

Time Period	*Number of Workers Needed*
9 AM – 11 AM	22
11 AM – 1 PM	30
1 PM – 3 PM	25
3 PM – 5 PM	23

Part-timers earn Rs 28 per day on the average, while full-timers earn Rs 90 per day in salary and benefits on the average. The management wants to set a schedule that would minimize total manpower costs.

Formulate this problem as an LP model to minimize total daily manpower cost.

LP model formulation Let

y = full-time workers
x_j = part-time workers starting at 9 am, 11 am and 1 pm, respectively (j = 1, 2, 3)

The LP model

Minimize (total daily manpower cost) $Z = 90y + 28(x_1 + x_2 + x_3)$
subject to the constraints

(i) $y + x_1 \geq 22$ [9 am – 11 am], (ii) $\frac{1}{2}y + x_1 + x_2 \geq 30$ [11 am – 1 pm],

(iii) $\frac{1}{2}y + x_2 + x_3 \geq 25$ [1 pm – 3 pm], (iv) $y + x_3 \geq 23$ [3 pm – 5 pm],

(v) $y \leq 24$ [Full-timers available], (iv) $4(x_1 + x_2 + x_3) \leq 0.50(22 + 30 + 25 + 23)$
[Part-timers' hours cannot exceed 50% of total hours required each day which is the sum of the workers needed each hour]

and $y, x_j \geq 0$ for all j.

Example 2.35 The security and traffic force, on the eve of Republic Day, must satisfy the staffing requirements as shown in the table. Officers work 8-hour shifts starting at each of the 4-hour intervals as shown below. How many officers should report for duty at the beginning of each time period in order to minimize the total number of officers needed to satisfy the requirements?

Formulate this problem as an LP model so as to determine the minimum number of officers required on duty at beginning of each time period.

Time	*Number of Officers Required*
0:01 – 4:00	5
4:01 – 8:00	7
8:01 – 12:00	15
12:01 – 16:00	7
16:01 – 20:00	12
20:01 – 24:00	9

LP model formulation x_i = number of officers who start in shift i (i = 1, 2, 3, . . ., 6)

The LP model

Minimize (number of officers required on duty) $Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$
subject to the constraints

(i) $x_1 + x_2 \geq 7$, (ii) $x_2 + x_3 \geq 15$, (iii) $x_3 + x_4 \geq 7$,

(iv) $x_4 + x_5 \geq 12$, (v) $x_5 + x_6 \geq 9$, (vi) $x_6 + x_1 \geq 5$

and $x_j \geq 0$, for all j.

CONCEPTUAL QUESTIONS

1. (a) What is linear programming? What are its major assumptions and limitations? [*Delhi Univ., MBA, Nov. 2005*]
   (b) Two of the major limitations of linear programming are: assumption of 'additivity' and 'single objective'. Elaborate by giving appropriate examples. [*Delhi Univ., MBA, Nov. 2009*]
2. Linear programming has no real-life applications'. Do you agree with this statement? Discuss. [*Delhi Univ., MBA, 2004*]
3. In relation to the LP problem, explain the implications of the following assumptions of the model:
   (i) Linearity of the objective function and constraints,
   (ii) Continuous variables,
   (iii) Certainty.
4. What is meant by a feasible solution of an LP problem?
5. 'Linear programming is one of the most frequently and successfully applied operations research technique to managerial decisions.' Elucidate this statement with some examples. [*Delhi Univ., MBA, 2008*]
6. (a) What are the advantages and limitations of LP models?
   (b) Discuss and describe the role of linear programming in managerial decision-making, bringing out limitations, if any. [*Delhi Univ., MBA, 2003*]
7. Regardless of the way one defines linear programming, certain basic requirements are necessary before this technique can be employed to business problems. What are these basic requirements in formulation? Explain briefly.
8. Discuss in brief linear programming as a technique for resource utilization. [*Delhi Univ., MBA* (*HCA*)*, 2004*]
9. What are the four major types of allocation problems that can be solved using the linear programming technique? Briefly explain each with an example.
10. Give the mathematical and economic structure of linear programming problems. What requirements should be met in order to apply linear programming?
11. Discuss and describe the role of linear programming in managerial decision-making bringing out limitations, if any. [*Delhi Univ., MBA, 2004, 2009*]

SELF PRACTICE PROBLEMS

Problems on Production

1. A company sells two different products A and B, making a profit of Rs 40 and Rs 30 per unit, respectively. They are both produced with the help of a common production process and are sold in two different markets. The production process has a total capacity of 30,000 man-hours. It takes three hours to produce a unit of A and one hour to produce a unit of B. The market has been surveyed and company officials feel that the maximum number of units of A that can be sold is 8,000 units and that of B is 12,000 units. Subject to these limitations, products can be sold in any combination. Formulate this problem as an LP model to maximize profit.
2. The manager of an oil refinery must decide on the optimal mix of two possible blending processes of which the input and output per production run are given as follows:

Process (*units*)	*Input* (*units*)		*Output*	
	Crude A	*Crude B*	*Gasoline X*	*Gasoline Y*
1	5	3	5	8
2	4	5	4	4

The maximum available amount of crude A and B are 200 units and 150 units, respectively. Market requirements show that at least 100 units of gasoline X and 80 units of gasoline Y must be produced. The profit per production run from process 1 and process 2 are Rs 300 and Rs 400, respectively. Formulate this problem as an LP model to maximize profit.

3. A firm places an order for a particular product at the beginning of each month and that product is received at the end of the month. The firm sells during the month from the stocks and it can sell any quantity.

The prices at which the firm buys and sells vary every month. The following table shows the projected buying and selling prices for the next four months:

Month	*Selling Price* (*Rs*) (*During the Month*)	*Purchase Price* (*Rs*) (*Beginning of the Month*)
April	–	75
May	90	75
June	60	60
July	75	–

As on April 1 the firm has no stocks on hand, and does not wish to have any stocks at the end of July. The firm has a warehouse of limited size, which can hold a maximum of 150 units of the product.

Formulate this problem as an LP model to determine the number of units to buy and sell each month so as to maximize the profits from its operations. [*Delhi Univ., MBA, 2007*]

4. A manufacturer produces three models (I, II and III) of a certain product. He uses two types of raw materials (A and B) of which 4,000 and 6,000 units, respectively, are available. The raw material requirements per unit of the three models are as follows:

The labour time of each unit of model I is twice that of model II and three times that of model III. The entire labour force of the factory can produce equivalent of 2,500 units of model I. A market survey indicates that the minimum demand of the three models is: 500, 500 and 375 units, respectively. However, the ratios of the number of units produced must be equal to 3 : 2 : 5. Assume that the profit per unit of models I, II and III are Rs 60, 40 and 100, respectively. Formulate this problem as an LP model to determine the number of units of each product which will maximize profit.

5. Consider a small plant which makes two types of automobile parts, say A and B. It buys castings that are machined, bored and polished. The capacity of machining is 25 per hour for A and 24 per hour for B, capacity of boring is 28 per hour for A and 35 per hour for B, and the capacity of polishing is 35 per hour for A and 25 per hour for B. Castings for part A cost Rs 2 and sell for Rs 5 each and those for part B cost Rs 3 and sell for Rs 6 each. The three machines have running costs of Rs 20, Rs 14 and Rs 17.50 per hour. Assuming that any combination of parts A and B can be sold, formulate this problem as an LP model to determine the product mix which would maximizes profit.
6. On October 1, a company received a contract to supply 6,000 units of a specialized product. The terms of contract require that 1,000 units of the product be shipped in October; 3,000 units in November and 2,000 units in December. The company can manufacture 1,500 units per month on regular time and 750 units per month in overtime. The manufacturing cost per item produced during regular time is Rs 3 and the cost per item produced during overtime is Rs 5. The monthly storage cost is Re 1. Formulate this problem as an LP model so as to minimize total costs. [*Delhi Univ., MBA, 1999*]
7. A wine maker has a stock of three different wines with the following characteristics:

Wine	*Proofs*	*Acids%*	*Specific Gravity*	*Stock* (*gallons*)
A	27	0.32	1.70	20
B	33	0.20	1.08	34
C	32	0.30	1.04	22

A good dry table wine should be between 30 and 31 degree proof, it should contain at least 0.25% acid and should have a specific gravity of at least 1.06. The wine maker wishes to blend the three types of wine to produce as large a quantity as possible of a satisfactory dry table wine. However, his stock of wine A must be completely used in the blend because further storage would cause it to deteriorate. What quantities of wines B and C should be used in the blend. Formulate this problem as an LP model.

8. ABC foods company is developing a low-calorie high-protein diet supplement called Hi-Pro. The specifications of Hi-Pro have been established by a panel of medical experts. These specifications along with the calorie, protein and vitamin content of three basic foods, are given in the following table:

Nutritional Elements	*Units of Nutritional Elements* (*Per 100 gm Serving of Basic Foods*)			*Basic Foods Hi-Pro Specifications*
	1	*2*	*3*	
Calories	350	250	200	300
Proteins	250	300	150	200
Vitamin A	100	150	75	100
Vitamin C	75	125	150	100
Cost per serving (Rs)	1.50	2.00	1.20	

What quantities of foods 1, 2, and 3 should be used? Formulate this problem as an LP model to minimize cost of serving.

[*Delhi Univ., MBA* (*HCA*)*, 2009*]

9. Omega leather goods company manufactures two types of leather soccer balls X and Y. Each type of ball requires work by two types of employees – semi-skilled and skilled. Basically, the semi-skilled employees use machines, while the skilled employees stitch the balls. The available time (per week) for manufacturing each type of employee and the time requirement for each type of ball are given below:

Type of Employee	Manufacturing Time Requirement (hr)		Time Available (hr/week)
	Ball X	Ball Y	
Semi-skilled	2	3	80
Skilled	4	6	150

The cost of an hour of semi-skilled labour is Rs 5.50 and that of an hour of skilled labour is Rs 8.50. To meet the weekly demand requirements, at least 15 balls of type X and at least 10 balls of type Y must be manufactured. Formulate this problem as an LP model so as to minimize cost of production.

10. A pharmaceutical company has developed a new pill to be taken by smokers that will nauseate them if they smoke. This new pill is a combination of four ingredients that are costly and in limited supply. The available supply and costs are as follows:

Ingredient	Supply Availability (kg)	Cost (Rs/kg)
1	22	28
2	18	25
3	20	52
4	24	26

Blending requirements for this new pill are as follows:

(i) Ingredient 1 must be at least 45 per cent of the total quantity, but cannot exceed 60 per cent of the total.
(ii) Ingredients 2 and 3 must each comprise at least 10 per cent of the mixture, but their combined percentage cannot exceed 25 per cent of the total quantity.
(iii) Ingredient 4 must not be more than 50 per cent of the total quantity. Additionally, at least 25 kg of the pill must be produced.

Formulate this problem as an LP model to determine optimum blending of ingredients.

[*Delhi Univ., MBA* (*HCA*)*, 2008*]

11. A paint manufacturing company manufactures paints at two of its plants. Firm orders have been received from three large contractors. The firm has determined that the following shipping cost data is appropriate for these contractors with respect to its two plants:

Contractor	Order Size (gallon)	Shipping Cost/Gallon (Rs)	
		From Plant 1	From Plant 2
A	750	1.80	2.00
B	1,500	2.60	2.20
C	1,500	2.10	2.25

Each gallon of paint must be blended and tinted. The company's costs with respect to these two operations at both of the plants are as follows:

Plant/Operation		Hours Required per Gallon	Cost/hour (Rs)	Hours Available
Blending	Plant 1	0.10	3.80	300
Tinting		0.25	3.20	360
Blending	Plant 2	0.15	4.00	600
Tinting		0.20	3.10	720

Formulate this problem as an LP model.

12. Vitamins A and B are found in foods F_1 and F_2. One unit of food F_1 contains three units of Vitamin A and four units of Vitamin B. One unit of food F_2 contains six units of Vitamin A and three units of Vitamin B, One unit of food F_1 and F_2 cost Rs 4 and 5, respectively. The minimum daily requirement (for a person) of Vitamins A and B is 80 and 100 units, respectively. Assuming that anything in excess of the daily minimum requirement of A and B is not harmful. Formulate this problem as an LP model to find out the optimum mixture of food F_1 and F_2 at the minimum cost which meets the daily minimum requirement of Vitamins A and B. [*Delhi Univ., MBA* (*HCA*)*, 2009*]

13. The Omega Data Processing Company performs three types of activities: payrolls, account receivables, and inventories. The profit and time requirements for keypunch, computation and office printing for a 'standard job' are shown in the following table:

Job	Profit/Standard Job (Rs)	Time Requirement (Min.)		
		Keypunch	Computation	Print
Payroll	275	1,200	20	100
A/c Receivable	125	1,400	15	60
Inventory	225	800	35	80

Omega guarantees overnight completion of all standard jobs. Any job scheduled during the day can be completed during the day or night. Any job scheduled during the night, however, must be completed during the night. The capacity for both day and night are shown in the following table:

Capacity (Min.)	Keypunch	Computation	Print
Day	4,200	150	400
Night	9,200	250	650

Formulate this problem as an LP model to determine the 'mix' of standard jobs that should be accepted during the day and night. [*Delhi Univ., MBA, 2008*]

14. PQR coffee company mixes South Indian, Assamese and imported coffee for making two brands of coffee: plains X and plains XX. The characteristics used in blending the coffee include strength, acidity and caffeine. The test results of the available supplies of South Indian, Assamese and imported coffee are shown in the following table:

Price per kg (Rs)	Strength Index	Acidity Index	Per cent Caffeine	Supply Available (kg)
South Indian, 30	6	4.0	2.0	40,000
Assamese, 40	8	3.0	2.5	20,000
Imported, 35	5	3.5	1.5	15,000

The requirements for plains X and plains XX coffees are given in the following table:

Plains Coffee	Price per kg (Rs)	Minimum Strength Caffeine	Maximum Acidity (kg)	Maximum Per Cent	Quantity Demanded
X	45	6.5	3.8	2.2	35,000
XX	55	6.0	3.5	2.0	25,000

Assume that 35,000 kg of plains X and 25,000 kg of plains XX, are to be sold. Formulate this problem as an LP model to maximize sales.

15. A manufacturer of metal office equipments makes desks, chairs, cabinets and book cases. The work is carried out in the three major manufacturing departments: Metal stamping, Assembly and Finishing. Exhibits, A, B the and C give the requisite data of the problem.

Exhibit A

Department	Time Required per Unit of Product (hrs) Products				Available Time per Week (hrs)
	Desk	Chair	Cabinet	Bookcase	
Stamping	4	2	3	3	800
Assembly	10	6	8	7	1,200
Finishing	10	8	8	8	800

Exhibit B

Department	Cost (Rs) of Operation per Unit of Product			
	Desk	Chair	Cabinet	Bookcase
Stamping	15	8	12	12
Assembly	30	18	24	21
Finishing	35	28	25	21

Exhibit C: Selling price (Rs) per unit of product

Desk	:	175	Chair	:	95
Cabinet	:	145	Bookcase	:	130

In order to maximize weekly profits, what should be all production programme? Assuming that the items produced can be sold, which department needs to be expanded for increasing profits? Formulate this problem as an LP model.

16. The PQR stone company sells stone procured from any of three adjacent quarries. The stone sold by the company conforms to the following specification:

    Material X equal to 30%
    Material Y equal to or less than 40%
    Material *Z* between 30% and 40%

    Stone from quarry A costs Rs 10 per tonne and has the following properties:

    Material X : 20%, Material Y : 60%, Material *Z* : 20%

    Stone from quarry B costs Rs 12 per tonne and has the following properties:

    Material X : 40%, Material Y : 30%, Material Z : 30%

    Stone from quarry C costs Rs 15 per tonne and has the following properties:

    Material X : 10%, Material Y : 40%, Material Z : 50%

    From what quarries should the PQR stone company procure rocks in order to minimize cost per tonne of rock? Formulate this problem as an LP model. [*Delhi Univ., MBA, 2006*]

17. A manufacturer produces three models (I, II and III) of a certain product. He uses two types of raw material: A and B of which 4,000 and 6,000 units are available, respectively. The raw material requirements per unit of the three models are given below:

Raw Material	Requirements per Unit of Given Model		
	I	*II*	*III*
A	2	3	5
B	4	2	7

The labour time for each unit of model I is twice that of model II and three times that of model III. The entire labour force of the factory can produce the equivalent of 2,500 units of model I. A market survey indicates that the minimum demand for the three models is 500, 500 and 375 units, respectively. However, the ratios of the number of units produced must be equal to 3 : 2 : 5. Assume that the profit per unit of models I, II and III is Rs 60, Rs 40 and Rs 100, respectively. Formulate this problem as an LP model to determine the number of units of each product that will the maximize amount of profit.

18. A company manufactures two models of garden rollers: X and Y. When preparing the 2008 budget, it was found that the limitations on capacity were represented by the following weekly production maxima:

Model	Foundry	Machine-shop	Contribution per Model (Rs)
X	100	200	120
Y	240	150	90

In addition, the material required for model X was in short supply and sufficient only for 140 units per week, guaranteed for the year. Formulate this problem as an LP model to determine the optimal combination of output.

19. A company manufacturing television and radio sets has four major departments: chassis, cabinet, assembly and final testing. The monthly capacities of these are as follows:

	Television		Radio
Chassis	1,500	or	4,500
Cabinet	1,000	or	8,000
Assembly	2,000	or	4,000
Testing	3,000	or	9,000

The contribution of a television set is Rs 500 and that of a radio set Rs 250. Assume that the company can sell any quantity of either product. Formulate this problem as an LP model to determine the optimal combination of television and radio sets.

20. A company wants to plan production for the ensuing year so as to minimize the combined cost of production and inventory storage. In each quarter of the year, demand is anticipated to be 65, 80, 135 and 75 respectively. The product can be manufactured during regular time at a cost of Rs 16 per unit produced, or during overtime at a cost of Rs 20 per unit. The table given below gives data pertinent to production capacities. The cost of carrying one unit in inventory per quarter is Rs 2. The inventory level at the beginning of the first quarter is zero.

Quarter	Capacities (units)		Quarterly Demand
	Regular Time	Overtime	
1	80	10	65
2	90	10	80
3	95	20	135
4	70	10	75

Formulate this problem as an LP model so as to minimize the production plus storage costs for the entire year. [*Delhi Univ., MBA, 2005*]

Problems on Marketing

21. Suppose a media specialist has to decide how to allocate advertising in three media vehicles. Let x_i be the number of messages carried in the media, i = 1, 2, 3. The unit costs of a message in the three media are Rs 1,000, Rs 750 and Rs 500. The total budget available for the campaign is Rs 2,00,000 period of a year. The first media is a monthly magazine and it is desired to advertise not more than one insertion in one issue. At least six messages should appear in the second media. The number of messages in the third media should strictly lie between 4 and 8. The expected effective audience for unit message in the media vehicles is shown below:

Vehicle	Expected Effective Audience
1	80,000
2	60,000
3	45,000

Formulate this problem as an LP model to determine the optimum allocation that would maximize total effective audience.

22. An advertising company is planning an advertisement campaign for a new product recently introduced in the market. It is decided to insert advertisements in three leading magazines. The

company has made a careful analysis of three available media, and has compiled the following set of relevant data:

Data Item	*Magazines*		
	1	*2*	*3*
Reader characteristics			
(a) Age: 25–35 yrs	70%	80%	40%
(b) Education level: Graduation and above	80%	60%	50%
(c) Income: Rs 5,000 and above	60%	70%	40%
Minimum number of advertisements	20	10	5
Maximum number of advertisements	50	40	30
Cost per advertisement (Rs)	1,000	700	500
Readership	3,20,000	5,00,000	2,00,000

Additionally, the company has specified that the relative importance of the reader characteristics should be weighted as follows:

Reader Characteristics	*Weightage*
Age: 25–35 yrs	0.4
Graduate and above	0.4
Income ≥ Rs 5,000	0.2

At this point in time the company has Rs 5,00,000 to spend. Formulate this problem as an LP model to maximize the effective exposure level.

23. The owner of Metro Sports wishes to determine the number of advertisements to be placed in the selected three monthly magazines A, B and C. His objective is to advertise in such a way that total exposure to principal buyers of expensive sports goods is maximized. The percentage of readers for each magazine is known. Exposure to any particular magazine is the number of advertisements placed multiplied by the number of principal buyers. The following data may be used:

	Magazines		
	A	*B*	*C*
Readers	1 lakh	0.6 lakh	0.4 lakh
Principal buyers	10 %	15 %	7 %
Cost per advertisement (Rs)	5,000	4,500	4,250

The budget amount is at the most Rs 2,00,000 for the advertisements. The owner has already decided that magazine A should have no more than six advertisements and that B and C each should have at least two advertisements. Formulate this problem as an LP model to determine the number of advertisements that should be placed in each magazine.

24. The XYZ company is preparing a proposal for an advertising campaign for a client who is a publisher of law books. An optimal allocation of advertising funds to maximize the total number of exposures has to be made for the client. The relevant characteristics of the three alternative publications are shown in the following table:

	Home Beautiful (*Rs*)	*Home and Garden* (*Rs*)	*Care* (*Rs*)
Cost/advertisement	600	800	450
Max. number of ads	12	24	12
Min. number of ads	3	6	2
Characteristics			
Homeowner	80%	70%	20%
Income: Rs 10,000 or more	70%	80%	60%
Occupation: gardener	15%	20%	40%
Audience size	6,00,000	8,00,000	3,00,000

The relative importance of the three characteristics is: Homeowner, 0.4; Income, 0.2; Gardener, 0.4. The advertising budget is Rs 2,00,000. Formulate this problem as an LP model to find the most effective number of exposures in each magazine.

Problems on Finance

25. A gambler plays a game that requires dividing bet money among four different choices. The game has three outcomes. The following table gives the corresponding gain (or loss) per rupee deposited in each of the four choices for the three outcomes:

Outcome	*Gain (or loss) per Rupee*		*Deposited in Given Choice*	
	1	*2*	*3*	*4*
1	– 3	4	– 7	15
2	5	– 3	9	4
3	3	– 9	10	– 8

Assume that the gambler has a total of Rs 500 with which he may play only once. The exact outcome of the game is not known in advance and in the face of this uncertainty the gambler decides to make the allocation that would maximize the minimum return. Formulate this problem as an LP model.

26. An investor has money-making activities A_1, A_2, A_3 and A_4. He has only one lakh rupees to invest. In order to avoid excessive investment, no more than 50 per cent of the total investment can be placed in activity A_2 and/or activity A_3. Activity A_1 is very conservative, while activity A_4 is speculative. To avoid excessive speculation, at least Re 1 must be invested in activity A_1 for every Rs 3 invested in activity A_4. The data on the return on investment is as follows:

Activity	*Anticipated Return on Investment* (*%*)
A_1	10
A_2	12
A_3	14
A_4	16

The investor wishes to know how much to invest in each in order activity to maximize the total return on the investment. Formulate this problem as an LP model.

27. The board of directors of a company has given approval for the construction of a new plant. The plant will require an investment of Rs 50 lakh. The required funds will come from the sale of a proposed bond issue and by taking loans from two financial corporations. For the company, it will not be possible to sell more than Rs 20 lakh worth of bonds at the proposed rate of 12%. Financial corporation A will give loan up to Rs 30 lakh at an interest rate of 16% but insists that the amount of bond debt plus the amount owned to financial corporation B be no more than twice the amount owed to financial corporation A. Financial corporation B will loan the same amount as that loaned by financial corporation A but it would do so at an interest rate of 18%. Formulate this problem as an LP model to determine the amount of funds to be obtained from each source in a manner that minimizes the total annual interest charges.

28. An investor wishes to diversify his portfolio and make due allowance for long-term potentialities, but at the same time wishes to maximize his current dividend income. He has considered various securities in which he might invest, and has classified them into four types:

Type A : Relatively high element of risk, with commensurately high dividend and considerable growth potential.

Type B : Speculative stock with considerable risk, high dividends, but less growth potential than type A.

Type C : Stock with little risk, considerable growth potential, but relatively low dividend income at present.

Type D : Stock with little risk, not much growth potential, and fairly high dividends.

Because of the element of risk, the investor wishes to restrict purchases of types A and B to not more than 30% of his investment.

To enhance prospects for long-term growth of his investments, he wishes to have at least 40% of his total outlay in types A and C. Within these restrictions, he wishes to maximize his current dividend income. Total investment is Rs 1,00,000. Dividend returns on the four types of investments are A: 6%, B: 7%, C: 3%, D: 5%. Formulate this problem as an LP model to suggest the total investment to be allocated.

29. The Agro Promotion Bank is trying to select an investment portfolio for a cotton farmer. The bank has chosen a set of five investment alternatives, with subjective estimates of rates of return and risk as follows:

Investment	*Annual Rate of Return*	*Risk*
Tax-free municipal bonds	6.0	1.3
Corporate bonds	8.0	1.5
High grade common stock	5.0	1.9
Mutual fund	7.0	1.7
Real estate	15.0	2.7

The bank officer incharge of the portfolio would like to maximize the average annual rate of return on the portfolio. However, the wealthy investor has specified that the average risk of the portfolio should not exceed 2.0. The investor and does not want more than 20% of the investment to be put into real estate. Formulate this problem as an LP model.

30. Raj, a retired government officer, has recently received his retirement benefits, viz., provident fund, gratuity, etc. He is contemplating how much money he should invest in various alternatives open to him so as to maximize return on his investment. The investment alternatives are: government securities, fixed deposits of a public limited company, equity shares, time deposits in a bank, and house construction. He has made a subjective estimate of the risk involved on a five-point scale. The data on the return on investment, the number of years for which the funds will be blocked to earn this return on investment and the subjective risk involved are as follows:

	Return (%)	*Number of Years*	*Risk*
Government securities	6	15	1
Company deposits	13	3	3
Time deposits	10	5	2
Equity share	20	6	5
House construction	25	10	1

He is wondering as to what percentage of funds he should invest in each alternative so as to maximize the return on investment. He has decided that the risk should not be more than 4, and funds should not be locked up for more than 15 years. He would necessarily invest at least 25% in house construction. Formulate this problem as an LP model.

31. A dealer of used scooters wishes to stock up his lot to maximize his profit. He can select scooters A, B and C which are valued on wholesale at Rs 5,000, Rs 7,000 and Rs 8,500 respectively. These can be sold at Rs 6,000, Rs 8,500 and Rs 10,500, respectively. For each type of scooter, the probabilities of sale are:

Type of scooter	:	A	B	C
Prob. of sale in 90 days	:	0.7	0.8	0.6

For every two scooters of B-type he should buy one scooter of type A or type C. If he has Rs 1,00,000 to invest, what should he buy in order to maximize his expected gain. Formulate this problem as an LP model.

32. A transport company is considering the purchase of new vehicles for transportation between Delhi airport and hotels in the city. There are three vehicles under consideration – station wagons, mini buses and large buses. The purchase price would be Rs 2,45,000 for each station wagon, Rs 3,50,000 for a mini bus and Rs 5,00,000 for a large bus. The board of directors has authorized a maximum amount of Rs 50,00,000 for these purchases. Because of the heavy air travel involved, the new vehicles would be utilized at maximum capacity, regardless of the type of vehicles purchased. The expected net annual profit would be Rs 15,000 for the station wagon, Rs 35,000 for the mini bus, and Rs 45,000 for the large bus. The company has hired 30 new drivers for the new vehicles. They are qualified drivers for all the three types of vehicles. The maintenance department has the capacity to handle an additional 80 station wagons. A mini bus is equivalent to 5/3 station wagons and each large bus is equivalent to two station wagons in terms of their use of the maintenance department. Formulate this problem as an LP model to determine optimal number of each type of vehicle to be purchased in order to maximize profit.

[*Delhi Univ., MBA, Oct. 2000*]

33. The managers of several cattle feed lots are interested in determining how many of each of several types of livestock feeds should be purchased in order to satisfy the nutritional requirements for their livestock. They wish to purchase such food in a manner that minimizes the cost of feeding their livestock. Relevant costs and nutritional data are as below:

Required Nutrient	*Units of Nutritional Element*				*Minimum Nutrient Requirements*
	Alfa	*Corn*	*Soyabean*	*Sorghum*	
A 40	50	30	60	500	
B 30	60	35	40	750	
C 25	30	25	50	600	
Cost per unit (Rs)	1.00	1.25	0.95	1.35	

Formulate this problem as an LP model.

34. Old hens can be bought at Rs 100 each and young ones at Rs 250 each. The old hens lay 3 eggs per week and the young ones 5 eggs per week, each egg being worth 50 paise. A hen costs Rs 20 per week to be fed. There are only Rs 8,000 available to be spent on purchasing the hens and at the most 20 hens can be accommodated in the space. Formulate this problem as an LP model to determine each kind of hen that should be bought in order to yield the maximum profit per week.

35. A pension fund manager is considering investing in two shares A and B. It is estimated that:

(i) Share A will earn a dividend of 12 per cent per annum and share B, 4 per cent per annum.

(ii) Growth in the market value in one year of share A will be 10 paise per Re l invested and in B, 40 paise per Re 1 invested.

He requires to invest the maximum total sum which will give:

(i) dividend income of at least Rs 600 per annum; and

(ii) growth in one year of at least Rs 1,000 on the initial investment.

Formulate this problem as an LP model to compute the minimum sum in order to be invested to meet the manager's objective.

36. A scrap metal dealer has received an order from a customer for at least 2,000 kg of scrap metal. The customer requires that at least 1,000 kg of the shipment of the metal be high quality copper that can be melted down and further used to produce copper tubings. Furthermore, the customer will not accept delivery of the order if it contains more than 175 kg of metal that he deems unfit for commercial use, i.e. metal that contains an excessive amount of impurity and cannot be melted down and defined profitably.

The dealer can purchase scrap metal from two different suppliers in unlimited quantities with the following percentage (in terms of weight) of high quality copper and unfit scrap:

	Supplier A	*Supplier B*
Copper	25%	75%
Unfit scrap	5%	10%

The cost per kg of metal purchased from supplier A and supplier B is Re 1 and Rs 4, respectively. Formulate this problem as an LP model so as to determine the optimal quantities of metal that the dealer should purchase from each of the two suppliers in order to minimize total the purchase cost.

[*Delhi Univ., MBA, 2008*]

37. A company needs 50 new machines. The machines have an economic life of two years and can be purchased for Rs 4,500 or be leased for Rs 2,800 per year. The purchased machines, at the end of two years, have no salvage value. Company has Rs 1,00,000 in uncommitted funds that can be used for the purchase or the lease of machines at the beginning of year 1. The company can obtain a loan of upto Rs 2,00,000 at 18 per cent interest per year. According to the terms of loans, the company has to repay the amount borrowed plus the interest at the end of each year. Each machine can earn Rs 3,000 per year. The earnings from the first year can be used to lease costs and the repayment of debt at the start of the second year. The company wants to minimize the total cost of using 50 machines over a two-year period. The objective is to minimize the costs of purchasing machines or leasing machines during the years 1 and 2, and to minimize the interest payments on funds borrowed to obtain the machines. Formulate this problem as a linear programming problem.

[*Delhi Univ., MBA, 2009*]

38. A trucking company with Rs 40,00,000 to spend on new equipment is contemplating three types of vehicles. Vehicle A has a 10 tonne payload and is expected to average 35 km per hour. It costs Rs 80,000. Vehicle B has a 20-tonne payload and is expected to average 30 km per hour. It costs Rs 1,30,000. Vehicle C is a modified form of vehicle B; it carries sleeping quarters for one driver and then reduces its capacity to 18 tonnes and raises the cost to Rs 1,50,000. Vehicle A requires a crew of one average man, and if driven on three shifts per day, could be run for an average of 18 hours per day. Vehicles B and C require a crew of two men each, while B would be driven 18 hours per day with three shifts, C however would average 21 hours per day. The company has 150 drivers available each day and would find it very difficult to obtain further crews. Maintenance facilities are such that the total number of vehicles must not exceed 30. How many vehicles of each type should be purchased if the company wishes to maximize its capacity in tonne-kms per day? Formulate this problem as an LP model. [*Delhi Univ., MBA, 2009*]

HINTS AND ANSWERS

1. Let x_1, x_2 = number of units of products A and B to be produced, respectively.

   Max $Z = 40x_1 + 30x_2$

   subject to $3x_1 + x_2 \le 3{,}000$ (Man-hours)

   $x_1 \le 8{,}000$; $x_2 \le 1{,}200$ (Marketing)

   and $x_1, x_2 \ge 0$.

2. Let x_1, x_2 = number of productive runs of process 1 and 2, respectively.

   Max $Z = 300x_1 + 400x_2$

   subject to $\left.\begin{aligned} 5x_1 + 4x_2 &\le 200 \\ 3x_1 + 5x_2 &\le 150 \end{aligned}\right\}$ (Max amount of crude A and B)

   $\left.\begin{aligned} 5x_1 + 4x_2 &\ge 100 \\ 8x_1 + 4x_2 &\ge 80 \end{aligned}\right\}$ (Market requirement of gasoline X and Y)

   and $x_1, x_2 \ge 0$.

3. Let x_i = number of units purchased per month (i = 1, 2, 3 – April, May, June)

   y_i = number of units sold per month (i = 1, 2, 3 – May, June, July)

   Max $Z = (90y_1 + 60y_2 + 75y_3) - (75x_1 + 75x_2 + 60x_3)$

   subject to $y_1 \le x_1 \le 150$

   $y_2 \le x_1 + x_2 - y_1 \le 150$

   $y_3 \le x_1 + x_2 + x_3 - y_1 - y_2 \le 150$

   $x_1 + x_2 + x_3 = y_1 + y_2 + y_3$

   and $x_i, y_i \ge 0$ for all i.

4. Let x_1, x_2 and x_3 = number of units of models I, II and III, respectively to be manufactured

   Max $Z = 60x_1 + 40x_2 + 100x_3$

   subject to $\left.\begin{aligned} 2x_1 + 3x_2 + 5x_3 &\le 4{,}000 \\ 4x_1 + 2x_2 + 7x_3 &\le 6{,}000 \end{aligned}\right\}$ (Raw material requirement)

   $x_1 + \frac{1}{2}x_2 + \frac{1}{3}x_3 \le 2{,}500$ (Production limitation)

   $x_1 \ge 500$; $x_2 \ge 500$; $x_3 \ge 375$ (Market demand)

   $\frac{1}{3}x_1 = \frac{1}{2}x_2$; $\frac{1}{2}x_2 = \frac{1}{5}x_3$ (Ratios of production)

5. Let x_1 and x_2 = parts of A and B per hour manufactured, respectively.

   $$\text{Max } Z = (5x_1 + 6x_2) - \left(\frac{20}{25} + \frac{14}{28} + \frac{17.50}{35} + 2.00\right)x_1 - \left(\frac{20}{24} + \frac{14}{35} + \frac{17.50}{25} + 3.00\right)x_2$$

   $$= 1.20x_1 + 1.40x_2$$

   subject to (i) $\frac{x_1}{25} + \frac{x_2}{24} \le 1$; (ii) $\frac{x_1}{28} + \frac{x_2}{35} \le 1$; (iii) $\frac{x_1}{35} + \frac{x_2}{25} \le 1$ (Manufacturing capacity)

   and $x_1, x_2 \ge 0$.

6. Let x_{ijk} = number of units manufactured in month i (1, 2, 3 – Oct., Nov., Dec.) during shift j (j = 1, 2 – regular, overtime) and shipped in month k (k = 1, 2, 3 – Oct., Nov., Dec.)

   Min $Z = 3x_{111} + 5x_{121} + 4x_{112} + 6x_{122} + 5x_{113} + 7x_{123} + 3x_{212} + 5x_{222} + 4x_{213} + 6x_{223} + 3x_{313} + 5x_{323}$

   subject to $\left.\begin{aligned} x_{111} + x_{112} + x_{113} &\le 1{,}500 \\ x_{212} + x_{213} &\le 1{,}500 \\ x_{313} &\le 1{,}500 \end{aligned}\right\}$ (Regular time)

$$\left.\begin{aligned} x_{121} + x_{122} + x_{123} &\le 750 \\ x_{222} + x_{223} &\le 750 \\ x_{323} &\le 750 \end{aligned}\right\} \text{(Overtime)}$$

$$x_{111} + x_{121} = 1{,}000$$
$$x_{121} + x_{122} + x_{212} + x_{222} = 3{,}000$$
$$x_{113} + x_{123} + x_{213} + x_{223} + x_{313} + x_{323} = 2{,}000$$

and $x_{ijk} \ge 0$ for all i, j, k.

7. Let x_1, x_2 = number of gallons of wine B and C in the blend, respectively.

Max $Z = 20 + x_1 + x_2$

subject to $30 \le \dfrac{(20 \times 27) + 33x_1 + 32x_2}{20 + x_1 + x_2} \le 31$ (Resultant degrees proof of blend)

$$\frac{(20 \times 0.32) + 0.2x_1 + 0.3x_2}{20 + x_1 + x_2} \ge 0.25 \quad \text{(Acidity)}$$

$$\frac{(20 \times 1.07) + 1.08x_1 + 1.04x_2}{20 + x_1 + x_2} \ge 1.06 \quad \text{(Specific gravity)}$$

$$x_1 \le 34 \quad \text{(Quality)}$$

and $x_1, x_2 \ge 0$.

8. Let x_1, x_2 and x_3 = quantity of foods 1, 2 and 3 to be used, respectively.

Min $Z = 1.50x_1 + 2.00x_1 + 1.20x_3$

subject to
$$x_1 + y_1 \le 12,$$
$$350x_1 + 250x_2 + 200x_3 \ge 300$$
$$250x_1 + 300x_2 + 150x_3 \ge 200$$
$$100x_1 + 150x_2 + 75x_3 \ge 100$$
$$75x_1 + 125x_2 + 150x_3 \ge 100$$

and $x_1, x_2, x_3 \ge 0$.

9. Let x_1 and x_2 = number of soccer balls of types X and Y, respectively.

Min Z = (2 hrs) (Rs 5.50/hr) x_1 + (4 hrs) (Rs 8.50/hr) x_1 + (3 hrs) (Rs 5.50/hr) x_2 + (6 hrs) (Rs 8.50/hr) $x_2 = 45x_1 + 67.50x_2$

subject to
$2x_1 + 3x_2 \le 80$ (Semi-skilled hours)
$4x_1 + 6x_2 \le 150$ (Skilled hours)
$x_1 \le 15$ (Ball X)
$x_2 \le 10$ (Ball Y)

and $x_1, x_2 \ge 0$.

10. Let x_j = number of kg of ingredient j (j = 1, 2, 3, 4) used in the mixture

Min $Z = 28x_1 + 25x_2 + 52x_3 + 26x_4$

subject to
$$\left.\begin{aligned} x_1 \le 22;\quad x_2 \le 18 \\ x_3 \le 20;\quad x_4 \le 24 \end{aligned}\right\} \text{(Supplies)}$$

$$\left.\begin{aligned} 0.55x_1 - 0.45x_2 - 0.45x_3 - 0.45x_4 \ge 0 \\ 0.40x_1 - 0.60x_2 - 0.60x_3 - 0.60x_4 \ge 0 \\ -0.10x_1 + 0.90x_2 + 0.90x_3 - 0.10x_4 \ge 0 \\ -0.25x_1 + 0.75x_2 + 0.75x_3 - 0.25x_4 \le 0 \\ -0.50x_1 - 0.50x_2 - 0.50x_3 + 0.50x_4 \le 0 \end{aligned}\right\}$$

(Mixing requirements)

$x_1 + x_2 + x_3 + x_4 \ge 25$ (Total requirement)

and $x_1, x_2, x_3, x_4 \ge 0$.

11. Let x_{ij} = gallons of paint manufactured at plant i (i = 1, 2) and shipped to contractor j (j = 1, 2, 3) Min (production cost + shipping cost)

$$Z = (x_{11} + x_{12} + x_{13})\ \{(0.10)\ (3.80) + (0.25)\ (3.20)\} + (x_{21} + x_{22} + x_{23})\ \{(0.15)\ (4.00) + (0.20)\ (3.10)\} + 1.80x_{11} + 2.00x_{21} + 2.60x_{12} + 2.20x_{22} + 2.10x_{13} + 2.25x_{23}$$
$$= 2.98x_{11} + 3.78x_{12} + 3.28x_{13} + 3.22x_{21} + 3.42x_{22} + 3.47x_{23}$$

subject to
$$\left.\begin{aligned} x_{11} + x_{21} &= 750 \\ x_{12} + x_{22} &= 1{,}500 \\ x_{13} + x_{23} &= 1{,}500 \end{aligned}\right\} \text{(Order size)}$$

$$\left.\begin{aligned} (x_{11} + x_{12} + x_{13})(0.10) &\le 300 \ \text{Plant 1} \\ (x_{21} + x_{22} + x_{23})(0.15) &\le 600 \ \text{Plant 2} \end{aligned}\right\} \text{(Blending)}$$

$$\left.\begin{aligned} (x_{11} + x_{12} + x_{13})(0.25) &\le 360 \ \text{Plant 1} \\ (x_{21} + x_{22} + x_{23})(0.20) &\le 720 \ \text{Plant 2} \end{aligned}\right\} \text{(Tinting)}$$

and $x_{ij} \ge 0$ for all i and j.

12. Let x_1 and x_2 = number of vitamin units purchased of food F_1 and F_2, respectively.

Min $Z = 4x_1 + 5x_2$

subject to (i) $3x_1 + 6x_2 \ge 80$; (ii) $4x_1 + 3x_2 \ge 100$

and $x_1, x_2 \ge 0$.

13. Let x_{ij} = number of jobs accepted during day and night

Max $Z = 275\ (x_{11} + x_{12}) + 125\ (x_{21} + x_{22}) + 225\ (x_{31} + x_{32})$

subject to
$$1{,}200\ (x_{11} + x_{12}) + 1{,}400\ (x_{21} + x_{22}) + 800\ (x_{31} + x_{32}) \le 13{,}400$$
$$100\ (x_{11} + x_{12}) + 60\ (x_{21} + x_{22}) + 80\ (x_{31} + x_{32}) \le 1{,}050$$
$$1{,}200x_{12} + 1{,}400x_{22} + 800x_{32} \le 9{,}200$$
$$100x_{12} + 60x_{22} + 80x_{32} \le 650$$

and $x_{ij} \ge 0$ for all i and j.

14. *For plain coffee X*

x_{11}, x_{12} and x_{13} = quantity (in kg) of the three coffees, respectively.

For plain coffee XX

x_{21}, x_{22} and x_{23} = quantity (in kg) of the three coffees, respectively.

Max $Z = 45\ (30x_{11} + 40x_{12} + 35x_{13}) + 55\ (30x_{21} + 40x_{22} + 35x_{23})$

subject
$$\left.\begin{aligned} 6x_{11} + 8x_{12} + 5x_{13} &\ge 6.5 \\ 4x_{11} + 3x_{12} + 3.5x_{13} &\le 3.8 \\ 2x_{11} + 2.5x_{12} + 1.5x_{13} &\le 2.2 \\ x_{11} + x_{12} + x_{13} &= 35{,}000 \end{aligned}\right\} \text{(Plain coffee } X\text{)}$$

$$\left.\begin{aligned} 6x_{21} + 8x_{22} + 5x_{23} &\ge 6.0 \\ 4x_{21} + 3x_{22} + 3.5x_{23} &\le 3.5 \\ 2x_{21} + 2.5x_{22} + 1.5x_{23} &\le 2.0 \\ x_{21} + x_{22} + x_{23} &= 25{,}000 \end{aligned}\right\} \text{(Plain coffee } XX\text{)}$$

$x_{11} + x_{21} \le 40{,}000;\ x_{12} + x_{22} \le 20{,}000;$

$x_{13} + x_{23} \le 15{,}000$

and $x_{ij} \ge 0$ for all i and j.

15. Let x_1, x_2, x_3 and x_4 = quantities of four products to be manufactured, respectively.

$$\text{Max } Z = [175 - (15 + 30 + 35)]\,x_1 + [95 - (8 + 18 + 28)]\,x_2 + [145 - (12 + 24 + 25)]\,x_3 + [130 - (12 + 21 + 21)]\,x_4$$
$$= 95x_1 + 41x_2 + 84x_3 + 76x_4$$

subject to
$$4x_1 + 2x_2 + 3x_3 + 3x_4 \le 800,$$
$$10x_1 + 6x_2 + 8x_3 + 7x_4 \le 1{,}200$$
$$10x_1 + 8x_2 + 8x_3 + 8x_4 \le 800$$

and $x_1, x_2, x_3, x_4 \ge 0$

16. The data of the problem can be summarized as given below:

	Quarry A	*Quarry B*	*Quarry C*	*Specifications*
Material *X*	20%	40%	10%	= 30%
Material *Y*	60%	30%	40%	≤ 40%
Material *Z*	20%	30%	50%	between 30% and 40%
Cost/tonne (Rs)	10	12	15	

Let x_1, x_2, and x_3 = number of tonnes procured from quarry A, B and C, respectively.

Min (total cost) $Z = 10x_1 + 12x_2 + 15x_3$

subject to
$$2x_1 + 4x_2 + x_3 = 3 \quad \text{(Material } X\text{)}$$
$$6x_1 + 3x_2 + 4x_3 \le 4 \quad \text{(Material } Y\text{)}$$
$$2x_1 + 3x_2 + 5x_3 \le 4,\quad 2x_1 + 3x_2 + 5x_3 \ge 3 \quad \text{(Material } Z\text{)}$$

17. Let x_1, x_2 and x_3 = number of units of types I, II and III model, respectively.

Max (total profit) $Z = 60x_1 + 40x_2 + 1{,}000x_3$

subject to
$$2x_1 + 3x_2 + 5x_3 \le 4{,}000 \quad \text{(Raw material } A\text{)}$$
$$4x_1 + 2x_2 + 7x_3 \le 6{,}000 \quad \text{(Raw material } B\text{)}$$
$$x_1 + \frac{x_2}{2} + \frac{x_3}{3} \le 2{,}500 \quad \text{(Labour force)}$$
$$\frac{x_1}{3} = \frac{x_2}{2};\ \frac{x_2}{2} = \frac{x_3}{5} \quad \text{(Number of units produced)}$$
$$x_1 \ge 500;\quad x_2 \ge 500;\quad x_3 \ge 375 \quad \text{(Market demand)}$$

and $x_1, x_2, x_3 \ge 0$.

18. Let x_1, and x_2 = number of units of models X and Y, respectively.

Max $Z = 120x_1 + 90x_2$

subject to
$$\frac{x_1}{100} + \frac{x_2}{240} \le 1;\quad \frac{x_1}{200} + \frac{x_2}{150} \le 1;\ x_1 \le 140$$

and $x_1, x_2 \ge 0$.

19. Let x_1 and x_2 = number of television and radio sets to be manufactured, respectively.

Max $Z = 500x_1 + 250x_2$

subject to
$$(1/1{,}500)\,x_1 + (1/4{,}500)\,x_2 \le 1,$$
$$(1/1{,}000)\,x_1 + (1/8{,}000)\,x_2 \le 1,$$
$$(1/2{,}000)\,x_1 + (1/4{,}000)\,x_2 \le 1,$$
$$(1/3{,}000)\,x_1 + (1/9{,}000)\,x_2 \le 1,$$

and $x_1, x_2 \ge 0$.

21. Let x_1, x_2 and x_3 = number of messages carried in media 1, 2 and 3, respectively.

Max $Z = 80{,}000x_1 + 60{,}000x_2 + 45{,}000x_3$

subject to
$$1{,}000x_1 + 750x_2 + 500x_3 \le 2{,}00{,}000$$
$$x_1 \le 12;\quad x_2 \ge 6,\quad 4 \le x_3 \le 8,$$

and $x_1, x_2 \ge 0$.

22. Let x_1, x_2 and x_3 = number of advertisements in magazine 1, 2 and 3, repectively.

$$\text{Max } Z = 3{,}20{,}000\,(0.7 \times 0.4 + 0.8 \times 0.4 + 0.6 \times 0.2)\,x_1 + 5{,}00{,}000\,(0.8 \times 0.4 + 0.6 \times 0.4 + 0.7 \times 0.2)\,x_2 + 2{,}00{,}000\,(0.4 \times 0.4 + 0.5 \times 0.4 + 0.4 \times 0.2)\,x_3$$
$$= 2{,}30{,}400x_1 + 35{,}000x_2 + 88{,}000x_3$$

subject to
$$1{,}000x_1 + 700x_2 + 500x_3 \le 5{,}00{,}000$$
$$20 \le x_1 \le 50;\quad 10 \le x_1 \le 40;\quad 5 \le x_3 \le 30$$

and $x_j \ge 0$ for all j.

23. Let x_1, x_2 and x_3 = number of insertions in magazines A, B and C, respectively.

Max (total exposure) $Z = (10\% \text{ of } 1{,}00{,}000)x_1 + (15\% \text{ of } 60{,}000)x_2 + (7\% \text{ of } 40{,}000)x_3$

subject to
$$5{,}000x_1 + 4{,}500x_2 + 4{,}250x_3 \le 1{,}00{,}000$$
$$x_1 \le 6;\quad x_2 \ge 2;\quad x_3 \ge 2$$

and $x_1, x_2, x_3 \ge 0$.

24. Let x_j = number of advertisements in media j (j = 1, 2, 3).

Media	Effectiveness coefficient
1	0.80 (0.4) + 0.70 (0.2) + 0.15 (0.4) = 0.52
2	0.70 (0.4) + 0.80 (0.2) + 0.20 (0.4) = 0.52
3	0.20 (0.4) + 0.60 (0.2) + 0.40 (0.4) = 0.36

$$\text{Max } Z = 0.52\,(6{,}00{,}000)x_1 + 0.52\,(8{,}00{,}000)x_2 + 0.36\,(3{,}00{,}000)x_3$$

subject to
$$600x_1 + 800x_2 + 450x_3 \le 2{,}00{,}000$$
$$x_1 \le 12;\quad x_2 \le 24;\quad x_3 \le 12$$
$$x_1 \ge 3;\quad x_2 \ge 6;\quad x_3 \ge 2$$

and $x_1, x_2, x_3 \ge 0$.

25. y = minimum expected gain per rupee deposited in the given choice j (j = 1, 2, 3, 4) by the gambler

x_j = amount of bet money used among four different choices, respectively (j = 1, 2, 3, 4)

Max $Z = y$

subject to
$$-3x_1 + 4x_2 - 7x_3 + 15x_4 \ge y$$
$$5x_1 - 3x_2 + 9x_3 + 4x_4 \ge y$$
$$3x_1 - 9x_2 + 10x_3 - 8x_4 \ge y$$
$$x_1 + x_2 + x_3 + x_4 \le 500$$

and x_j may be negative, zero or positive.

26. Let x_1, x_2, x_3 and x_4 = percentage of the total fund that should be invested in activities A_1, A_2, A_3 and A_4, respectively.

Max $Z = 0.10x_1 + 0.12x_2 + 0.14x_3 + 0.16x_4$

subject to $x_1 + x_2 + x_3 + x_4 \le 1{,}00{,}000$ (total money invested)

$x_2 + x_3 \le 0.50\ (x_1 + x_2 + x_3 + x_4)$

$x_1 \ge \frac{1}{3}x_4$

and $x_j \ge 0$ for all j.

27. Let x_1 = bond debt to be obtained.
x_2 and x_3 = loan to be obtained from financial corporations A and B, respectively.

Min $Z = 0.12x_1 + 0.16x_2 + 0.18x_3$

subject to $x_1 + x_2 + x_3 = 50$

$x_1 \le 20; \quad x_2 \le 30$

$x_1 + x_3 \le 2x_2; \quad x_1 \le x_2$

and $x_1, x_2 \ge 0.$

28. Let x_1, x_2, x_3 and x_4 = amount of money to be invested in A, B, C and D securities, respectively.

Max (current dividend return)

$Z = 0.06x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4$

subject to $x_1 + x_2 + x_3 + x_4 \le 1{,}00{,}000$

$x_1 + x_2 \le 30{,}000; \quad x_1 + x_3 \ge 40{,}000$

and $x_1, x_2, x_3, x_4 \ge 0.$

31. Let x_1, x_2, x_3, x_4 and x_5 = percentage of the total fund that should be invested in all given five schemes, respectively.

Max $Z = 6x_1 + 13x_2 + 10x_3 + 20x_4 + 25x_5$

subject to $15x_1 + 3x_2 + 5x_3 + 6x_4 + 10x_5 \le 15$

$x_1 + 3x_2 + 2x_3 + 5x_4 + x_5 \le 4$

$x_5 \ge 0.25$

$x_1 + x_2 + x_3 + x_4 + x_5 = 1$

and $x_j \ge 0$ for all j.

32. Let x_1, x_2, x_3 = number of scooters of types A, B and C, respectively.

Max $Z = 0.7\ (6{,}000 - 5{,}000)x_1 + 0.8\ (8{,}500 - 7{,}000)x_2 + 0.6\ (10{,}500 - 8{,}500)x_3$

subject to $5{,}000x_1 + 2 \times 7{,}000x_2 \le 1{,}00{,}000$

$2 \times 7{,}000x_2 + 8{,}500x_3 \le 1{,}00{,}000$

and $x_1, x_2, x_3 \ge 0.$

33. Let x_1, x_2, x_3 = number of station wagons, minibuses and larges buses, respectively to be purchased.

Max Z (total profit) = $15{,}000x_1 + 35{,}000x_2 + 45{,}000x_3$

subject to $x_1 + x_2 + x_3 \le 30$ (Availability of drivers)

$2{,}45{,}000x_1 + 3{,}50{,}000x_2 + 5{,}00{,}000x_3 \le 50{,}00{,}000$ (Budget)

$x_1 + \frac{3}{5}x_2 + \frac{1}{2}x_3 \le 80$ (Maintenance capacity)

and $x_1, x_2, x_3 \ge 0.$

Note: 1 *S.W.* = (3/5) *M.B.,* because (5/3) *S.W.* = 1 *M.B.*; and 1 *S.W.* = (3/5) *M.B.*, because (5/3) *S.W.* = 1 *M.B.*

34. Let x_j = number of units of food type j (j = 1, 2, 3, 4) used.

Min (total cost) $Z = 1.00x_1 + 1.25x_2 + 0.95x_3 + 1.35x_4$

subject to $40x_1 + 50x_2 + 30x_3 + 60x_4 \ge 500$

$30x_1 + 60x_2 + 35x_3 + 40x_4 \ge 750$

$25x_1 + 30x_2 + 25x_3 + 50x_4 \ge 600$

and $x_1, x_2, x_3, x_4 \ge 0.$

35. Let x_1 and x_2 = number of old hens and young hens bought, respectively.

Max $Z = 0.5\ (3x_1 + 5x_2) - (x_1 + x_2) = 0.5x_2 - 1.5x_1$

subject to $100x_1 + 250x_1 \ge 8{,}000; \quad x_1 + x_2 \le 20$

and $x_1, x_2 \ge 0.$

36. Let x_1 and x_2 = number of units of share A and B respectively.

to Min (total investment) $Z = x_1 + x_2$

subject to $0.12x_1 + 0.04x_2 \ge 600$ (Dividend constraint)

$0.10x_1 + 0.40x_2 \ge 1{,}000$ (Investment constraint)

and $x_1, x_2 \ge 0.$

37. Let x_1 and x_2 = volume of purchase from supplier A and B, respectively.

Min $Z = x_1 + 4x_2$

subject to (i) $x_1 + x_2 \ge 2{,}000$; (ii) $\frac{1}{4}x_1 + \frac{3}{4}x_2 \ge 1{,}000$

(iii) $\frac{1}{20}x_1 + \frac{1}{10}x_2 \ge 175$

and $x_1, x_2 \ge 0.$

CHAPTER SUMMARY

This chapter presents basic assumptions, limitations, components of any linear programming model and broad application areas of linear programming. The guidelines of mathematical modelling of any decision problem were explained followed by a large number of model building solved exercises in all functional areas of management and allied areas. These exercises are illustrative for students to deal with more complex and real-life problems.

CHAPTER CONCEPTS QUIZ

True or False

1. In a Linear Programming model, all parameter are assumed to be known as constant.
2. In LP model, any variable can assume to take only integer values or restricted to take discrete number of values.
3. Total contribution is used in place of profit in the objective function of maximization. problem because whole profit is not linearly related to sales volume.
4. An equation is more restrictive than an inequality.
5. All the variables in the solution of a linear programming problem are either positive or negative because of the existence of structural constraints.
6. Linear programming is a technique for finding the best uses of an organizations manpower, money and machinery.
7. Production planning is one of the application areas of the linear programming.
8. The effect of time and uncertainty are taken into consideration by linear programming model.
9. Linear Programming determines the economic and efficient way of locating manufacturing plants for physical distribution.
10. All variables in the linear programming problem must take one negative values.

Fill in the Blanks

11. Linear programming is a technique which attempts to determine how best to allocate _____ in order achieve some _____.
12. A linear programming technique improves the quality of _____.
13. In a linear programming, all relationships among decision variables are _____.
14. If two variables always take on values which are in the same proportion, the variables are _____ related.
15. _____ appearing in the models are assumed to be constant but _____ in real life situations.
16. Every linear programming problem includes _____ which relates variable in the problem to the goal of the firm and _____ which represent the limit on resource available to the firm.
17. Most of the constraints in the linear programming problem are expressed as _____.
18. Linear programming is used to allocate _____ to activities so as to optimize the value of objective function.
19. If the value of the variables are under the control of decision makers then variables are said to be _____ otherwise _____.
20. _____ of the decision variables is one of the assumption of the linear programming model.

Multiple Choice

21. The mathematical model of an LP problem is important because
    (a) it helps in converting the verbal description and numerical data into mathematical expression
    (b) decision-makers prefer to work with formal models
    (c) it captures the relevant relationship among decision factors
    (d) it enables the use of algebraic technique
22. Linear programming is a
    (a) constrained optimization technique
    (b) technique for economic allocation of limited resources
    (c) mathematical technique
    (d) all of the above
23. A constraint in an LP model restricts
    (a) value of objective function
    (b) value of a decision variable
    (c) use of the available resource
    (d) all of the above
24. The distinguishing feature of an LP model is
    (a) relationship among all variables is linear
    (b) it has single objective function and constraints
    (c) value of decision variables is non-negative
    (d) all of the above
25. Constraints in an LP model represents
    (a) limitations
    (b) requirements
    (c) balancing limitations and requirements
    (d) all of the above
26. Non-negativity condition is an important component of LP model because
    (a) variables value should remain under the control of the decision-maker
    (b) value of variables make sense and correspond to real-world problems
    (c) variables are interrelated in terms of limited resources
    (d) none of the above
27. Before formulating a formal LP model, it is better to
    (a) express each constraint in words
    (b) express the objective function in words
    (c) verbally identify decision variables
    (d) all of the above
28. Each constraint in an LP model is expressed as an
    (a) inequality with ³ sign
    (b) inequality with £ sign
    (c) equation with = sign
    (d) none of the above
29. Maximization of objective function in an LP model means
    (a) value occurs at allowable set of decisions
    (b) highest value is chosen among allowable decisions
    (c) neither of above
    (d) both (a) and (b)
30. Which of the following is not a characteristic of the LP model
    (a) alternative courses of action
    (b) an objective function of maximization type
    (c) limited amount of resources
    (d) non-negativity condition on the value of decision variables
31. The best use of linear programming technique is to find an optimal use of
    (a) money (b) manpower
    (c) machine (d) all of the above
32. Which of the following is not the characteristic of linear programming
    (a) resources must be limited
    (b) only one objective function
    (c) parameters value remains constant during the planning period
    (d) the problem must be of minimization type
33. Non-negativity condition in an LP model implies
    (a) a positive coefficient of variables in objective function
    (b) a positive coefficient of variables in any constraint
    (c) non-negative value of resources
    (d) none of the above
34. Which of the following is an assumption of an LP model
    (a) divisibility (b) proportionality
    (c) additivity (d) all of the above
35. Which of the following is a limitation associated with an LP Model
    (a) the relationship among decision variables in linear
    (b) no guarantee to get integer valued solutions
    (c) no consideration of effect of time and uncertainty on LP model
    (d) all of the above

Answers to Quiz

1. T	2. F	3. T	4. T	5. F	6. T	7. T
8. F	9. T	10. T	11. resources, objective		12. decisions	13. linear
14. linearly	15. parameters, unknown		16. objective function, constraints		17. inequalities	18. scarce resources
19. controllable and uncontrollable		20. certainty	21. (a)	22. (d)	23. (d)	24. (a)
25. (d)	26. (b)	27. (d)	28. (d)	29. (a)	30. (b)	31. (d)
32. (d)	33. (d)	34. (d)	35. (d)			

CASE STUDY

Case 2.1: Welltype Manufacturing

Welltype manufacturing company produces three types of typewriters. All the three models are required to be machined first and then assembled. The time required for the various models are as follows:

Types	*Manual Typewriters*	*Electronic Typewriters*	*Deluxe Electronic Typewriters*
Machine time (in hours)	15	12	14
Assembly time (in hours)	4	3	5

The total available machine time and assembly time are 3,000 hours and 1,200 hours, respectively. The data regarding the selling price and variable costs for the three types are:

Types	*Manual Typewriters*	*Electronic Typewriters*	*Deluxe Electronic Typewriters*
Selling price (Rs)	4,100	7,500	14,600
Labour, material and other variable costs (Rs)	2,500	4,500	9,000

The company sells all the three types on credit, but can only collect the amounts on the first of the following month. The labour, material and other variable expenses will have to be paid in cash. This company has taken a loan of Rs 40,000 from a co-operative bank and will have to repay it to the bank on 1 April 2008. The TNC Bank from whom this company has borrowed Rs 60,000 has expressed its approval to renew the loan.

Balance Sheet of the Company as on 31 March 2008

Liabilities	*Rs*	*Assets*	*Rs*
Equity Share Capital	1,50,000	Land	90,000
Capital Reserve	15,000	Building	70,000
General Reserve	1,10,000	Plant & Machinery	1,00,000
Profit & Loss A/c	25,000	Furniture & Fixtures	15,000
Long-term Loan	1,00,000	Vehicles	30,000
Loan from TNC Bank	60,000	Inventory	5,000
Loan from Co-operative Bank	40,000	Receivables	50,000
		Cash	1,40,000
Total	5,00,000	Total	5,00,000

The company will have to pay a sum of Rs 10,000 towards salary of top management executives and other fixed overheads for the month. Interest on long-term loans is to be paid every month at 24% per annum. Interest on loans from TNC and cooperative banks may be taken to be 1,200 for the month. Also this company has promised to deliver 2 manual typewriters and 8 deluxe electronic typewriters to one of its valued customers next month. Keep

in mind the fact that the level of operations in this company is subject to the availability of cash next month. This company will also be able to sell all types of typewriters in the market. The senior manager of this company desires to know as to how many units of each typewriter must be manufactured in the factory next month so as maximize the profits of the company. Advise the management of the company for manufacturing strategy with an aim to maximize profit.

Case 2.2: Shreya Advertizers

Shreya Advertizers – a marketing company dealing with laminated sheets 'Gloss' in the western zone covering Maharashtra, Gujarat and Madhya Pradesh, is considering to launch an advertisement campaign within a budget of Rs 2.5 lakh. On the basis of advertisement testing of the previous year, the company's research department has found that magazines and films are the ideal media for advertising laminated sheets. The company is not in a position to use the audiovisual medium due to limitation of funds. The magazines enjoying good recall in last year's campaign are *Stardust, Filmfare, Reader's Digest* and *Madhuri.* This is attributed to the effective visual impact made by the good reproduction of the advertisements both in colour, and black and white.

The characteristics of target audience for 'Gloss' and weightage for each characteristic are as follows:

	Characteristics	*Weightage (%)*
Age	15–34 yrs	20
Monthly income	Over Rs 5,000	70
Education	Above S.S.C.	10

The audience characteristics for the four magazines selected are given below:

Characteristics	*Stardust (%)*	*Filmfare (%)*	*Reader's Digest (%)*	*Madhuri (%)*
Age: 15–34yrs	75	45	56	80
Monthly income: Over Rs 5,000	52	43	47	25
Education: Above S.S.C.	83	53	72	34

The efficacy index for a black and white advertisement may be taken as 0.15 and that for a colour advertisement as 0.20. The cost per insertion of a black and white, and a colour advertisement and the readership for the four magazines are as follows:

Magazines	*Cost (Rs) per Insertion*		*Readership (in '000 Rs)*
	Black and White	*Colour*	
Stardust (Monthly)	4,500	8,400	189
Filmfare (Fortnightly)	4,200	8,400	256
Reader's Digest (Monthly)	6,400	9,600	136
Madhuri (Fortnightly)	3,300	6,600	205

It has also been found that for creating an impact at least 03 insertions are necessary in *Stardust* and *Reader's Digest*, while a minimum of 04 insertions will be required in the case of *Filmfare.*

Suggest an advertising strategy for the company to maximize the expected effective exposure.

Chapter 3

Linear Programming: The Graphical Method

"People realize that technology certainly is a tool. This tool can be used to enhance operations, improve efficiencies and really add value to academic research and teaching exercises."

– Peter Murray

PREVIEW

This chapter presents graphical solution method for solving any LP problem with only two decision variables. This method provides a conceptual basis for solving large and complex LP problems.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- solve an LP problem by the graphical method.
- understand various important terms such as extreme points, infeasibility, redundancy and multiple solutions and demonstrate them with the help of the graphical method.
- interpret the solution of an LP model.

CHAPTER OUTLINE

3.1 Introduction
3.2 Important Definitions
3.3 Graphical Solution Methods of LP Problem
3.4 Special Cases in Linear Programming
- Conceptual Questions
- Self Practice Problems
- Hints and Answers

❑ Chapter Summary
❑ Chapter Concepts Quiz
❑ Case Study

3.1 INTRODUCTION

An optimal as well as a feasible solution to an LP problem is obtained by choosing one set of values from several possible values of decision variables $x_1, x_2, \ldots, x_n$, that satisfies the given constraints simultaneously and also provides an optimal (maximum or minimum) value of the given objective function.

For LP problems that have only two variables, it is possible that the entire set of feasible solutions can be displayed graphically by plotting linear constraints on a graph paper in order to locate the best (optimal) solution. The technique used to identify the optimal solution is called the *graphical solution method (approach or technique)* for an LP problem with two variables.

Since most real-world problems have more than two decision variables, such problems cannot be solved graphically. However, graphical approach provides understanding of solving an LP problem algebraically, involving more than two variables.

Optimal solution to an LP problem is obtained by
(i) Extreme (corner) point, and
(ii) Iso-profit (cost) function line method

In this chapter, we shall discuss the following two graphical solution methods (or approaches):

(i) Extreme point solution method

(ii) Iso-profit (cost) function line method

to find the optimal solution to an LP problem.

3.2 IMPORTANT DEFINITIONS

Solution The set of values of decision variables x_j ($j = 1, 2, \ldots, n$) that satisfy the constraints of an LP problem is said to constitute the solution to that LP problem.

Feasible solution The set of values of decision variables x_j ($j = 1, 2, \ldots, n$) that satisfy all the constraints and non-negativity conditions of an LP problem simultaneously is said to constitute the feasible solution to that LP problem.

Infeasible solution The set of values of decision variables x_j ($j = 1, 2, \ldots, n$) that do not satisfy all the constraints and non-negativity conditions of an LP problem simultaneously is said to constitute the infeasible solution to that LP problem.

Basic solution For a set of m simultaneous equations in n variables ($n > m$) in an LP problem, a solution obtained by setting ($n - m$) variables equal to zero and solving for remaining m equations in m variables is called a basic solution of that LP problem.

The ($n - m$) variables whose value did not appear in basic solution are called *non-basic variables* and the remaining m variables are called *basic variables.*

Basic feasible solution A feasible solution to an LP problem which is also the basic solution is called the basic feasible solution. That is, all basic variables assume non-negative values. Basic feasible solution is of two types:

(a) *Degenerate* A basic feasible solution is called degenerate if the value of at least one basic variable is zero.

(b) *Non-degenerate* A basic feasible solution is called non-degenerate if value of all m basic variables is non-zero and positive.

Optimum basic feasible solution A basic feasible solution that optimizes (maximizes or minimizes) the objective function value of the given LP problem is called an optimum basic feasible solution.

Unbounded solution A solution that can increase or decrease infinitely the value of the objective function of the LP problem is called an unbounded solution.

3.3 GRAPHICAL SOLUTION METHODS OF LP PROBLEM

While obtaining the optimal solution to the LP problem by the graphical method, the statement of the following theorems of linear programming is used

- *The collection of all feasible solutions to an LP problem constitutes a convex set whose extreme points correspond to the basic feasible solutions.*

- *There are a finite number of basic feasible solutions within the feasible solution space.*
- *If the convex set of the feasible solutions of the system of simultaneous equations:* $\mathbf{Ax = b, x \geq 0,}$ *is a convex polyhedron, then at least one of the extreme points gives an optimal solution.*
- *If the optimal solution occurs at more than one extreme point, the value of the objective function will be the same for all convex combinations of these extreme points.*

Remarks 1. A convex set is a polygon and by 'convex' we mean that if any two points of a polygon are selected arbitrarily, a straight line segment joining these two points lies completely within the polygon.

2. Each corner (extreme or vertex) point of the feasible region (space or area) falls at the intersection of two constraint equalities.
3. The extreme points of the convex set provide the basic feasible solution to the LP problem.

3.3.1 Extreme Point Solution Method

Extreme point refers to the corner of the feasible region (or space), i.e. this point lies at the intersection of two constraint equations.

In this method, the coordinates of all corner (or extreme) points of the feasible region (space or area) are determined and then value of the objective function at each of these points is computed and compared. The coordinates of an extreme point where the optimal (maximum or minimum) value of the objective function is found represent solution of the given LP problem. The steps of the method are summarized as follows:

Step 1 : Develop an LP model State the given problem in the mathematical LP model as illustrated in the previous chapter.

Step 2 : Plot constraints on graph paper and decide the feasible region

(a) Replace the inequality sign in each constraint by an equality sign.

(b) Draw these straight lines on the graph paper and decide each time the area of feasible solutions according to the inequality sign of the constraint. Shade the common portion of the graph that satisfies all the constraints simultaneously drawn so far.

(c) The final shaded area is called the *feasible region* (*or solution space*) of the given LP problem. Any point inside this region is called *feasible solution* and this provides values of x_1 and x_2 that satisfy all the constraints.

Step 3 : Examine extreme points of the feasible solution space to find an optimal solution

(a) Determine the coordinates of each extreme point of the feasible solution space.

(b) Compute and compare the value of the objective function at each extreme point.

(c) Identify the extreme point that gives optimal (max. or min.) value of the objective function.

Extreme point method is one of the methods of finding the optimal solution to an LP problem by examining the profit (or cost) level at each corner point of feasible region.

3.3.2 Examples on Maximization LP Problem

Example 3.1 Use the graphical method to solve the following LP problem.

Maximize $Z = 15x_1 + 10x_2$

subject to the constraints

(i) $4x_1 + 6x_2 \leq 360$, (ii) $3x_1 + 0x_2 \leq 180$, (iii) $0x_1 + 5x_2 \leq 200$

and $x_1, x_2 \geq 0$.

Solution 1. The given LP problem is already in mathematical form.

2. Treat x_1 as the horizontal axis and x_2 as the vertical axis. Plot each constraint on the graph by treating it as a linear equation and it is then that the appropriate inequality conditions will be used to mark the area of feasible solutions.

Consider the first constraint $4x_1 + 6x_2 \leq 360$. Treat this as the equation $4x_1 + 6x_2 = 360$. For this find any two points that satisfy the equation and then draw a straight line through them. The two points are generally the points at which the line intersects the x_1 and x_2 axes. For example, when $x_1 = 0$ we get $6x_2 = 360$ or $x_2 = 60$. Similarly when $x_2 = 0$, $4x_1 = 360$, $x_1 = 90$.

These two points are then connected by a straight line as shown in Fig. 3.1(a). But the question is: *Where are these points satisfying* $4x_1 + 6x_2 \leq 360$. Any point above the constraint line violates the inequality condition. But any point below the line does not violate the constraint. Thus, the inequality and non-negativity condition can only be satisfied by the shaded area (feasible region) as shown in Fig. 3.1(a).

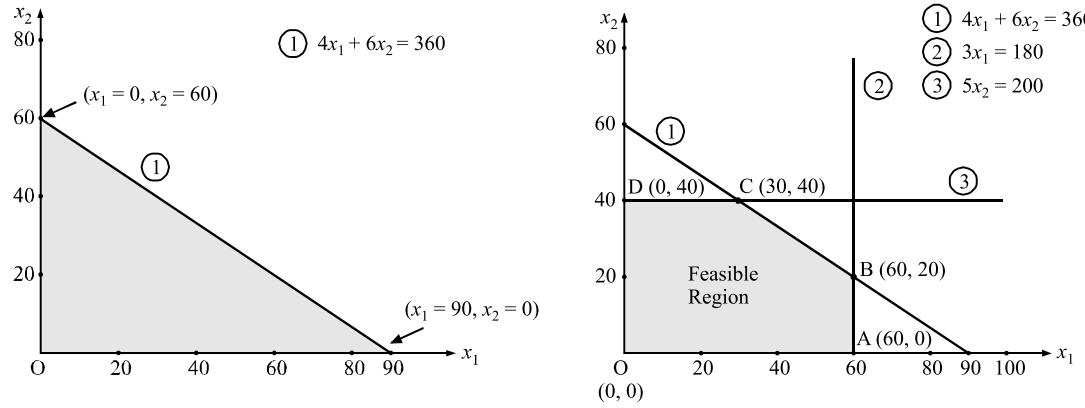


Fig. 3.1 Graphical Solution of LP Problem

Similarly, the constraints $3x_1 \leq 180$ and $5x_2 \leq 200$ are also plotted on the graph and are indicated by the shaded area as shown in Fig. 3.1(b).

Since all constraints have been graphed, the area which is bounded by all the constraints lines including all the boundary points is called the *feasible region* (*or solution space*). The feasible region is shown in Fig. 3.1(b) by the shaded area OABCD.

3. (i) Since the optimal value of the objective function occurs at one of the extreme points of the feasible region, it is necessary to determine their coordinates. The coordinates of extreme points of the feasible region are: O = (0, 0), A = (60, 0), B = (60, 20), C = (30, 40), D = (0, 40).

(ii) Evaluate objective function value at each extreme point of the feasible region as shown in the Table 3.1:

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 15x_1 + 10x_2$
O	(0, 0)	15(0) + 10(0) = 0
A	(60, 0)	15(60) + 10(0) = 900
B	(60, 20)	15(60) + 10(20) = 1,100
C	(30, 40)	15(30) + 10(40) = 850
D	(0, 40)	15(0) + 10(40) = 400

Table 3.1 Set of Feasible Solutions

(iii) Since objective function Z is to be maximized, from Table 3.1 we conclude that maximum value of Z = 1,100 is achieved at the point extreme B (60, 20). Hence the optimal solution to the given LP problem is: $x_1 = 60$, $x_2 = 20$ and Max Z = 1,100.

Remark To determine which side of a constraint equation is in the feasible region, examine whether the origin (0, 0) satisfies the constraints. If it does, then all points on and below the constraint equation towards the origin are feasible points. If it does not, then all points on and above the constraint equation away from the origin are feasible points.

Feasible region is the overlapping area of constraints that satisfies all of the constraints on resources.

Example 3.2 Use the graphical method to solve the following LP problem.

Maximize $Z = 2x_1 + x_2$

subject to the constraints

(i) $x_1 + 2x_2 \leq 10$, (ii) $x_1 + x_2 \leq 6$,

(iii) $x_1 - x_2 \leq 2$, (iv) $x_1 - 2x_2 \leq 1$

and $x_1, x_2 \geq 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.2. It may be noted

that we have not considered the area below the lines $x_1 - x_2 = 2$ and $x_1 - 2x_2 = 1$ for the negative values of x_2. This is because of the non-negativity condition, $x_2 \geq 0$.

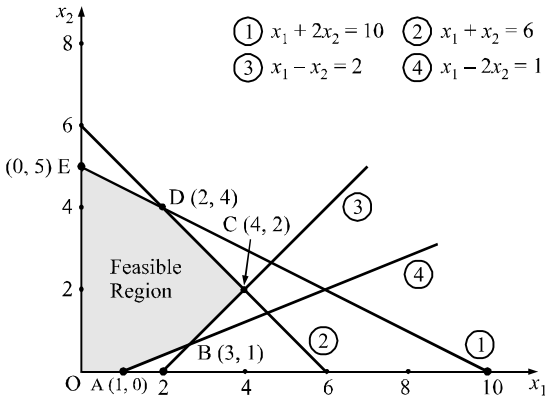


Fig. 3.2 Graphical Solution of LP Problem

The coordinates of extreme points of the feasible region are: O = (0, 0), A = (1, 0), B = (3, 1), C = (4, 2), D = (2, 4), and E = (0, 5). The value of objective function at each of these extreme points is shown in Table 3.2.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 2x_1 + x_2$
O	(0, 0)	2(0) + 1(0) = 0
A	(1, 0)	2(1) + 1(0) = 2
B	(3, 1)	2(3) + 1(1) = 7
C	(4, 2)	2(4) + 1(2) = 10
D	(2, 4)	2(2) + 1(4) = 8
E	(0, 5)	2(0) + 1(5) = 5

Table 3.2 Set of Feasible Solutions

The maximum value of the objective function $Z = 10$ occurs at the extreme point (4, 2). Hence, the optimal solution to the given LP problem is: $x_1 = 4$, $x_2 = 2$ and Max $Z = 10$.

Example 3.3 Solve the following LP problem graphically:

Maximize $Z = -x_1 + 2x_2$

subject to the constraints

(i) $x_1 - x_2 \leq -1$; (ii) $-0.5x_1 + x_2 \leq 2$

and $x_1, x_2 \geq 0$. [*Punjab Univ., BE (CS and E), 2006*]

Solution Since resource value (RHS) of the first constraint is negative, multiplying both sides of this constraint by –1, the constraint becomes: $-x_1 + x_2 \geq 1$. Plot on a graph each constraint by first treating them as a linear equation and mark the feasible region as shown in Fig. 3.3.

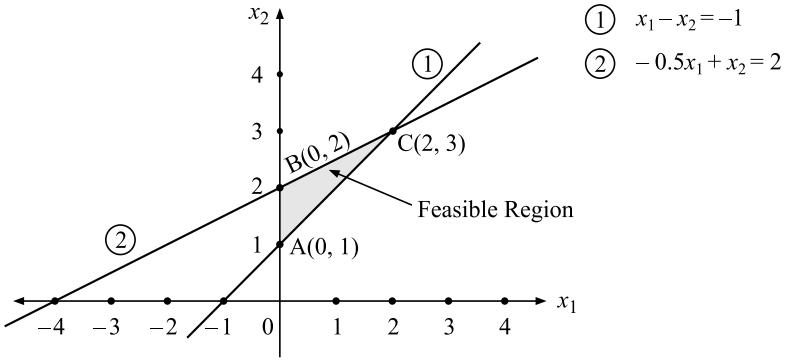


Fig. 3.3 Graphical Solution of LP Problem

The value of the objective function at each of the extreme points A(0, 1), B(0, 2) and C(2, 3) is shown in Table 3.3.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = -x_1 + 2x_2$	
A	(0, 1)	$0 + 2 \times 1 = 2$	
B	(0, 2)	$0 + 2 \times 2 = 4$	Multiple optimal
C	(2, 3)	$-1 \times 2 + 2 \times 3 = 4$	solution

Table 3.3 Set of Feasible Solutions

The maximum value of objective function Z = 4 occurs at extreme points B and C. This implies that every point between B and C on the line BC also gives the same value of Z. Hence, problem has multiple optimal solultions: $x_1 = 0,\ x_2 = 2$ and $x_1 = 2, x_2 = 3$ and Max Z = 4.

Example 3.4 The ABC Company has been a producer of picture tubes for television sets and certain printed circuits for radios. The company has just expanded into full scale production and marketing of AM and AM-FM radios. It has built a new plant that can operate 48 hours per week. Production of an AM radio in the new plant will require 2 hours and production of an AM-FM radio will require 3 hours. Each AM radio will contribute Rs 40 to profits while an AM-FM radio will contribute Rs 80 to profits. The marketing department, after extensive research has determined that a maximum of 15AM radios and 10 AM-FM radios can be sold each week.

(a) Formulate a linear programming model to determine the optimum production mix of AM and FM radios that will maximize profits.
(b) Solve this problem using the graphical method. [*Delhi Univ., MBA, 2002, 2008*]

Solution Let us define the following decision variables

x_1 and x_2 = number of units of AM radio and AM-FM radio to be produced, respectively.

Then LP model of the given problem is:

Maximize (total profit) $Z = 40x_1 + 80x_2$

subject to the constraints

(i) Plant : $2x_1 + 3x_2 \le 48$, (ii) AM radio : $x_1 \le 15$, (iii) AM-FM radio : $x_2 \le 10$

and $x_1, x_2 \ge 0$.

Plot on a graph each constraint by first treating it as a linear equation. Then use inequality condition of each constraint to mark the feasible region. The feasible solution space (or region) is shown in Fig. 3.4 by shaded area.

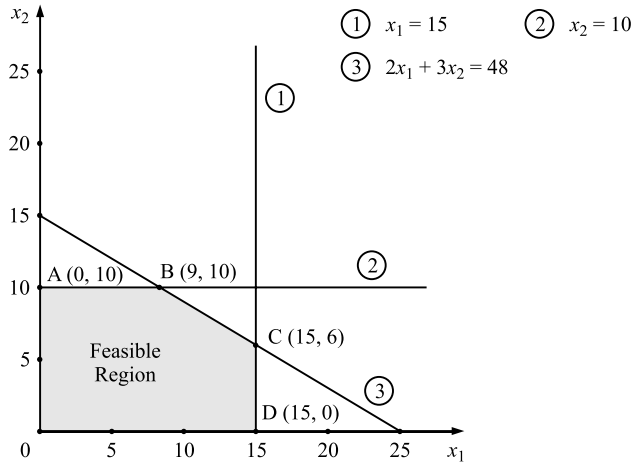


Fig. 3.4 Graphical Solution of LP Problems

The coordinates of extreme points of the feasible region are: O (0, 0), A (0, 10), B (9, 10), C (15, 6) and D (15, 0). The value of objective function at each of corner (or extreme) points is shown in Table 3.4.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 40x_1 + 80x_2$
O	(0, 0)	40(0) + 80(0) = 0
A	(0, 10)	40(0) + 80(10) = 800
B	(9, 10)	40(9) + 80(10) = **1,160**
C	(15, 6)	40(15) + 80(6) = 1,080
D	(15, 0)	40(15) + 80(0) = 600

Table 3.4
Set of Feasible Solutions

Since the maximum value of the objective function $Z = 1{,}160$ occurs at the extreme point (9, 10), the optimum solution to the given LP problem is: $x_1 = 9, x_2 = 10$ and Max. $Z =$ Rs 1,160.

Example 3.5 Anita Electric Company produces two products P_1 and P_2. Products are produced and sold on a weekly basis. The weekly production cannot exceed 25 for product P_1 and 35 for product P_2 because of limited available facilities. The company employs total of 60 workers. Product P_1 requires 2 man-weeks of labour, while P_2 requires one man-week of labour. Profit margin on P_1 is Rs. 60 and on P_2 is Rs. 40. Formulate this problem as an LP problem and solve that using graphical method.

Solution Let us define the following decision variables:

x_1 and x_2 = number of units of product P_1 and P_2, to be produced, respectively.

Then LP model of the given problem is:

Maximize $Z = 60x_1 + 40x_2$

subject to the constraints

(i) Weekly productions for $P_1 : x_1 \le 25$, (ii) Weekly production for $P_2 : x_2 \le 35$,

(iii) Workers: $2x_1 + x_2 = 60$

and $x_1, x_2 \ge 0$.

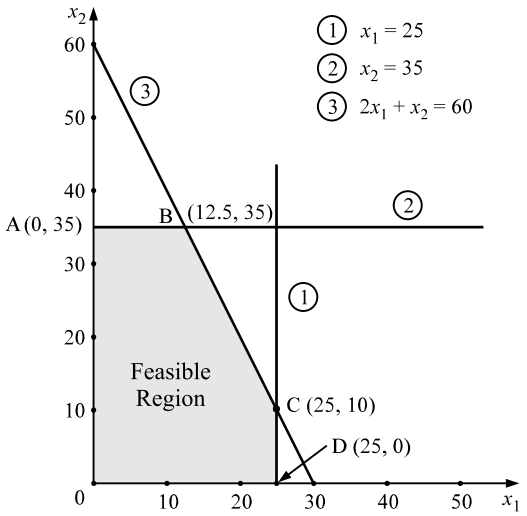


Fig. 3.5
Graphical Solution of LP Problem

Plot on a graph each constraint by first treating it as linear equation. Then use inequality condition of each constraint to mark the feasible region shown in Fig. 3.6.

The coordinates of extreme points of the feasible solution space (or region) are: A(0, 35), B(12.5, 35), C(25, 10) and D(25, 0). The value of the objective function at each of these extreme points is shown in Table 3.5.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 60x_1 + 40x_2$
A	(0, 35)	60(0) + 40(35) = 1,400
B	(12.5, 35)	60(12.5) + 40(35) = **2,150**
C	(25, 10)	60(25) + 40(10) = 1,900
D	(25, 0)	60(25) + 40(0) = 1,500

Table 3.5
Set of Feasible Solutions

The optimal (maximum) value, $Z = 2{,}150$ is obtained at the point B (12.5, 35). Hence, $x_1 = 12.5$ units of product P_1 and $x_1 = 35$ units of product P_2 should be produced in order to have the maximum profit, $Z =$ Rs. 2,150.

3.3.3 Examples on Minimization LP Problem

Example 3.6 Use the graphical method to solve the following LP problem.

Minimize $Z = 3x_1 + 2x_2$

subject to the constraints

(i) $5x_1 + x_2 \geq 10$, (ii) $x_1 + x_2 \geq 6$, (iii) $x_1 + 4x_2 \geq 12$

and $x_1, x_2 \geq 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.6. This region is bounded from below by extreme points *A*, *B*, *C* and *D*.

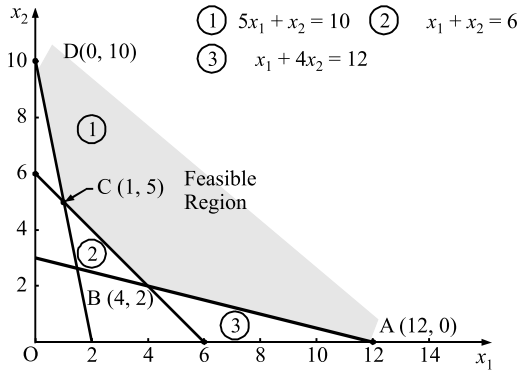


Fig. 3.6 Graphical Solution of LP Problem

The coordinates of the extreme points of the feasible region (bounded from below) are: A = (12, 0), B = (4, 2), C = (1, 5) and D = (0, 10). The value of objective function at each of these extreme points is shown in Table 3.6.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 3x_1 + 2x_2$
A	(12, 0)	3(12) + 2(0) = 36
B	(4, 2)	3(4) + 2(2) = 16
C	(1, 5)	3(1) + 2(5) = **13**
D	(0, 10)	3(0) + 2(10) = 20

Table 3.6 Set of Feasible Solutions

The minimum (optimal) value of the objective function $Z = 13$ occurs at the extreme point C (1, 5). Hence, the optimal solution to the given LP problem is: $x_1 = 1$, $x_2 = 5$, and Min $Z = 13$.

Example 3.7 Use the graphical method to solve the following LP problem.

Minimize $Z = -x_1 + 2x_2$

subject to the constraints

(i) $-x_1 + 3x_2 \leq 10$, (ii) $x_1 + x_2 \leq 6$, (iii) $x_1 - x_2 \leq 2$

and $x_1, x_2 \geq 0$.

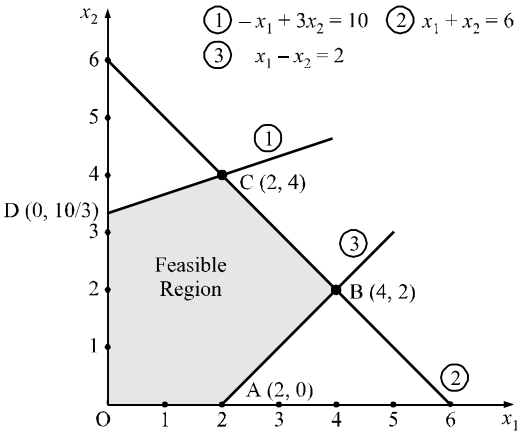


Fig. 3.7 Graphical Solution of LP Problem

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.7. It may be noted that the area below the lines $-x_1 + 3x_2 = 10$ and $x_1 - x_2 = 2$ is not desirable, due to the reason that values of x_1 and x_2 are desired to be non-negative, i.e. $x_1 \geq 0$, $x_2 \geq 0$.

The coordinates of the extreme points of the feasible region are: O = (0, 0), A = (2, 0), B = (4, 2), C = (2, 4), and D = (0, 10/3). The value of the objective function at each of these extreme points is shown in Table 3.7.

Table 3.7 Set of Feasible Solutions

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = -x_1 + 2x_2$
O	(0, 0)	– 1(0) + 2(0) = 0
A	(2, 0)	– 1(2) + 2(0) = –2
B	(4, 2)	– 1(4) + 2(2) = 0
C	(2, 4)	– 1(2) + 2(4) = 6
D	(0, 10/3)	– 1(0) + 2(10/3) = 20/3

The minimum (optimal) value of the objective function $Z = -2$ occurs at the extreme point A (2, 0). Hence, the optimal solution to the given LP problem is: $x_1 = 2$, $x_2 = 0$ and Min $Z = -2$.

Example 3.8 G.J. Breweries Ltd have two bottling plants, one located at 'G' and the other at 'J'. Each plant produces three drinks, whisky, beer and brandy named A, B and C respectively. The number of the bottles produced per day are shown in the table.

Drink	*Plant at*	
	G	*J*
Whisky	1,500	1,500
Beer	3,000	1,000
Brandy	2,000	5,000

A market survey indicates that during the month of July, there will be a demand of 20,000 bottles of whisky, 40,000 bottles of beer and 44,000 bottles of brandy. The operating cost per day for plants at G and J are 600 and 400 monetary units. For how many days each plant be run in July so as to minimize the production cost, while still meeting the market demand? Formulate this problem as an LP problem and solve that using graphical method. [*Pune Univ., MBA, 2009*]

Solution Let us define the following decision variables:

x_1 and x_2 = number of days of work at plant G and J, respectively.

Then LP model of the given problem is:

Minimize $Z = 600x_1 + 400x_2$

subject to the constraints

(i) Whisky : $1500x_1 + 1500x_2 = 20{,}000$ (ii) Beer : $3000x_1 + 1000x_2 = 40{,}000$,

(iii) Brandy : $2000x_1 + 5000x_2 = 44{,}000$

and $x_1, x_2 \geq 0$.

Plot on a graph each constraint by first treating it as linear equation. The feasible solution space (region) is shown in Fig. 3.8 by shaded area. This region is bounded from below by the extreme points A, B, and C. The coordinates of extreme points of the feasible solution space marked by shaded area are: A(22, 0), B(12, 4) and C(0, 40).

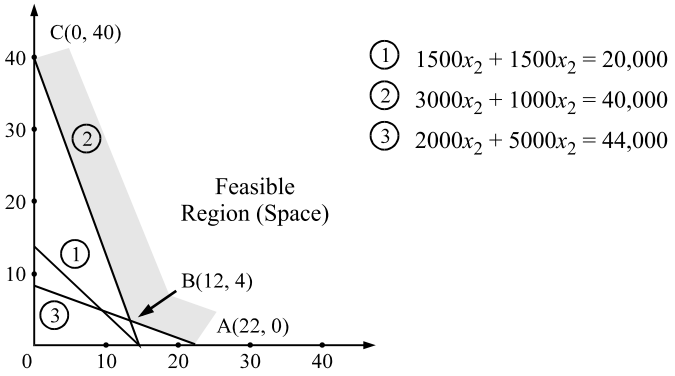


Fig. 3.8 Graphical Solution of LP Problem

The value of objective function at each of the extreme points is shown shown is Table 3.8.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 600x_1 + 400x_2$
A	(22, 0)	600(22) + 400(0) = 13,200
B	(12, 4)	600(12) + 400(4) = **8,800**
C	(0, 40)	600(0) + 400(40) = 16,000

Table 3.8 Set of Feasible Solutions

The minimum (optimal) value of objective function occurs at point B(12, 4). Hence, the Plant G should run for $x_1 = 12$ days and plant J for $x_2 = 4$ days to have a minimum production cost of Rs. 8,800.

Example 3.9 A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals and 1,400 calories. Two foods A and B are available at a cost of Rs. 4 and Rs. 3 per unit, respectively. If one of A contains 200 units of vitamins, 1 unit of mineral and 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 calories. Formulate this problem as an LP model and solve it by graphical method to find combination of foods to be used to have least cost?

Solution The data of the given problem can be summarized as shown below.

Food	*Units Content of*			*Cost per*
	Vitamins	*Mineral*	*Calories*	*Unit (Rs.)*
A	200	1	40	4
B	100	2	40	3
Minimum requirement	4,000	50	1,400	

Let us define the following decision variables:

x_1 and x_2 = number of units of food A and B to be used, respectively.

Then LP model of the given problem is:

Minimize (total cost) $Z = 4x_1 + 3x_2$

subject to the constraints

(i) Vitamins : $200x_1 + 100x_2 \geq 4{,}000$, (ii) Mineral : $x_1 + 2x_2 \geq 50$,

(iii) Calories : $40x_1 + 40x_2 \geq 1{,}400$

and $x_1, x_2 \geq 0$

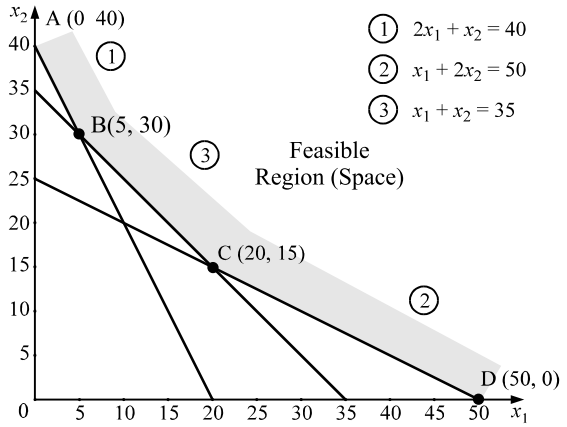


Fig. 3.9 Graphical Solution of LP Problem

Plot on a graph each constraint by treating it as a linear equation. Then use inequality sign of each constraint to mark the feasible solution space (region) by shaded area as shown in Fig. 3.9. The coordinates of the extreme points of the feasible solution space are: A(0, 40), B(5, 30), C(20, 15) and D(50, 0). The value of objective function at each of these extreme points is shown in Table 3.9.

Table 3.9
Set of Feasible Solutions

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 4x_1 + 3x_2$
A	(0, 40)	4(0) + 3(40) = 120
B	(5, 30)	4(5) + 3(30) = **110**
C	(20, 15)	4(20) + 3(15) = 125
D	(50, 0)	4(50) + 3(0) = 200

The minimum (optimal) value of the objective function $Z = 110$ occurs at the extreme point B(5, 30). Hence, to have least cost of Rs. 110, the diet should contain $x_1 = 5$ units of food A and $x_2 = 30$ units of food B.

3.3.4 Examples on Mixed Constraints LP Problems

Example 3.10 A firm plans to purchase at least 200 quintals of scrap containing high quality metal X and low quality metal Y. It decides that the scrap to be purchased must contain at least 100 quintals of metal X and not more than 35 quintals of metal Y. The firm can purchase the scrap from two suppliers (A and B) in unlimited quantities. The percentage of X and Y metals in terms of weight in the scrap supplied by A and B is given below.

Metals	*Supplier A*	*Supplier B*
X	25%	75%
Y	10%	20%

The price of A's scrap is Rs 200 per quintal and that of B is Rs 400 per quintal. The firm wants to determine the quantities that it should buy from the two suppliers so that the total cost is minimized.

[*Delhi Univ., MBA, 1998, 2001*]

Solution Let us define the following decision variables:

x_1 and x_2 = quantity (in quintals) of scrap to be purchased from suppliers A and B, respectively.

Then LP model of the given problem is:

Minimize $Z = 200x_1 + 400x_2$

subject to the constraints

(i) Maximum purchase : $x_1 + x_2 \geq 200$

(ii) Scrap containing : $\frac{x_1}{4} + \frac{3x_2}{4} \geq 100$ or $x_1 + 3x_2 \geq 400$ (Metal X)

$\frac{x_1}{10} + \frac{x_2}{5} \geq 35$ or $x_1 + 2x_2 \leq 350$ (Metal Y)

and $x_1, x_2 \geq 0$.

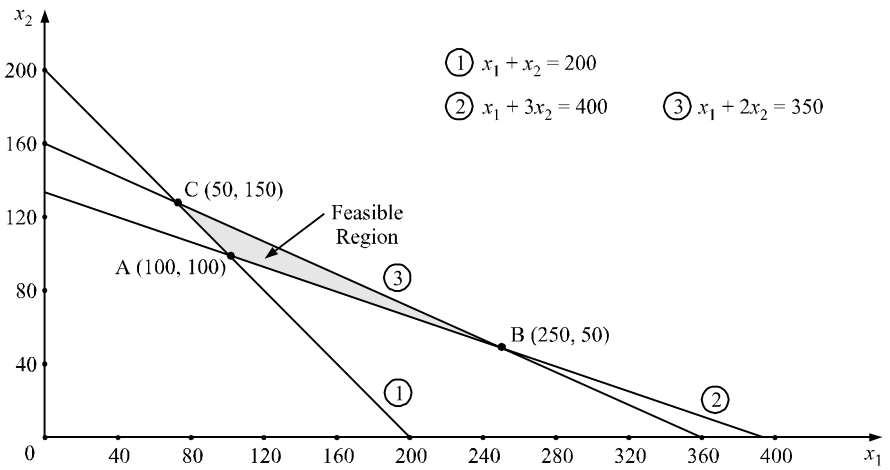


Fig. 3.10
Graphical Solution of LP Problem

Plot on a graph each constraint by treating it as a linear equation. Then use inequality sign of each constraint to mark the feasible solution space (or region) by shaded area as shown in Fig. 3.10. The feasible region is bounded by the corners, A, B and C.

The coordinates of the extreme points of the feasible region are: A = (100, 100), B = (250, 50), and C = (50, 150). The value of objective function at each of these extreme points is shown in Table 3.10.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 200x_1 + 400x_2$
A	(100, 100)	200(100) + 400(100) = **60,000**
B	(250, 50)	200(250) + 400(50) = 70,000
C	(50, 150)	200(50) + 400(150) = 70,000

Table 3.10 Set of Feasible Solutions

Since minimum (optimal) value of objective function, Z = Rs 60,000 occurs at the extreme point A (100, 100), firm should buy $x_1 = x_2 = 100$ quintals of scrap each from suppliers A and B.

Example 3.11 Use the graphical method to solve the following LP problem.

Maximize $Z = 7x_1 + 3x_2$

subject to the constraints

(i) $x_1 + 2x_2 \geq 3$ (ii) $x_1 + x_2 \leq 4$

(iii) $0 \leq x_1 \leq 5/2$ (iv) $0 \leq x_2 \leq 3/2$

and $x_1, x_2 \geq 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.11.

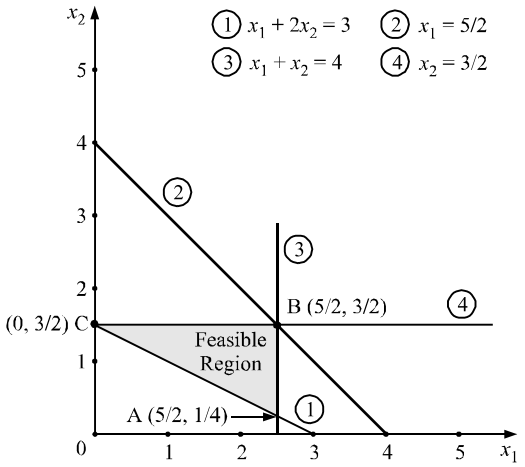


Fig. 3.11 Graphical Solution of LP Problem

The coordinates of the extreme points of the feasible region are: A = (5/2, 1/4), B = (5/2, 3/2), and C = (0, 3/2). The value of the objective function at each of these extreme points is shown in Table 3.11.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 7x_1 + 3x_2$
A	(5/2, 1/4)	7(5/2) + 3(1/4) = 73/4
B	(5/2, 3/2)	7(5/2) + 3(3/2) = **22**
C	(0, 3/2)	7(0) + 3(3/2) = 9/2

Table 3.11 Set of Feasible Solutions

The maximum (optimal) value of the objective function, $Z = 22$ occurs at the extreme point B (5/2, 3/2). Hence, the optimal solution to the given LP problem is: $x_1 = 5/2, x_2 = 3/2$ and Max $Z = 22$.

Example 3.12 Use the graphical method to solve the following LP problem.

Minimize $Z = 20x_1 + 10x_2$

subject to the constraints

(i) $x_1 + 2x_2 \leq 40$, (ii) $3x_1 + x_2 \geq 30$, (iii) $4x_1 + 3x_2 \geq 60$

and $x_1, x_2 \geq 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.12.

The coordinates of the extreme points of the feasible region are: A = (15, 0), B = (40, 0), C = (4, 18) and D = (6, 12). The value of the objective function at each of these extreme points is shown Table 3.12.

Table 3.12 Set of Feasible Solutions

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 20x_1 + 10x_2$
A	(15,0)	20(15) + 10(0) = 300
B	(40,0)	20(40) + 10(0) = 800
C	(4,18)	20(4) + 10(18) = 260
D	(6,12)	20(6) + 10(12) = 240

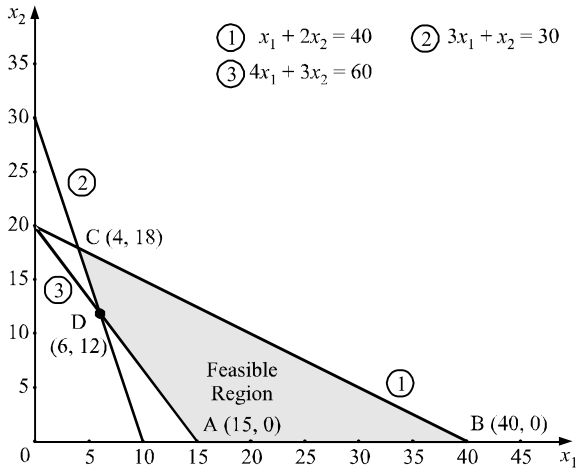


Fig. 3.12 Graphical Solution of LP Problem

The minimum (optimal) value of the objective function, $Z = 240$ occurs at the extreme point D (6, 12). Hence, the optimal solution to the given LP problem is: $x_1 = 6$, $x_2 = 12$ and Min $Z = 240$.

Example 3.13 Use the graphical method to solve the following LP problem.

Maximize $Z = 2x_1 + 3x_2$

subject to the constraints

(i) $x_1 + x_2 \le 30$ (ii) $x_2 \ge 3$ (iii) $0 \le x_2 \le 12$

(iv) $0 \le x_1 \le 20$ (v) $x_1 - x_2 \ge 0$

and $x_1, x_2 \ge 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region by shaded area as shown in Fig. 3.13.

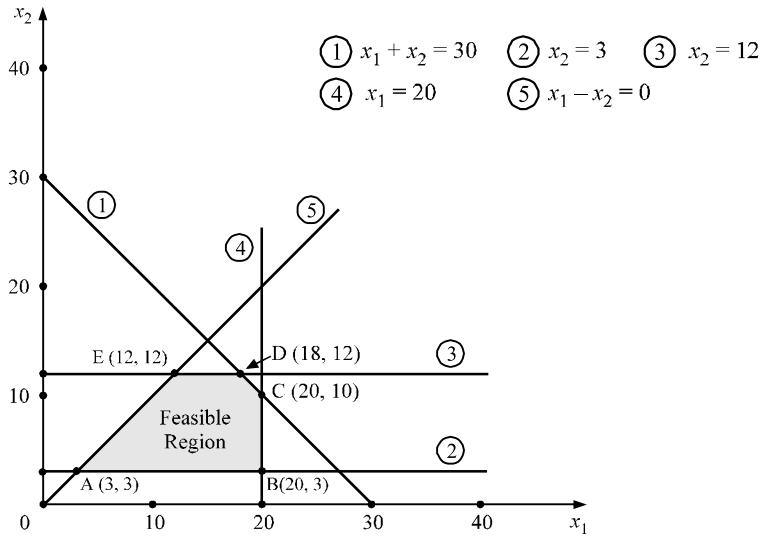


Fig. 3.13 Graphical Solution of LP Problem

The coordinates of the extreme points of the feasible region are : A = (3, 3), B = (20, 3), C = (20, 10), D = (18, 12) and E = (12, 12). The value of the objective function at each of these extreme points is shown in Table 3.13.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 2x_1 + 3x_2$
A	(3, 3)	2(3) + 3(3) = 15
B	(20, 3)	2(20) + 3(3) = 49
C	(20, 10)	2(20) + 3(10) = 70
D	(18, 12)	2(18) + 3(12) = **72**
E	(12, 12)	2(12) + 3(12) = 60

Table 3.13 Set of Feasible Solutions

The maximum value of the objective function, $Z = 72$ occurs at the extreme point D (18, 12). Hence, the optimal solution to the given LP problem is: $x_1 = 18$, $x_2 = 12$ and Max $Z = 72$.

Example 3.14 A firm makes two products X and Y, and has a total production capacity of 9 tonnes per day. Both X and Y require the same production capacity. The firm has a permanent contract to supply at least 2 tonnes of X and at least 3 tonnes of Y per day to another company. Each tonne of X requires 20 machine hours of production time and each tonne of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All of the firm's output can be sold. The profit made is Rs 80 per tonne of X and Rs 120 per tonne of Y. Formulate this problem as an LP model and solve it by using graphical method to determine the production schedule that yields the maximum profit.

[*Delhi Univ., MBA, 1999*]

Solution Let us define the following decision variables:

x_1 and x_2 = number of units (in tonnes) of products X and Y to be manufactured, respectively.

Then the LP model of the given problem can be written as

Maximize (total profit) $Z = 80x_1 + 120x_2$

subject to the constraints

(i) $x_1 + x_2 \le 9$ (Production capacity)

(ii) $x_1 \ge 2$; $x_2 \ge 3$ (Supply)

(iii) $20x_1 + 50x_2 \le 360$ (Machine hours)

and $x_1, x_2 \ge 0$

For solving this LP problem graphically, plot on a graph each constraint by first treating it as a linear equation. Then use the inequality sign of each constraint to mark the feasible region as shown in Fig. 3.14.

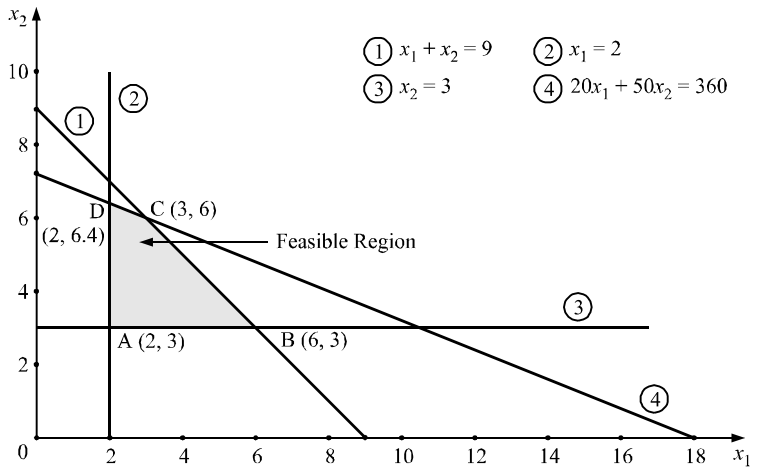


Fig. 3.14 Graphical Solution of LP Problem

The coordinates of the extreme points of the feasible region are: A = (2, 3), B = (6, 3), C = (3, 6), and D = (2, 6.4). The value of the objective function at each of these extreme points is shown in Table 3.14.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 80x_1 + 120x_2$
A	(2, 3)	80(2) + 120(3) = 520
B	(6, 3)	80(6) + 120(3) = 840
C	(3, 6)	80(3) + 120(6) = **960**
D	(2, 6.4)	80(2) + 120(6.4) = 928

Table 3.14 Set of Feasible Solutions

The maximum (optimal) value of the objective function, $Z = 960$ occurs at the extreme point C (3, 6). Hence the company should produce, $x_1 = 3$ tonnes of product X and $x_2 = 6$ tonnes of product Y in order to yield a maximum profit of Rs 960.

Example 3.15 The standard weight of a special purpose brick is 5 kg and it contains two basic ingredients B_1 and B_2. B_1 costs Rs 5 per kg and B_2 costs Rs 8 per kg. Strength considerations dictate that the brick should contain not more than 4 kg of B_1 and a minimum of 2 kg of B_2. Since the demand for the product is likely to be related to the price of the brick, graphically find out the minimum cost of the brick satisfying the above conditions.

Solution Let us define the following decision variables.

x_1 and x_2 = amount (in kg) of ingredient B_1 and B_2 contained in the brick, respectively.

Then the LP model of the given problem can be written as:

Minimize (total cost) $Z = 5x_1 + 8x_2$

subject to the constraints

(i) $x_1 \le 4$, (ii) $x_2 \ge 2$, (iii) $x_1 + x_2 = 5$ (Brick strength)

and $x_1, x_2 \ge 0$

For solving this LP problem graphically, plotting on a graph each constraint by first treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region as shown in Fig. 3.15.

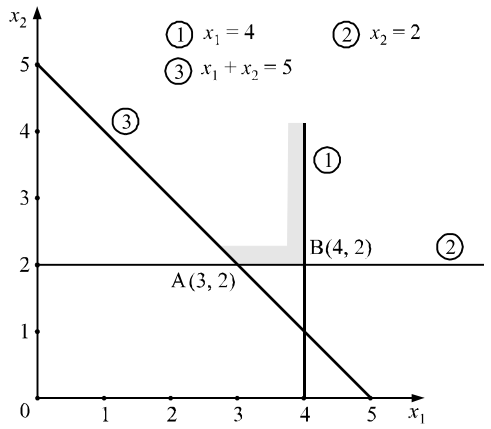


Fig. 3.15 Graphical Solution of LP Problem

It may be noted that in Fig. 3.15 there is no unique (or single) feasible region. The value of the objective function at extreme points A = (3, 2) and B = (4, 2) is given in Table 3.15.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 5x_1 + 8x_2$
A	(3, 2)	5(3) + 8(2) = **31**
B	(4, 2)	5(4) + 8(2) = 36

Table 3.15 Set of Feasible Solutions

The minimum value of objective function $Z = 31$ occurs at the extreme point A (3, 2). Hence, the optimal product mix is: 3 kg of ingredient B_1 and 2 kg of ingredient B_2 of a special purpose brick to achieve the minimum cost of Rs 31.

Example 3.16 The manager of an oil refinery must decide on the optimal mix of two possible blending processes of which the inputs and outputs per production run are as follows:

Process (units)	*Input (units)*		*Output (units)*	
	Grade A	*Grade B*	*Gasoline X*	*Gasoline Y*
1	5	3	5	8
2	4	5	4	4

The maximum amounts available of crudes A and B are 200 units and 150 units, respectively. Market requirements show that at least 100 units of gasoline X and 80 units of gasoline Y must be produced. The profits per production run for process 1 and process 2 are Rs 300 and Rs 400, respectively. Formulate this problem as an LP model and solve it using graphical method to determine the production run for process 1 and 2. [*Gujarat Univ., MBA, 1999*]

Solution Let us define the following decision variables

x_1 and x_2 = number of production run for process 1 and process 2, respectively.

The LP model of the given problem can be written as:

Maximize (total profit) $Z = 300x_1 + 400x_2$

subject to the constraints

$$\text{(i)}\ \left.\begin{aligned} 5x_1 + 4x_2 &\le 200 \\ 3x_1 + 5x_2 &\le 150 \end{aligned}\right\}\text{(Input)}$$

$$\text{(ii)}\ \left.\begin{aligned} 5x_1 + 4x_2 &\ge 100 \\ 8x_1 + 4x_2 &\ge 80 \end{aligned}\right\}\text{(Output)}$$

and $x_1, x_2 \ge 0$

For solving this LP problem graphically, plotting on a graph each constraint by treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region as shown in Fig. 3.16.

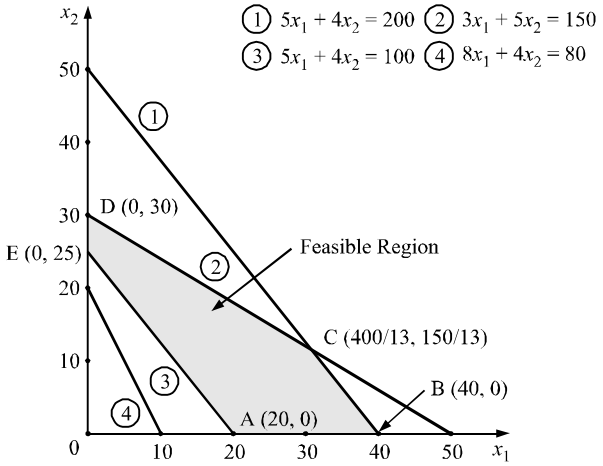


Fig. 3.16 Graphical Solution of LP Problem

The coordinates of extreme points of the feasible region are: A = (20, 0), B = (40, 0), C = (400/13, 150/13), D = (0, 30) and E = (0, 25). The value of the objective function at each of these extreme points is give in Table 3.16.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 300x_1 + 400x_2$
A	(20, 0)	300(20) + 400(0) = 6,000
B	(40, 0)	300(40) + 400(0) = 12,000
C	(400/13, 150/13)	300(400/13) + 400(150/13) = **1,80,000/13**
D	(0, 30)	300(0) + 400(30) = 12,000
E	(0, 25)	300(0) + 400(25) = 10,000

Table 3.16 Set of Feasible Solutions

The maximum (optimal) value of the objective function occurs at the extreme point (400/13, 150/13). Hence, the manager of the oil refinery should produce, $x_1 = 400/13$ units under process 1 and $x_2 = 150/13$ units under process 2 in order to achieve the maximum profit of Rs 1,80,000/13.

Example 3.17 A manufacturer produces two different models – X and Y of the same product. Model X makes a contribution of Rs 50 per unit and model Y, Rs 30 per unit, towards total profit. Raw materials r_1 and r_2 are required for production. At least 18 kg of r_1 and 12 kg of r_2 must be used daily. Also at most 34 hours of labour are to be utilized. A quantity of 2 kg of r_1 is needed for model X and 1 kg of r_1 for model Y. For each of X and Y, 1 kg of r_2 is required. It takes 3 hours to manufacture model X and 2 hours to manufacture model Y. How many units of each model should be produced in order to maximize the profit?

Solution Let us define the following decision variables:

x_1 and x_2 = number of units of model X and Y to be produced, respectively.

Then the LP model of the given problem can be written as:

Maximize (total profit) $Z = 50x_1 + 30x_2$

subject to the constraints

$$\left.\begin{array}{ll} \text{(i)} & 2x_1 + x_2 \geq 18 \\ & x_1 + x_2 \geq 12 \end{array}\right\} \text{(Raw material)}$$

$$\text{(ii)} \quad 3x_1 + 2x_2 \leq 34 \qquad \text{(Labour hours)}$$

and $x_1, x_2 \geq 0$

For solving this LP problem graphically, plotting on a graph each constraint by treating it as a linear equation. Then use the inequality sign of each constraint to mark the feasible region (shaded area) as shown in Fig. 3.17.

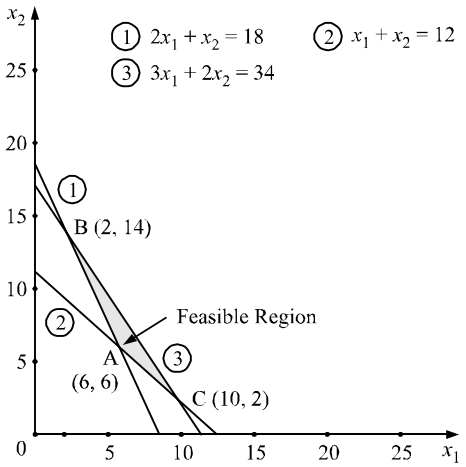


Fig. 3.17 Graphical Solution of LP Problem

The coordinates of extreme points of the feasible region are: A = (6, 6), B = (2, 14), and C = (10, 2). The value of the objective function at each of these points is given in Table 3.17.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 50x_1 + 30x_2$
A	(6, 6)	50(6) + 30(6) = 480
B	(2, 14)	50(2) + 30(14) = 520
C	(10, 2)	50(10) + 30(2) = **560**

Table 3.17 Set of Feasible Solutions

Since the maximum (optimal) value of $Z = 560$ occurs at the point C (10, 2), the manufacturer should produce $x_1 = 10$ units of model X and $x_2 = 2$ units of Y in order to yield the maximum profit of Rs 560.

Example 3.18 An advertising agency wishes to reach two types of audiences – customers with annual income greater than one lakh rupees (target audience A) and customers with annual income of less than one lakh rupees (target audience B). The total advertising budget is Rs 2,00,000. One programme of TV advertising costs Rs 50,000; one programme of radio advertising costs Rs 20,000. For contract reasons, at least three programmes ought to be on TV and the number of radio programmes must be limited to 5. Surveys indicate that a single TV programme reaches 4,50,000 prospective customers in target audience A and 50,000 in target audience B. One radio programme reaches 20,000 prospective customers in target audience A and 80,000 in target audience B. Determine the media mix so as to maximize the total reach.

[Delhi Univ., MBA, 2008]

Solution Let us define the following decision variables:

x_1 and x_2 = number of programmes to be released on TV and radio, respectively.

Then the LP model of the given problem can be expressed as:

Maximize $Z = (4{,}50{,}000 + 50{,}000)\, x_1 + (20{,}000 + 80{,}000)\, x_2$

subject to the constraints

(i) $50{,}000x_1 + 20{,}000x_2 \le 2{,}00{,}000$ or $5x_1 + 2x_2 \le 20$ (Budget available)

(ii) $x_1 \ge 3$; $x_2 \le 5$ (Programmes)

and $x_1, x_2 \ge 0$

For solving this LP problem by graphical method, plot each constraint on a graph first treating it as a liner equation. Then use inequality sign of each constraint to mark feasible solution region (shaded area) as shown in Fig. 3.18.

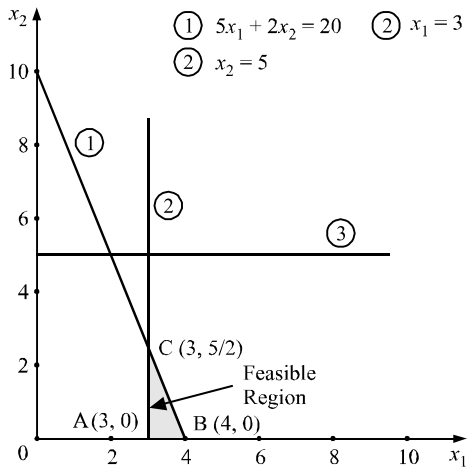


Fig. 3.18 Graphical Solution of LP Problem

The coordinates of the extreme points of feasible region are: A = (3, 0), B = (4, 0), and C = (3, 5/2). The value of the objective function at each of these three corner points is given in Table 3.18.

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 5{,}00{,}000x_1 + 1{,}00{,}000x_2$
A	(3, 0)	5,00,000(3) + 1,00,000(0) = 15,00,000
B	(4, 0)	5,00,000(4) + 1,00,000(0) = **20,00,000**
C	(3, 5/2)	5,00,000(3) + 1,00,000(5/2) = 17,50,000

Table 3.18 Set of Feasible Solutions

Since the maximum (optimal) value of $Z = 20{,}00{,}000$ occurs at the point B (4, 0), the agency must release, $x_2 = 4$ programmes on TV and $x_2 = 0$ (no programme on radio) in order to achieve the maximum reach of $Z = 20{,}00{,}000$ audiences.

3.3.5 Iso-profit (Cost) Function Line Method

According to this method, the optimal solution is found by using the *slope* of the objective function line (or equation). An iso-profit (or cost) line is a collection of points that give solution with the same value of objective function. By assigning various values to Z, we get different profit (cost) lines. Graphically many such lines can be plotted parallel to each other. The steps of iso-profit (cost) function method are as follows:

Step 1: Identify the feasible region and extreme points of the feasible region.

Step 2: Draw an iso-profit (iso-cost) line for an arbitrary but small value of the objective function without violating any of the constraints of the given LP problem. However, it is simple to pick a value that gives an integer value to x_1 when we set $x_2 = 0$ and vice-versa. A good choice is to use a number that is divided by the coefficients of both variables.

Iso-profit (or cost) function line is a straight line that represents all non-negative combinations of x_1 and x_2 variable values for a particular profit (or cost) level.

Step 3: Move iso-profit (iso-cost) lines parallel in the direction of increasing (decreasing) objective function values. The farthest iso-profit line may intersect only at one corner point of feasible region providing a single optimal solution. Also, this line may coincide with one of the boundary lines of the feasible area. Then *at least* two optimal solutions must lie on two adjoining corners and others will lie on the boundary connecting them. However, if the iso-profit line goes on without limit from the constraints, then an unbounded solution would exist. This usually indicates that an error has been made in formulating the LP model.

Step 4: An extreme (corner) point touched by an iso-profit (or cost) line is considered as the optimal solution point. The coordinates of this extreme point give the value of the objective function.

Example 3.19 Consider the LP problem

Maximize $Z = 15x_1 + 10x_2$

subject to the constraints

(i) $4x_1 + 6x_2 \le 360$, (ii) $3x_1 \le 180$, (iii) $5x_2 \le 200$

and $x_1, x_2 \ge 0$.

Solution: Plot each constraint on a graph first treating it as a linear equation. Then use inequality sign of each constraint to mark feasible solution region (shaded area) as shown in Fig. 3.19.

A family of lines (shown by dotted lines) that represents various values of objective function is shown in Fig. 3.19. Such lines are referred as iso-profit lines.

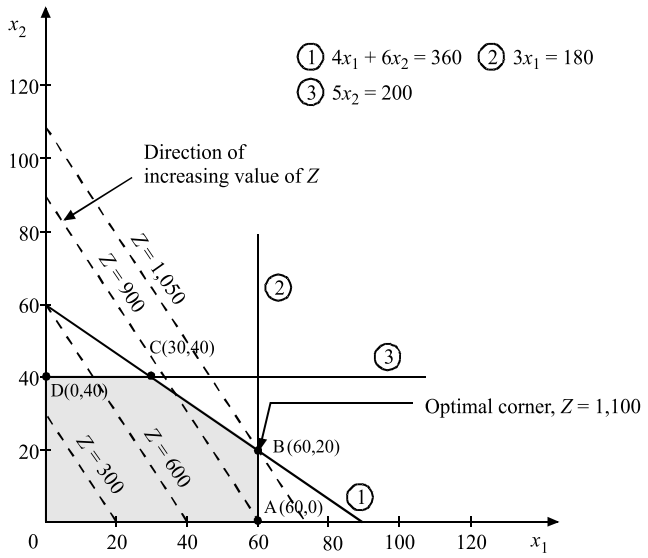


Fig. 3.19 Optimal Solution (Iso-profit Function Approach)

In Fig. 3.19, a value of $Z = 300$ is arbitrarily selected. The iso-profit (objective) function equation then becomes: $15x_1 + 10x_2 = 300$. This equation is also plotted in the same way as other equality constraints plotted before. This line is then moved upward until it first intersects a corner (or corners) in the feasible region (corner B). The coordinates of corner point B can be read from the graph or can be computed as the intersection of the two linear equations.

The coordinates $x_1 = 60$ and $x_2 = 0$ of corner point B satisfy the given constraints and the total profit obtained is $Z = 1{,}100$.

3.3.3 Comparison of Two Graphical Solution Methods

After having plotted the constraints of the given LP problem to locate the feasible solution region (area) one of the two graphical solution methods may be used to get optimal value of the given LP problem.

Extreme Point Method	*Iso-Profit (or Cost) Method*
(i) Identify coordinates of each of the extreme (or corner) points of the feasible region by either drawing perpendiculars on the x-axis and the y-axis or by solving two intersecting equations.	(i) Determine the slope (x_1, x_2) of the objective function and then join intercepts to identify the profit (or cost) line.
(ii) Compute the profit (or cost) at each extreme point by substituting that point's coordinates into the objective function.	(ii) In case of maximization, maintain the same slope through a series of parallel lines, and move the line upward towards the right until it touches the feasible region at only one point. But in case of minimization, move downward towards left until it touches only one point in the feasible region.
(iii) Identify the optimal solution at that extreme point with highest profit in a maximization problem or lowest cost in a minimization problem.	(iii) Compute the coordinates of the point touched by the iso-profit (or cost) line on the feasible region. (iv) Compute the profit or cost.

3.4 SPECIAL CASES IN LINEAR PROGRAMMING

3.4.1 Alternative (or Multiple) Optimal Solutions

We have seen that the optimal solution of any linear programming problem occurs at an extreme point of the feasible region and that the solution is unique, i.e. no other solution yields the same value of the objective function. However, in certain cases, a given LP problem may have more than one solution yielding the same optimal objective function value. Each of such optimal solutions is termed as *alternative optimal solution.*

Alternative optimal solutions are arrived only when slope of the objective function is same as the slope of a constraint in the LP problem.

There are two conditions that should be satisfied for an alternative optimal solution to exist:

(i) The slope of the objective function should be the same as that of the constraint forming the boundary of the feasible solutions region, and

(ii) The constraint should form a boundary on the feasible region in the direction of optimal movement of the objective function. In other words, the constraint should be an active constraint.

Remark The constraint is said to be *active* (*binding* or *tight*), if at the point of optimality, the left-hand side of a constraint equals the right-hand side. In other words, an equality constraint is always active, and inequality constraint may or may not be active.

Geometrically, an *active* constraint is one that passes through one of the extreme points of the feasible solution space.

Example 3.20 Use the graphical method to solve the following LP problem.

Maximize $Z = 10x_1 + 6x_2$

subject to the constraints

(i) $5x_1 + 3x_2 \le 30$, (ii) $x_1 + 2x_2 \le 18$

and $x_1, x_2 \ge 0$.

Solution The constraints are plotted on a graph by first treating these as equations and then their inequality signs are used to identify feasible region (shaded area) as shown in Fig. 3.20. The extreme points of the region are O, A, B and C.

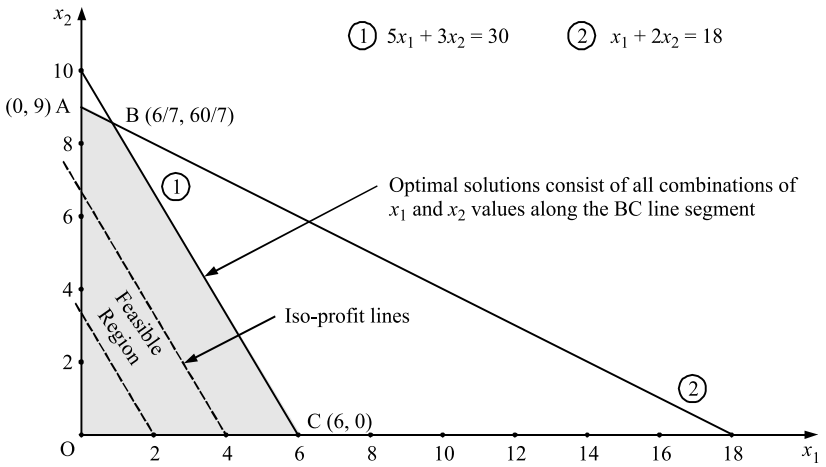


Fig. 3.20 Graphical Solution — Multiple Optima

Since objective function (iso-profit line) is parallel to the line BC (first constraint : $5x_1 + 3x_2 = 30$), which also falls on the boundary of the feasible region. Thus, as the iso-profit line moves away from the origin, it coincides with the line BC of the constraint equation line that falls on the boundary of the feasible region. This implies that an optimal solution of LP problem can be obtained at any point between B and C including extreme points B and C on the same line. Therefore, several combinations of values of x_1 and x_2 give the same value of objective function.

The value of variables x_1 and x_2 obtained at extreme points B and C should only be considered to establish that the solution to an LP problem will always lie at an extreme point of the feasible region.

The value of objective function at each of the extreme points is shown in Table 3.19.

Table 3.19 Set of Feasible Solutions

Extreme Point	*Coordinates* (x_1, x_2)	*Objective Function Value* $Z = 10x_1 + 6x_2$
O	(0, 0)	10(0) + 6(0) = 0
A	(0, 9)	10(0) + 6(9) = 54
B	(6/7, 60/7)	10(6/7) + 6(60/7) = 60
C	(6, 0)	10(6) + 6(0) = 60

Since value (maximum) of objective function, Z = 60 at two different extreme points B and C is same, therefore two alternative solutions: $x_1 = 6/7$, $x_2 = 60/7$ and $x_1 = 6$, $x_2 = 0$ exist.

Unbounded solution exists when the value of a decision variables can be made infinitely large without violating any of the LP problem constraints.

Remark If slope of a constraint is parallel to the slope of the objective function, but it does not fall on the boundary of the feasible region, the multiple solutions will not exist. This type of a constraint is called *redundant constraint* (a constraint whose removal does not change the feasible region.)

3.4.2 Unbounded Solution

Sometimes an LP problem may have an infinite solution. Such a solution is referred as an *unbounded solution*. It happens when value of certain decision variables and the value of the objective function (maximization case) are permitted to increase infinitely, without violating the feasibility condition.

It may be noted that there is a difference between unbounded *feasible region* and *unbounded solution* to a LP problem. It is possible that for a particular LP problem the feasible region may be unbounded but LP problem solution may not be unbounded, i.e. an unbounded feasible region may yield some definite value of the objective function. In general, an unbounded LP problem solution exists due to improper formulation of the real-life problem.

Example 3.21 Use the graphical method to solve the following LP problem:

Maximize $Z = 3x_1 + 4x_2$

subject to the constraints

(i) $x_1 - x_2 = -1$ (ii) $-x_1 + x_2 \le 0$

and $x_1, x_2 \ge 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use the inequality condition of each constraint to mark the feasible region (shaded area) as shown in Fig. 3.21.

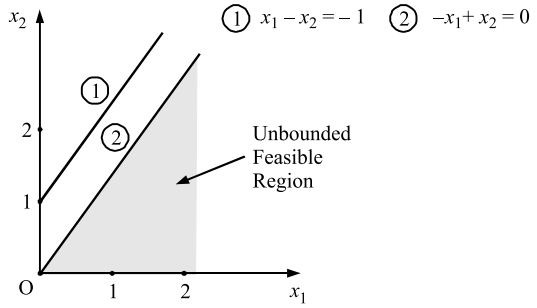


Fig. 3.21 Unbounded Solution

It may be noted from Fig. 3.21 that there exist an infinite number of points in the convex region for which the value of the objective function increases as we move from the extreme point (origin), to the right. That is, the value of variables x_1 and x_2 can be made arbitrarily large and accordingly the value of objective function Z will also increase. Thus, the LP problem has an unbounded solution.

Example 3.22 Use graphical method to solve the following LP problem:

Maximize $Z = 3x_1 + 2x_2$

subject to the constraints

(i) $x_1 - x_2 \geq 1$ (ii) $x_1 + x_2 \geq 3$

and $x_1, x_2 \geq 0$.

Solution Plot on a graph each constraint by first treating it as a linear equation. Then use the inequality sign of each constraint to mark the feasible region (shaded area) as shown in Fig. 3.22.

It may be noted that the shaded region (solution space) is unbounded from above. The two corners of the region are, A = (0, 3) and B = (2, 1). The value of the objective function at these corners is: Z(A) = 6 and Z(B) = 8.

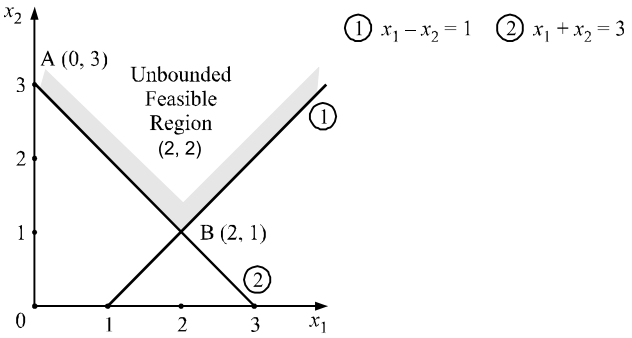


Fig. 3.22 Graphical Solution of LP Problem

Since the given LP problem is of maximization, there exist a number of points in the shaded region for which the value of the objective function is more than 8. For example, the point (2, 2) lies in the region and the objective function value at this point is 10 which is more than 8. Thus, as value of variables x_1 and x_2 increases arbitrarily large, the value of Z also starts increasing. Hence, the LP problem has an unbounded solution.

Example 3.23 Use graphical method to solve the following LP problem:

Maximize $Z = 5x_1 + 4x_2$

subject to the constraints

(i) $x_1 - 2x_2 \geq 1$, (ii) $x_1 + 2x_2 \geq 3$

and $x_1, x_2 \geq 0$ [*PT Univ., BE, 2002*]

Solution Constraints are plotted on a graph as usual as shown in Fig. 3.23. The solution space (shaded area) is bounded from below and unbounded from above.

The two extreme points of the solution space are, A(0, 3/2) and B(2, 1/2). The value of objective function at these points is Z(A) = 6 and Z(B) = 12. Since the given LP problem is of maximization, there exists a number of points in the solution space where the value of objective function is much more than 12. Hence, the unique value of Z cannot be found. The problem, therefore, has an unbounded solution.

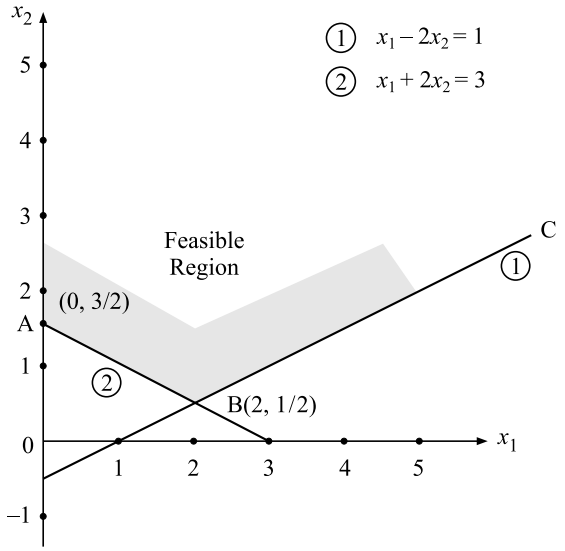


Fig. 3.23 Unbounded Solution

Example 3.24 Use graphical method to solve the following LP problem.

Maximize $Z = -4x_1 + 3x_2$

subject to the constraints

(i) $x_1 - x_2 \leq 0$, (ii) $x_1 \leq 4$

and $x_1, x_2 \geq 0$. [*Punjab Univ., B Com, 2009*]

Solution The solution space satisfying the constraints is shown shaded in Fig 3.24. The line $x_1 - x_2 = 0$ is drawn by joining origin point (0, 0) and has a slope of 45°. Also as $x_1 - x_2 \leq 0$, i.e. $x_1 \leq x_2$, the solution space due to this line is in the upward direction.

An infeasible solution lies outside the feasible region, it violates one or more of the given constraints.

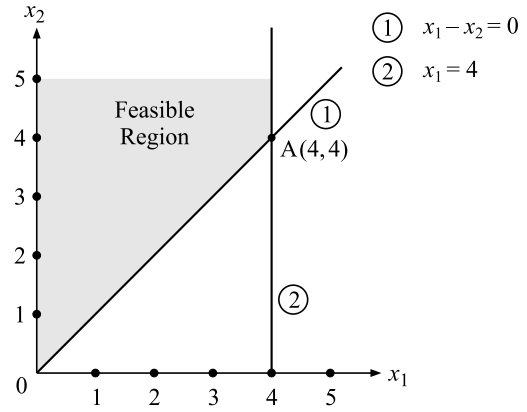


Fig. 3.24 Unbounded Solution

Since objective function is of maximization, therefore, the value of Z can be made arbitrarily large. Hence, this LP problem has an unbounded solution. Value of variable x_1 is limited to 4, while value of variable x_2 can be increased indefinitely.

3.4.3 Infeasible Solution

An infeasible solution to an LP problem arises when there is no solution that satisfies all the constraints simultaneously. This happens when there is no unique (single) feasible region. This situation arises when a LP model that has conflicting constraints. Any point lying outside the feasible region violates one or more of the given constraints.

Example 3.25 Use the graphical method to solve the following LP problem:

Maximize $Z = 6x_1 - 4x_2$

subject to the constraints

(i) $2x_1 + 4x_2 \le 4$ (ii) $4x_1 + 8x_2 \ge 16$

and $x_1, x_2 \ge 0$

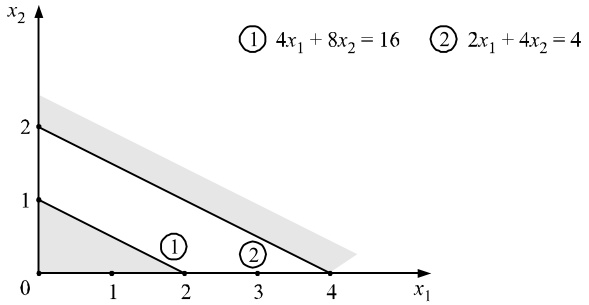


Fig. 3.25 An Infeasible Solution

Solution The constraints are plotted on graph as usual as shown in Fig. 3.25. Since there is no unique feasible solution space, therefore a unique set of values of variables x_1 and x_2 that satisfy all the constraints cannot be determined. Hence, there is no feasible solution to this LP problem because of the conflicting constraints.

Example 3.26 Use the graphical method to solve the following LP problem:

Maximize $Z = x_1 + \dfrac{x_2}{2}$

subject to the constraints

(i) $3x_1 + 2x_2 \le 12$ (ii) $5x_1 = 10$

(iii) $x_1 + x_2 \ge 8$ (iv) $-x_1 + x_2 \ge 4$

and $x_1, x_2 \ge 0$

Solution The constraints are plotted on graph as usual and feasible regions are shaded as shown in Fig. 3.26. The three shaded areas indicate non-overlapping regions. All of these can be considered feasible solution space because they all satisfy some subsets of the constraints.

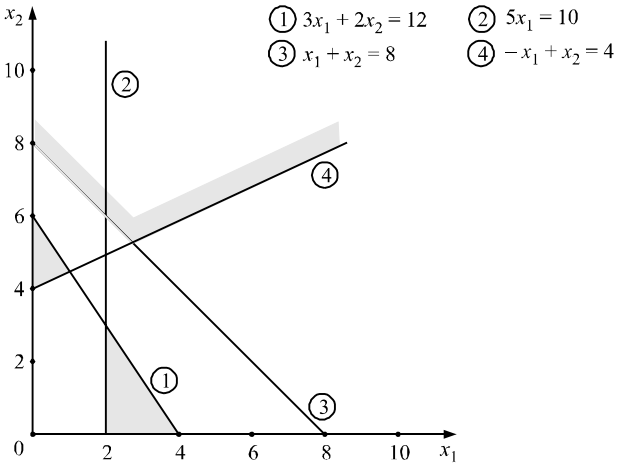


Fig. 3.26 An Infeasible Solution

However, there is no unique point (x_1, x_2) in these shaded regions that can satisfy all the constraints simultaneously. Thus, the LP problem has an infeasible solution.

Example 3.27 Use the graphical method to solve the following LP problem:

Maximize $Z = 3x + 2y$

subject to the constraints

(i) $-2x + 3y \le 9$, (ii) $3x - 2y \le -20$

and $x, y \ge 0$. [*Punjab Univ., BE (Elect.) 2006*]

Solution There are two solution spaces (shaded areas) shown in the Fig. 3.27. One of these solution space is satisfying the constraint $-2x + 3y \leq 9$ while the other is satisfying the constraint $3x - 2y \leq -20$. These two shaded regions in the first quadrant do not overlap and hence there is no point (x, y) common to both the shaded regions. This implies that the feasible solution to the problem does not exist. Consequently, this LP problem has one infeasible solution.

Redundancy is a situation in which one or more constraints do not affect the feasible solution region.

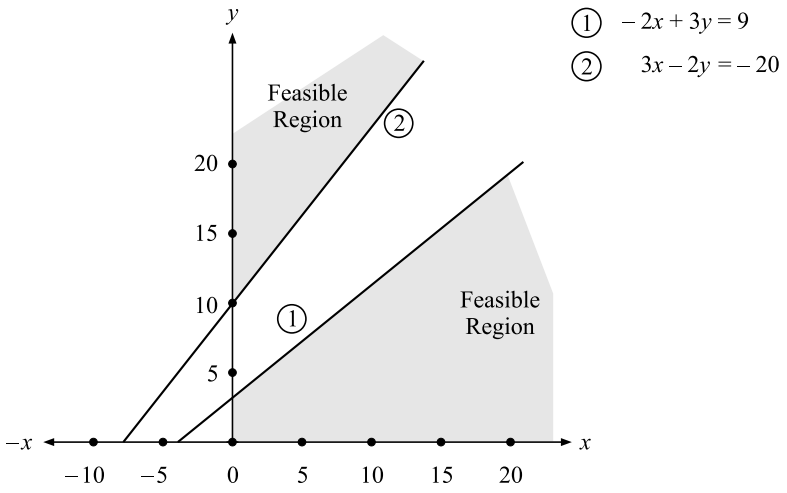


Fig. 3.27 No Feasible Solution

3.4.4 Redundancy

A redundant constraint is one that does not affect the feasible solution region (or space) and thus redundancy of any constraint does not cause any difficulty in solving an LP problem graphically. As shown in Fig. 3.16, constraint $8x_1 + 4x_2 \geq 80$ (Ex. 3.16) is redundant. In other words, a constraint is said to be redundant when it may be more binding (restrictive) than another.

CONCEPTUAL QUESTIONS

1. Explain graphical method of solving an LP problem.
2. Give a brief description of an LP problem with illustrations. How can it be solved graphically?
3. What is meant by the term 'feasible region'? Why must this be a well-defined boundary for the maximization problem?
4. Define iso-profit and iso-cost lines. How do these help us to obtain a solution to an LP problem?
5. Explain the procedure of generating extreme point solutions to an LP problem, pointing out the assumptions made, if any.
6. It has been said that each LP problem, that has a feasible region, has an infinite number of solutions. Explain.
7. You have just formulated a maximization LP problem and are preparing to solve it graphically. What criteria should you consider in deciding whether it would be easier to solve the problem by the extreme point enumeration method or the iso-profit line method.
8. Under what condition is it possible for an LP problem to have more than one optimal solution? What do these alternative optimal solutions represent?
9. What is an infeasible solution, and how does it occur? How is this condition recognized in the graphical method?
10. What is an unbounded solution, and how is this condition recognized in the graphical method?

SELF PRACTICE PROBLEMS

1. Comment on the solution of the following LP problems.

(i) Max $Z = 4x_1 + 4x_2$
subject to $x_1 + 2x_2 \leq 10$
$6x_1 + 6x_2 \leq 36$
$x_1 \leq 6$
and $x_1, x_2 \geq 0$

(ii) Max $Z = 5x_1 + 3x_2$
subject to $4x_1 + 2x_2 \leq 8$
$x_1 \geq 3$
$x_2 \geq 7$
and $x_1, x_2 \geq 0$

(iii) Max $Z = 4x_1 + 2x_2$
subject to $-x_1 + 2x_2 \leq 6$
$-x_1 + x_2 \leq 2$
and $x_1, x_2 \geq 0$

(iv) Min $Z = x_1 - 2x_2$
subject to $-2x_1 + x_2 \le 8$
$-x_1 + 2x_2 \le -24$
and $x_1, x_2 \ge 0$

(v) Max $Z = 6x_1 + 12x_2$
subject to $x_1 + 2x_2 \le 10$
$2x_1 - 5x_2 \le 20$
$x_1 + x_2 \le 15$
and $x_1, x_2 \ge 0$

(vi) Max $Z = 3x_1 + 5x_2$
subject to $x_1 + x_2 \ge 100$
$5x_1 + 10x_2 \le 400$
$6x_1 + 8x_2 \le 440$
and $x_1, x_2 \ge 0$

(vii) Max $Z = 1.75\,x_1 + 1.5\,x_2$
subject to $8x_1 + 5x_2 \le 320$
$4x_1 + 5x_2 \le 20$
$x_1 \ge 15;\ x_2 \ge 10$
and $x_1, x_2 \ge 0$

(viii) Max $Z = 3x_1 + 2x_2$
subject to $-2x_1 + 3x_2 \le 9$
$x_1 - 5x_2 \ge -20$
and $x_1, x_2 \ge 0$

(ix) Max $Z = 5x_1 + 3x_2$
subject to $3x_1 + 5x_2 \le 15$
$5x_1 + 2x_2 \le 10$
and $x_1, x_2 \ge 0$
[*Kerala, BSc (Engg.), 1995*]

(x) Max $Z = x_1 + x_2$
subject to $x_1 + x_2 \le 1$
$-3x_1 + x_2 \ge 3$
and $x_1, x_2 \ge 0$

2. Solve the following LP problems graphically and state what your solution indicates.

(i) Min $Z = 4x_1 - 2x_2$
subject to $x_1 + x_2 \le 14$
$3x_1 + 2x_2 \ge 36$
$2x_1 + x_2 \le 24$
and $x_1, x_2 \ge 0$

(ii) Min $Z = 3x_1 + 5x_2$
subject to $-3x_1 + 4x_2 \le 12$
$2x_1 - x_2 \ge -2$
$2x_1 + 3x_2 \ge 12$
$x_1 \le 4;\ x_2 \ge 2$
and $x_1, x_2 \ge 0$

(iii) Min $Z = 20x_1 + 10x_2$
subject to $x_1 + 2x_2 \le 40$
$3x_1 + x_2 \ge 30$
$4x_1 + 3x_2 \ge 60$
and $x_1, x_2 \ge 0$

(iv) Max $Z = x_1 + x_2$
subject to $x_1 - x_2 \ge 0$
$3x_1 - x_2 \le -3$
and $x_1, x_2 \ge 0$

3. A manufacturing firm produces two machine parts P_1 and P_2 of a machine. For this it makes use of milling and grinding machines. The different machining times required for each part, the machining times available on different machines and the profit on each machine part are as given below:

Machine	*Manufacturing Time Required (min)*		*Maximum Time Available per Week (min)*
	P_1	P_2	
Lathe	10	1.5	2,500
Milling machine	4	10	2,000
Grinding machine	1	1.5	450
Profit per unit (Rs)	50	100	

Determine the number of pieces of P_1 and P_2 to be manufactured per week in order to maximize profit.

4. A furniture manufacturer makes two products: chairs and tables. These products are processed using two machines – A and B. A chair requires 2 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours per day available on machine A and 30 hours on machine B. The profit gained by the manufacturer from a chair is Rs 2 and from a table is Rs 10. Solve this problem to find the daily production of each of the two products.

5. A company produces two types of leather belts, A and B. Belt A is of a superior quality and B is of an inferior quality. The profit from the two are 40 and 30 paise per belt, respectively. Each belt of type A requires twice as much time as required by a belt of type B. If all the belts were of type B, the company could produce 1,000 belts per day. But the supply of leather is sufficient only for 800 belts per day. Belt A requires a fancy buckle and only 400 of them are available per day. For belt B only 700 buckles are available per day. Solve this problem to determine how many units of the two types of belts the company should manufacture in order to have the maximum overall profit?

6. Consider a small plant which makes two types of automobile parts, say A and B. It buys castings that are machined, bored and polished. The capacity of machining is 25 per hour for A and 40 per hour for B, the capacity of boring is 28 per hour for A and 35 per hour for B, and the capacity of polishing is 35 per hour for A and 25 per hour for B. Casting for part A costs Rs 2 each and for part B it costs Rs 3 each. The company sells these for Rs 5 and Rs 6, respectively. The three machines have running costs of Rs 20, Rs 14 and Rs 17.50 per hour. Assuming that any combination of parts A and B can be sold, what product mix would maximize profit? Solve this problem using the graphical method.

7. A company machines and drills two castings X and Y. The time required to machine and drill one casting including machine set-up time is as follows:

Casting	*Machine Hours*	*Drilling Hours*
X	4	2
Y	2	5

There are two lathes and three drilling machines. The working week is of 40 hours; there is no lost time and overtime. Variables costs for both castings are Rs 120 per unit while total fixed costs amount to Rs 1,000 per week. The selling price of casting X is Rs 300 per unit and that of Y is Rs 360 per unit. There are no limitations on the number of X and Y casting that can be sold. The company wishes to maximize its profit. You are required to (i) formulate a linear programming model for the problem and (ii) solve this problem using the graphical method. [*Delhi Univ., MBA, 2007*]

8. A company possesses two manufacturing plants, each of which can produce three products: X, Y and Z from a common raw material. However, the proportions in which these products can be produced are different in each plant

and so are the plant's operating costs per hour. The data on production per hour and costs together with current orders in hand for each product, is as follows:

Plant	Products			Operating Cost per Hour (Rs)
	X	Y	Z	
A	2	4	3	9
B	4	3	2	10
Orders on hand	50	24	60	

You are required to use the graphical method to find out the number of production hours needed to fulfil the orders on hand at minimum cost.

9. A company manufactures two products: A and B. Product A yields a contribution of Rs 30 per unit and product B, Rs 40 per unit, towards the fixed costs. It is estimated that the sales of product A for the coming month will not exceed 20 units. Sales of product B have not yet been estimated but the company does have a contract to supply at least 10 units to a regular customer.

   There are 100 machine hours available for the coming month. It takes 4 hours to produce both – product A and product B. There are 180 labour hours available and products A and B require 4 hours and 6 hours of labour, respectively. Materials available are restricted to 40 units for the two products. One unit of each product requires one unit of material. The company wishes to maximize its profits. Use the graphical method, to find the optimum product mix.

10. A company manufactures two kinds of machines, each requiring a different manufacturing technique. The deluxe machine requires 18 hours of labour, 8 hours of testing and yields a profit of Rs 400. The standard machine requires 3 hours of labour, 4 hours of testing and yields a profit of Rs 200. There are 800 hours of labour and 600 hours of testing available each month. A marketing forecast has shown that the monthly demand for the standard machine is to be more than 150. The management wants to know the number of each model to be produced monthly that would maximize total profit. Formulate and solve this as a linear programming problem. [*Delhi Univ., MBA, 2003, 2008*]

11. A firm makes two types of furniture: chairs and tables. The contribution to profit by each product as calculated by the accounting department is – Rs 20 per chair and Rs 30 per table. Both products are to be processed on three machines M_1, M_2 and M_3. The time required in hours by each product and total time available in hours per week on each machine are as follows:

Machine	Chair	Table	Available Time (hrs)
M_1	3	3	36
M_2	5	2	50
M_3	2	6	60

How should the manufacturer schedule his production in order to maximize profit?

12. The production of a certain manufacturing firm involves a machining process that acquires raw materials and then converts them into (unassembled) parts. These parts are then sent to one of the two divisions for being assembly into the final product. Division 1 is used for product A, and Division 2 for product B. Product A requires 40 units of raw material and 10 hours of machine processing time. Product B requires 80 units of raw material and 4 hours of machine processing time. During the period, 800 units of raw material and 80 hours of machine processing time are available. The capabilities of the two assembly divisions during the period are 6 and 9 units, respectively. The profit contribution per unit to profit and overhead (fixed costs) is of Rs 200 for each unit of product A and of Rs 120 for each unit of product B. With this information, formulate this problem as a linear programming model and determine the optimal level of output for the two products using the graphic method.

13. A publisher of textbooks is in the process of bringing a new book in the market. The book may be bound by either cloth or hard paper. Each cloth-bound book sold contributes Rs 24 towards the profit and each paperbound book contributes Rs 23. It takes 10 minutes to bind a cloth cover, and 9 minutes to bind a paperback. The total available time for binding is 80 hours. After considering a number of market surveys, it is predicted that the cloth-cover sales will be anything more than 10,000 copies, but the paperback sales will not be more than 6,000 copies. Formulate this problem as a LP problem and solve it graphically.

14. PQR Feed Company markets two feed mixes for cattle. The feed mix, Fertilex, requires at least twice as much wheat as barley. The second mix, Multiplex, requires at least twice as much barley as wheat. Wheat costs Rs 1.50 per kg, and only 1,000 kg are available this month. Barley costs Rs 1.25 per kg and 1,200 kg is available: Fertilex sells for Rs 1.80 per kg up to 99 kg and each additional kg over 99 sells for Rs 1.65. Multiplex sells at Rs 1.70 per kg up to 99 kg and each additional kg over 99 sells for Rs 1.55 per kg. Bharat Farms is sure to buy any and all amounts of both mixes that PQR Feed Company will mix. Formulate this problem as a LP problem to determine the product mix so as to maximize the profits.

15. On October 1st, a company received a contract to supply 6,000 units of a specialized product. The terms of contract requires that 1,000 units be shipped in the north of October; 3,000 units in November and 2,000 units in December. The company can manufacture 1,500 units per month on regular time and 750 units per month on overtime. The manufacturing cost per item produced during regular time is Rs 3 and the cost per item produced during overtime is Rs 5. The monthly storage cost is Re 1. Formulate this problem as a linear programming problem so as to minimize the total cost.

16. A small-scale manufacturer has production facilities for producing two different products. Each of the products requires three different operations: grinding, assembly and testing. Product I requires 15, 20 and 10 minutes to grind, assemble and test, respectively, on the other hand product II requires 7.5, 40 and 45 minutes for grinding, assembling and testing, respectively. The production run calls for at least 7.5 hours of grinding time, at least 20 hours of assembly and at least 15 hours of testing time. If manufacturing product I costs Rs 60 and product II costs Rs 90, determine the number of units of each product the firm should produce in order to minimize the cost of operation.

17. A firm is engaged in breeding pigs. The pigs are fed on various products grown in the farm. Because of the need to ensure certain nutrient constituents, it is necessary to additionaly buy one or two products (call them A and B).

    The content of the various products (per unit) in the nutrient constituents (e.g. vitamins, proteins, etc.) is given in the following table:

Nutrient	Nutrient Content in Product		Minimum Amount of Nutrient
	A	B	
M_1	36	6	108
M_2	3	12	36
M_3	20	10	100

The last column of the above table gives the minimum quantity of nutrient constituents M_1, M_2 and M_3 that must be given to

the pigs. If products A and B cost Rs 20 and Rs 40 per unit, respectively, how much of each of these two products should be bought so that the total cost is minimized? Solve this LP problem graphically.

18. A company manufacturing television sets and radios has four major departments: chassis, cabinet, assembly and final testing. The monthly capacities of these are as follows:

Department	*Television Capacity*		*Radio Capacity*
Chassis	1,500	or	4,500
Cabinet	1,000	or	8,000
Assembly	2,000	or	4,000
Testing	3,000	or	9,000

The contribution of television is Rs 150 each and the contribution of radio is Rs 250 each. Assume that the company can sell any quantity of either product. Formulate this problem as an LP problem and solve it to determine the optimal combination of output.

19. The commander of a small tank has been ordered to win and occupy a valley located in a river delta area. He has 4 heavy tanks and 10 light tanks. Each heavy tank requires 4 men to operate, whereas the light tank requires 2 men. The total number of men available is 29. The fire power of the heavy tank is three times that of the light tank. Yet the commander feels that he should use more light tanks than heavy ones since the light ones are more effective against guerillas.
    (i) Formulate this problem as an LP model so as to determine the number of tanks in each type to be sent into the combat, keeping in mind that the commander wishes to maximize the total fire power.
    (ii) If you could persuade the commander that the rule 'more light tanks than heavy tanks' may not be applicable due to the fact that guerilla warfare is absent in the area, how many tanks of each type could be used?

20. A timber company cuts raw timber – oak and pine logs – into wooden boards. Two steps are required to produce boards from logs. The first step involves removing the bark from the logs. Two hours are required to remove the bark from 1,000 feet of oak logs and three hours per 1,000 feet of pine logs. After the logs have been debarked, they must be cut into boards. It takes 2.4 hours per 1,000 feet of oak logs to be cut into boards and 1.2 hours per 1,000 feet of pine logs. The bark removing machines can operate for up to 60 hours per week, while for cutting machines this number is limited to 48 hours per week. The company can buy a maximum of 18,000 feet of raw oak logs and 12,000 feet of raw pine logs each week. The profit per 1,000 feet of processed logs is Rs 1,800 and Rs 1,200 for oak and pine logs, respectively. Formulate this problem as an LP model and solve it to determine how many feet of each type of log should be processed each week in order to maximize the profit.

21. Upon completing the construction of his house, Mr Sharma discovers that 100 square feet of plywood scrap and 80 square feet of white pine scrap are in unusable form, which can be used for the construction of tables and bookcases. It takes 16 square feet of plywood and 16 square feet of white pine to construct a book case. By selling the finished products to a local furniture store, Mr Sharma can realize a profit of Rs 25 on each table and Rs 20 on each bookcase. How can he most profitably use the leftover wood? Use the graphical method to solve this LP problem.

22. A manufacturer produces electric hand saws and electric drills, for which the demand exceeds his capacity. The production cost of a saw is Rs 6 and the production cost of a drill Rs 4. The shipping cost is 20 paise for a saw and 30 paise for a drill. A saw sells for Rs 9 and a drill sells for Rs 5.50. The budget allows a maximum of Rs 2,400 for production costs and Rs 120 for shipping costs. Formulate this problem as an LP model and solve it to determine the number of saws and drills that should be produced in order to maximize the excess of sales over production and shipping costs.

23. A company produces two types of pens, say A and B. Pen A is of a superior quality and pen B is of an inferior quality. Profits on pen A and B are Rs 5 and Rs 3 per pen, respectively. The raw material required for producing single pen A is twice as that of pen B. The supply of raw material is sufficient only for 1,000 pens of B per day. Pen A requires a special clip and only 400 such clips are available per day. For pen B, only 700 clips are available per day. Solve this LP problem graphically to find the product mix so that the company can make the maximum profit. [*Delhi Univ., MBA, Nov. 1998*]

24. Two products A and B are to be manufactured. One single unit of product A requires 2.4 minutes of punch press time and 5 minutes of assembly time. The profit for product A is Re 0.60 per unit. One single unit of product B requires 3 minutes of punch press time and 2.5 minutes of welding time. The profit for product B is Re 0.70 per unit. The capacity of the punch press department available for these products is 1,200 minutes/week. The welding department has an idle capacity of 600 minutes/week and the assembly department has the capacity of 1,500 minutes/week. Formulate this problem as an LP model to determine the quantities of products A and B that would yield the maximum.

25. A rubber company is engaged in producing three different kinds of tyres A, B and C. These three different tyres are produced at two different plants of the company, which have different production capacities. In a normal 8-hour working day, Plant 1 produces 50, 100 and 100 tyres of types A, B and C, respectively. Plant 2, produces 60, 60 and 200 tyres of types A, B and C, respectively. The monthly demand for types A, B and C is 2,500, 3,000 and 7,000 units, respectively. The daily cost of operation of Plants 1 and 2 is Rs 2,500 and Rs 3,500, respectively. Formulate this problem as an LP model and solve it to determine how the company can minimize the number of days on which it operates, per month, at the two plants, so that the total cost is also minimized, while the demand is also met.

26. Kishore Joshi mixes pet food in his basement on a small scale. He advertises two types of pet food: Diet-Sup and Gro-More. Contribution from Diet-Sup is Rs 1.50 a bag and from Gro-More Rs 1.10 a bag. Both are mixed from two basic ingredients – a protein source and a carbohydrate source. Diet-Sup and Gro-More require ingredients in these amounts:

	Protein	*Carbohydrate*
Diet-Sup (7 kg bag)	4 kg	3 kg
Gro-More (3 kg bag)	2 kg	1 kg

Kishore has the whole weekend ahead of him, but will not be able to procure more ingredients over the weekend. He checks his bins and finds he has 700 kg of protein source and 500 kg of carbohydrate in the house. How many bags of each food should he mix in order to maximize his profits? Formulate this problem as an LP model and solve it to determine the optimum food-mix that would maximize the profit.

HINTS AND ANSWERS

1. (i) Alternative solutions (ii) Infeasible solution
(iii) Unbounded solution
(iv) Unbounded solution (v) Redundant constraints are second and third (vi) Infeasible solution
(vii) No feasible solution (viii) Unbounded solution
(ix) $x_1 = 20/19$, $x_2 = 45/19$ and Max $Z = 535/19$
(x) No feasible solution

2. (i) $x_1 = 8$, $x_2 = 6$ and Min $Z = 20$
(ii) $x_1 = 3$, $x_2 = 2$ and Min $Z = 19$
(iii) $x_1 = 6$, $x_2 = 12$ and Min $Z = 240$

3. Let x_1 and x_2 = number of pieces of P_1 and P_2 to be manufactured, respectively.

Max $Z = 50x_1 + 100x_2$
subject to (i) $10x_1 + 15x_2 \le 2{,}500$
(ii) $4x_1 + 10x_2 \le 2{,}000$
(iii) $x_1 + 15x_2 \le 450$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 187.5$, $x_2 = 125$ and Max Z = Rs 21,875.

4. Let x_1 and x_2 = number of chairs and tables produced, respectively.

Max $Z = 2x_1 + 10x_2$
subject to (i) $2x_1 + 5x_2 \le 16$; (ii) $6x_1 \le 30$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 0$, $x_2 = 3.2$ and Max Z = Rs 32.

5. Let x_1 and x_2 = number of belts of types A and B produced, respectively.

Max $Z = 0.40x_1 + 0.30x_2$
subject to (i) $x_1 + x_2 \le 800$; (ii) $2x_1 + x_2 \le 1{,}000$
(ii) $x_1 \le 400$; (iv) $x_2 \le 700$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 200$, $x_2 = 600$ and Max = Rs 260.

6. Let x_1 and x_2 = number of parts manufactured of automobile A and B, respectively.

$$\text{Max } Z = \{5-2+20/25+14/28+17.5/35\}x_1 + \{6-3+20/40+14/35+17.5/25\}x_2 = 4.80x_1 + 4.60x_2$$

subject to

(i) $\frac{x_1}{25} + \frac{x_2}{40} \le 1$, (ii) $\frac{x_1}{28} + \frac{x_2}{35} \le 1$, (iii) $\frac{x_1}{35} + \frac{x_2}{25} \le 1$

and $x_1, x_2 \ge 0$

7. Let x_1 and x_2 = number of units of casting X and Y to be manufactured, respectively.

Max $Z = (300x_1 + 360x_2) - (120x_1 + 120x_2) - 1{,}000$
$= 180x_1 + 240x_2 - 1{,}000$
subject to (i) $4x_1 + 2x_2 \le 2 \times 40$
(ii) $2x_1 + 5x_2 \le 3 \times 40$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 10$, $x_2 = 20$ and Max Z = 5,600.

8. Let x_1 and x_2 = number of production hours for plants A and B needed to complete the orders, respectively.

Min $Z = 9x_1 + 10x_2$
subject to (i) $2x_1 + 4x_2 \ge 50$; (ii) $4x_1 + 3x_2 \ge 24$
(iii) $3x_1 + 2x_2 \ge 60$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 35/2$ and $x_2 = 15/4$ and Min Z = Rs 195.

9. Let x_1 and x_2 = number of units of products A and B, respectively.

Max $Z = 30x_1 + 40x_2$
subject to (i) $x_1 \le 20$; $x_2 \ge 10$, (ii) $4x_1 + 4x_2 \le 100$,
(iii) $4x_1 + 6x_2 \le 180$, (iv) $x_1 + x_2 \le 40$
and $x_1, x_2 \ge 0$

11. Let x_1 and x_2 = number of chairs and tables to be made, respectively.

Max $Z = 20x_1 + 30x_2$
subject to (i) $3x_1 + 3x_2 \le 36$; (ii) $5x_1 + 2x_2 \le 50$
(iii) $2x_1 + 6x_2 \le 60$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 3$, $x_2 = 9$ and Max Z = Rs 330.

12. Let x_1 and x_2 = number of units to be produced of products A and B, respectively.

Max $Z = 200x_1 + 120x_2$
subject to (i) $40x_1 + 80x_2 \le 800$, (ii) $10x_1 + 04x_2 \le 80$,
(iii) $x_1 \le 6$; (iv) $x_2 \le 9$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 5$, $x_2 = 7.5$ and Max Z = Rs 1,900.

17. Let x_1 and x_2 = number of units of products A and B to be bought, respectively

Min (total cost) $Z = 20x_1 + 40x_2$
subject to (i) $36x_1 + 6x_2 \ge 108$; (ii) $3x_1 + 12x_2 \ge 36$
(iii) $20x_1 + 10x_2 \ge 100$
and $x_1, x_2 \ge 0$

Ans. $x_1 = 4$, $x_2 = 2$ and Min Z = Rs 160.

18. Let x_1 and x_2 = number of units of television and radio tubes produced, respectively.

Max (total profit) $Z = 150x_1 + 250x_2$
subject to (i) $x_1/1{,}500 + x_2/4{,}500 \ge 1$
(ii) $x_1/1{,}000 + x_2/8{,}000 \ge 1$
(iii) $x_1/2{,}000 + x_2/2{,}000 \ge 1$
(iv) $x_1/3{,}000 + x_2/9{,}000 \ge 1$
and $x_1, x_2 \ge 0$

20. Let x_1 and x_2 = number of feet of raw oak logs and raw pine logs to be processed each week, respectively

Max $Z = 1{,}800x_1 + 1{,}200x_2$
subject to (i) $2x_1 + 3x_2 \le 60$, (ii) $2.4x_1 + 1.2x_2 \le 48$,
(iii) $x_1 \le 18$, (iv) $x_2 \le 12$
and $x_1, x_2 \ge 0$

CHAPTER SUMMARY

The graphical solution approaches (or methods) provide a conceptual basis for solving large and complex LP problems. A graphical method is used to reach an optimal solution to an LP problem that has a number of constraints binding the objective function.

Both extreme point methods and the iso-profit (or cost) function line method are used for graphically solving any LP problem that has only two decision variables.

CHAPTER CONCEPTS QUIZ

True or False

1. The graphical method of solving linear programming problem is useful because of its applicability to many real life situations.
2. The problem of infeasibility in linear programming can only be solved by making additional resources available which in turn changes the constraints of the problem.
3. A linear programming problem is unbounded because constraints are incorrectly formulated.
4. If a LP problem has more than one solution yielding the same objective functional value, then such optimal solutions are known as alternative optimal solution.
5. Since the constraints to a linear programming are always linear, we can graph them by locating only two different points on a line.
6. An optimal solution does not necessarily use up all the limited resources available.
7. The intersection of any two constraints is an extreme point which is a corner of the feasible region.
8. When there are more than one optimal solution to the problem then the decision-maker will be unable to judge the best optimal solution among them.
9. If all the constraints are $\geq$ inequalities in a linear programming problem whose objective function is to be maximized, then the solution of the problem is unbounded.
10. The problem caused by redundant constraints is that two isoprofit lines may not be parallel to each other.

Fill in the Blanks

11. Constraints in a linear programming requiring all variables to be zero or positive are known as ..... constraints and rest from limited resources are referred as ..... constraints.
12. Instead of maximizing profit in linear programming problem, we ensure linearity of the objective function by maximizing .....
13. The ..... points of the convex set give the basic feasible solution to the linear programming.
14. A basic feasible solution is said to be ..... if the values of all ..... variables are nonzero and positive.
15. If an optimal solution to a linear programming problem exists, it will lie at ..... of the feasible solution.
16. A constraint in linear programming must be expressed as a linear ..... or a linear ..... .
17. When more than one solution best meets the objective of the linear programming problem then it is said to have ..... solution.
18. ..... occurs when no value of the variable is able to satisfy all the constraints in linear programming problem simultaneously.
19. An existence of objective of the problem for any firm is one of the major ..... of the linear programming problem.
20. Any two isoprofit or isocost lines for a given linear programming problem are ..... to each other.

Multiple Choice

21. The graphical method of LP problem uses
    (a) objective function equation
    (b) constraint equations
    (c) linear equations
    (d) all of the above
22. A feasible solution to an LP problem
    (a) must satisfy all of the problem's constraints simultaneously
    (b) need not satisfy all of the constraints, only some of them
    (c) must be a corner point of the feasible region
    (d) must optimize the value of the objective function
23. Which of the following statements is true with respect to the optimal solution of an LP problem
    (a) every LP problem has an optimal solution
    (b) optimal solution of an LP problem always occurs at an extreme point
    (c) at optimal solution all resources are completely used
    (d) if an optimal solution exists, there will always be at least one at a corner
24. An iso-profit line represents
    (a) an infinite number of solutions all of which yield the same profit
    (b) an infinite number of solution all of which yield the same cost
    (c) an infinite number of optimal solutions
    (d) a boundary of the feasible region
25. If an iso-profit line yielding the optimal solution coincides with a constraint line, then
    (a) the solution is unbounded
    (b) the solution is infeasible
    (c) the constraint which coincides is redundant
    (d) none of the above
26. While plotting constraints on a graph paper, terminal points on both the axes are connected by a straight line because
    (a) the resources are limited in supply
    (b) the objective function is a linear function
    (c) the constraints are linear equations or inequalities
    (d) all of the above
27. A constraint in an LP model becomes redundant because
    (a) two iso-profit line may be parallel to each other
    (b) the solution is unbounded
    (c) this constraint is not satisfied by the solution values
    (d) none of the above
28. If two constraints do not intersect in the positive quadrant of the graph, then
    (a) the problem is infeasible
    (b) the solution is unbounded
    (c) one of the constraints is redundant
    (d) none of the above
29. Constraints in LP problem are called active if they
    (a) represent optimal solution
    (b) at optimality do not consume all the available resources

(c) both of (a) and (b)
(d) none of the above

30. The solution space (region) of an LP problem is unbounded due to
(a) an incorrect formulation of the LP model
(b) objective function is unbounded
(c) neither (a) nor (b)
(d) both (a) and (b)

31. While solving a LP model graphically, the area bounded by the constraints is called
(a) feasible region (b) infeasible region
(c) unbounded solution (d) none of the above

32. Alternative solutions exist of an LP model when
(a) one of the constraints is redundant
(b) objective function equation is parallel to one of the constraints
(c) two constraints are parallel
(d) all of the above

33. While solving a LP problem, infeasibility may be removed by
(a) adding another constraint
(b) adding another variable
(c) removing a constraint
(d) removing a variable

34. If a non-redundant constraint is removed from an LP problem, then
(a) feasible region will become larger
(b) feasible region will become smaller
(c) solution will become infeasible
(d) none of the above

35. If one of the constraint of an equation in an LP problem has an unbounded solution, then
(a) solution to such LP problem must be degenerate
(b) feasible region should have a line segment
(c) alternative solutions exist
(d) none of the above

Answers to Quiz

1. T 2. F 3. F 4. T 5. T 6. T 7. T 8. F 9. T 10. T
11. non-negative, structural 12. Contribution 13. extreme 14. non-degenerative, basic,
15. an extreme point 16. equation, inequality 17. alternative 18. infeasibility
19. requirement 20. parallel.
21. (d) 22. (a) 23. (d) 24. (a) 25. (d) 26. (c) 27. (d) 28. (a) 29. (a) 30. (c)
31. (a) 32. (b) 33. (c) 34. (a) 35. (b)

CASE STUDY

Case 3.1: Raman Traveller

Mr. Raman, a local travel agent is planning a charter trip to a famous sea resort. The eight-day seven-night package includes the round-trip fare, surface transportation, boarding and lodging and selected tour options. The charter trip is restricted to 200 persons and past experience indicates that there will not be any problem in arranging 200 passangers. The problem faced by the travel agent is to determine the number of Deluxe Standard and Economy tour packages to offer for this charter. In all three of these plans, each differ in terms of their seating and service for the flight, quality of accommodation, meal plans and tour options. The following table summarizes the estimated prices of the three packages and the corresponding expenses of the travel agent. The travel agent has hired an aircraft for a flat fee of Rs 2,00,000 for the entire trip. The per person price and costs for the three packages are as follows:

Tour Plan	*Price (Rs)*	*Hotel Costs (Rs)*	*Meals and Other Expenses (Rs)*
Deluxe	10,000	3,000	4,750
Standard	7,000	2,200	2,500
Economy	6,500	1,900	2,200

In planning the trip, the following considerations have come up that need to be taken into account:

(i) At least 10 per cent of the packages must be of the deluxe type.
(ii) At least 35 per cent but not more than 70 per cent must be of the standard type.
(iii) At least 30 per cent must be of the economy type.
(iv) The maximum number of deluxe packages available in any aircraft is restricted to 60.
(v) The hotel desires that at least 120 of the tourists should be on the deluxe standard packages together.

Mr. Raman wishes to determine the number of packages to offer in each type of trip so as to maximize the total profit.

Case 3.2: Jain Clothing Stores

Mr. Vipin Jain, owner of Clothing Stores, is planning annual sale of shirts and pants. Mr Jain is planning to use two different forms of advertising, viz., radio and newspaper ads, in order to promote the sale. Based on past experience, Mr Jain feels confident that each newspaper advertisement will reach 40,000 shirt customers and 80,000 pant customers. Each radio advertisement, he believes, will reach 30,000 shirt customers and 20,000 pant customers. The cost of each newspaper advertisement is Rs 30,000 and the cost of each radio spot is Rs 45,000. The advertising agency will prepare the advertisements and it will require 5 man-hours of preparation for each newspaper advertisement and 15 man-hours of preparation for each radio spot. Mr Jain's sales manager says that a minimum of 75 man-hours should be spent on preparation of advertising in order to fully utilize the services of advertising agency. Mr Jain feels that in order to have a successful sale, the advertising must reach at least 3,60,000 shirt customers and at least 4,00,000 pant customers.

You as newly appointed management trainee are expected to suggest the media planning for advertisement at a minimum cost and still attain the objective of Mr. Jain.